
XDraw Library

Version 4.0

Reference Manual

Nova Software Labs

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CHAPTER 1: Introduction

The **NSL Draw Library** extends the capabilities of the OSF/Motif toolkit. It isolates the developer from the basic XLib drawing primitives by providing higher level interaction thus reducing time to market drastically.

The **Draw Library** implements drawing primitives with Motif widgets and gadgets, thus staying within a well accepted standard.

The XnsIDraw widget is an XmManager widget which applies constraint resources to its children. With these constraints, you can define attachment points on objects of your drawing and links that connect these attachment points. If an object is moved, any links associated to it are automatically updated.

The Draw Library makes animation of sophisticated drawings easy thanks to its convenience functions. Animations can be programmed directly in C in your application, or through the XDrawMaker drawing editor. In XDrawMaker, you can write animation scripts using the FACE language that was developed for XFaceMaker.

The Draw Library is used in a wide range of applications, including drawing editors, industrial process control, network monitoring, flight control, etc. All the widgets and gadgets in the library conform to the OSF/Motif style guidelines.

The **NSL Draw Library** includes the following libraries.

XnsIDraw NSL's graphic gadget and widget library and the associated convenience functions.

XnsITiff Image support for Tiff format

XnsIDf FACE language interpreter for the Draw library.

This volume contains all the information needed to work with the XnsIDraw library:

- instructions on how to build applications using the Draw Library.
- the reference pages for the XnsIDraw widget and the gadgets in standard Motif reference page format.
- the reference pages for the convenience functions.

Keyboard Conventions

Key combinations are expressed as follows:

Format	Example	Meaning
Key1--Key2	Ctrl--A	Hold down the C ontrol key as you press, then release key A . The s hift key is not used.
Key1 Key2	Esc a	Press and release the E scape key, then press and release key a .
	Esc A	Press and release the E scape key, then press and release key a while pressing the s hift key.

Table 1: Keyboard Conventions

Notice how in the case of the **ESC** modifier the two keys are pressed in succession, whereas in the case of the **Ctrl** modifier the key is pressed *while* the **Ctrl** key is held down.

Typesetting Conventions

The following typesetting conventions are used in this manual:

Type style	Used for
typewriter	Anything you should type as it appears, code, and key combinations.
<i>italic</i>	Mouse buttons and emphasis.
<i>bold slant</i>	Menu items and editing commands.

Table 2: Typesetting Conventions

Technical Support

Technical Support is available:

- Monday to Friday (except for French national holidays).
- 9.00 am 12.30 pm – 2.00 pm to 5:30 pm (MET). (5 pm on Friday).
- From French and English speaking support staff.
- By phone (33) [0]1 44 08 70 80 (ask for the Technical Support Department) or (33) [0]1 44 08 70 85.
- By electronic mail addressed to `support@nsl.fr`
- By FAX (33) [0]1 44 08 70 81 addressed to Technical Support including your return FAX number, and a telephone number where you can be reached.

Related Documents

The following reference books may be useful.

James Getty, Robert Scheifler, Robert Newman, *Xlib – C Language X Interface*, MIT X Consortium Standard, X Version 11, Release 5, Massachusetts Institute of Technology, Cambridge, Massachusetts

KL Group, *XRT/graph for Motif, Programmer's Guide & Reference Manual*, Version 2.2

KL Group, *XRT/3d for Motif, Programmer's Guide & Reference Manual*, Version 1.0

Joel McCormack, Paul Asente, Ralph R. Swick, *X Toolkit Intrinsics – C Language X Interface*, X Version 11, Release 4, Massachusetts Institute of Technology, Cambridge, Massachusetts

Nova Software Labs, *XFaceMaker User's Guide*, Version 3.3

Nova Software Labs, *XDrawMaker User's Guide*, Version 2.2

Open Software Foundation, *OSF/Motif Programmer's Guide*, Release 1.2

Open Software Foundation, *OSF/Motif Programmer's Reference*, Release 1.2

Open Software Foundation, *OSF/Motif Style Guide*, Release 1.2

O'Reilly & Associates, *The Definitive Guides to the X Window System*

Introduction

CHAPTER 2: Overview

This chapter describes the features of the **XnslDraw** library and the **XDrawMaker** editor. The **XnslDraw** library consists of the following main components:

- m The XnslDraw widget.
- m The XnslDrawText widget.
- m The XnslDraw primitive gadget classes.
- m The XnslDraw library functions.

In the product distribution, you will find examples of how to do things with the XnslDraw library. Please refer to the **demos** directory.

2.1 The XnslDraw Widget

The **XnslDraw** widget is a container widget (a subclass of the Motif Manager widget) able to contain any widget. It provides interactive editing capabilities. The **XnslDraw** widget can operate in two modes:

Edit Mode: Selection is enabled; pointer events are not dispatched to children. Once a widget is created with an **XnslDraw** widget as parent, the user can select, move and resize this widget. Selection is indicated by handles drawn on the selected widgets.

Run Mode: Events are dispatched to children. Children have their normal behavior.

2.1.1 Editing

Selection:

A feature of the **XnslDraw** widget is the capability of selecting and modifying children. The user can select a child by clicking on it or by dragging a rubber band, to enclose the widgets which are to be selected. The selection is indicated by handles on the selected widgets. For non-Draw widgets, the handles are placed at the corners of the widget. For Draw widgets, the handle locations are widget dependent. The selection may contain a special widget: the edit widget. This widget is the last clicked on widget (Figure 2.1, “The Selection Mechanism”).

Callbacks are called before and after a widget is selected. Before a selection, a **SelectVerifyCallback** is called. The application can then allow the widget to be selected or not. After a widget is selected, a **SelectCallback** is called to notify the application. A special callback is provided for the Edit Object. It is activated when a widget becomes the edit widget.

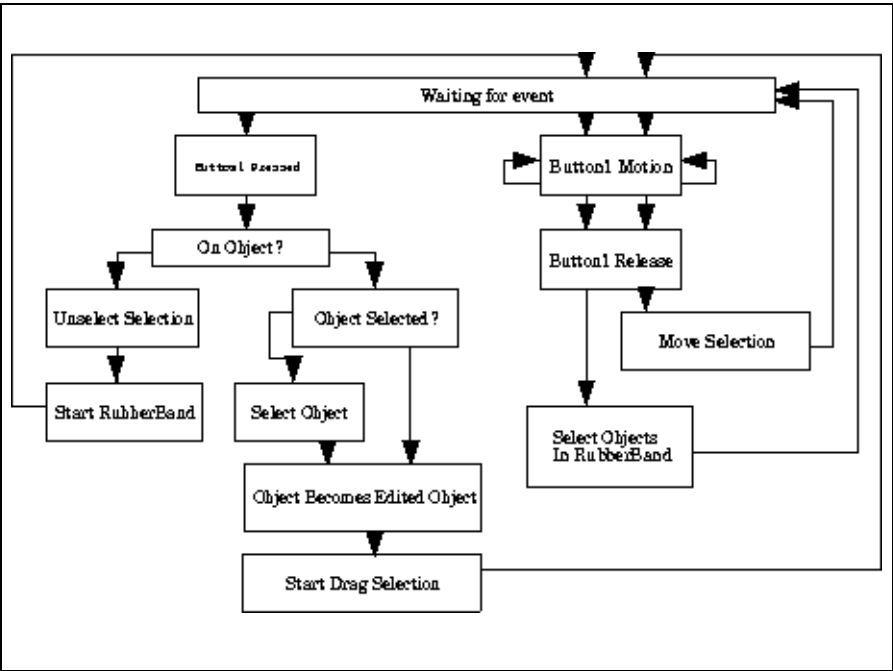


Figure 2.1: The Selection Mechanism

Grid:

The **XnsIDraw** widget includes a snap-on-grid mechanism. The grid is an array of points to which widget points are constrained if the grid is active. The grid can be shown on the screen or not shown. User input can be snapped to the grid or not.

Collect Reactors:

The XnsIDraw widget has a set of resources which provide a general mechanism for interactive user input. A set of functions associated to these resources handles the acquisition of the coordinates of a point, the drawing of the ghost images of rectangles or line segments which follow mouse motion, widget rotation, etc. When the action is complete, a callback is called to notify the application.

Stacking reorder:

The **XnsIDraw** widget provides functions to modify the stacking order of its children. These function include:

- m XnsIDrawCirculateUp
- m XnsIDrawCirculateDown
- m XnsIDrawSendToBack
- m XnsIDrawBringToFront

Functions are also available to modify the stacking order of children inside a group.

Zoom

A zoom feature is available with the **XnsIDraw** widget. A function is provided to zoom a part of the drawing. This function computes a scale factor to be applied to all widgets within the zoom area. The scale factor value is such that the zoomed area occupies as much of the viewing window as possible. The **XnsIDraw** widget memorizes each zoom operation. Unzooming returns the drawing to the previous zoom level.

Double Buffer

Double buffering is useful to avoid the flashing effects that may occur during animation and/or window redisplay. A double buffer is a non visible pixmap in which the drawings are done. If a part of the visible window is to be updated, this part is copied from the buffer to the window.

Slots

The **XnsIDraw** widget can add slots to its children. Slots are rectangular regions which are used to plug in a link. These slots can be displayed as a rectangle or as a pixmap. Each slot can have a value attached to it. Slot placement is automatic.

2.2 The Draw Primitive Object Classes

The following figure gives the hierarchy of the Draw Library Object Set. The main class is the **XmDrawObject** class. This class is the base class for all Draw widgets. The **XmDrawRect** class is the base class for rectangular widgets. The **XmDrawPoly** class is the base class for polygonal widgets. Only widgets subclass of this class can be used as links. This class also supports the arrow functionality.

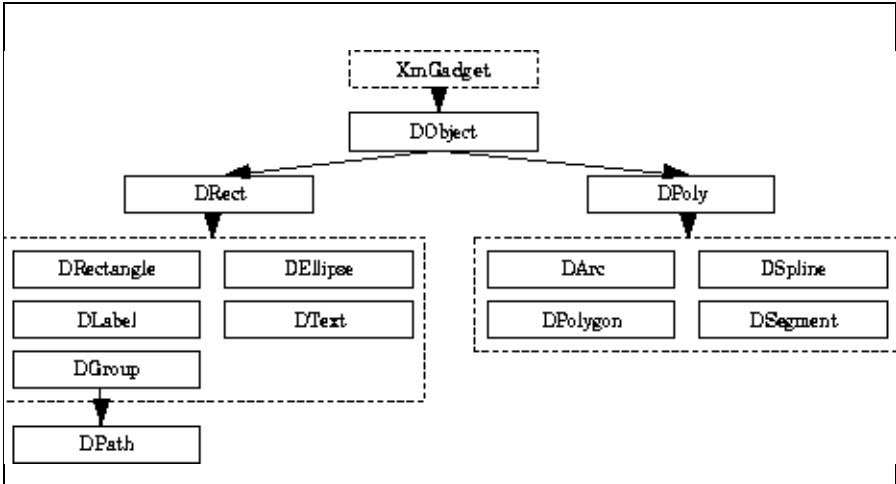


Figure 2.2: The Widget 'Classes

2.2.1 The Widget Set

Standard widgets are usually defined by providing the position and dimension of the widget. Unlike standard widgets, the Draw widgets can be defined by directly giving a set of coordinates. These coordinates are a set of floating point numbers which define the shape of the widget. The core resources **XmNx**, **XmNy**, **XmNwidth** and **XmNheight** are computed from these points.

Most of the draw widgets need a list of points: for example, a line needs a start and an end point. These points can be given in various units (inches, millimeters...).

- m **Line**: defined by two points,
- m **Rectangle**: defined by a top left corner position, and a width, height dimension.
- m **Polygon**: defined by a set of points. The points are connected to form a Polyline.
- m **Spline**: defined by a set of control points.
- m **Segment**: defined by a set of points, but the segments are drawn by disjointed pairs.
- m **Ellipse**: defined by a top left corner position, a width, a height, a start and an end angle.
- m **Arc**: defined by three points to form a quarter of an ellipse.
- m **Label**: defined by a top left corner position, a width, a height and a static string.
- m **Text**: The text widget is editable. It handles many display features such as underlining, multiple fonts, background and foreground colors on text segments.
- m **Group**: defined by a list of **Draw** widgets. When grouped, a set of widgets becomes a new widget.
- m **Path**: defined by a list of polygonal **Draw** widgets. As in groups, the set of widgets becomes a new widget, but the widgets are joined to form a continuous path.

2.2.2 Attributes

Geometry

All **XnsIDraw** widgets can be rotated, scaled and translated. Two kinds of scale are provided: a prescale and a postscale. The prescale is applied before the rotation and the postscale is applied after the rotation. A center is also part of the attributes. Rotation and scaling are done with respect to this center.

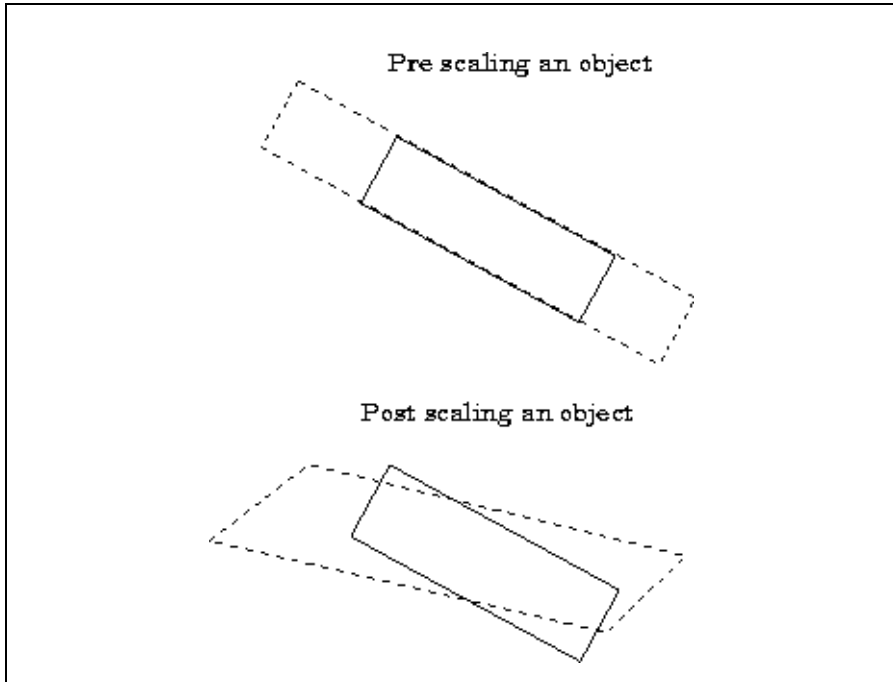


Figure 2.3: Pre Scale and Post Scale

Graphics:

Draw widgets have graphic attributes. This includes standard features such as foreground color, background color, line width, Some of these attributes are directly taken from an X graphics context:

- m foreground
- m background
- m function
- m lineWidth
- m font
- m stipple
- m dashes
- m xTsOrigin
- m yTsOrigin

Additional attributes are available:

m `closeMode`: Some widgets can be closed. The modes available are:

- Open: The widget is not closed.
- Center Closed: The widget is closed through the center.
- Object Closed: The close mode is widget dependent.

m `fillMode`: Objects can be filled or not. They also can be partially filled.

m `fillDir`: When an object is partially filled, this attribute is used to determine the direction from which fill starts. A widget can be filled from top, bottom, left, or right.

m `pctFill`: This attribute is used to partially fill a widget. It specifies the percentage of the widget's surface that should be filled.

m `opaque`: A widget can be transparent or not. When a widget is opaque, its enclosing rectangle is filled with the background color.

m `background_pixmap`: The background of a widget can also be filled with a color pixmap.

2.2.3 Links

Each polygonal widget can be used as a link. A widget becomes a link if link related resources have sensible values. These resources are:

m A source widget.

m A destination widget.

m A source slot.

m A destination slot.

When these resources are assigned to a widget, the widget becomes a link. Moving or resizing a source or destination widget updates the link. The link can be automatically updated or callback updated. Automatic updating is done by superimposing the first point of the widget with the source slot and its last point with the destination slot. Callback updating is done by calling a callback. In this case, the application can compute the points of the link.

2.2.4 Scripts

Objects can be activated through two resources:

- m **XnslNdrawActivateCallback**: This callback resource is triggered when the user clicks in the overall rectangle surrounding the widget.
- m **XnslNdrawActivateCallbackList**: This resource is a list of **FACE** scripts. Each script has an event condition and a location condition. The form of such a script is:

```
on <Event> [ and <location> ]
<Script body>
end
```

The script is launched if the user event matches the script event and if the event occurred in the specified location *and* the XnslDraw widget is in **Run** mode.

The event condition can be the name of an action (e.g. Button1Down) or can be a translation specification (e.g. “<KeyPress> a”). Note that the quotation marks must be included when specifying translations.

2.2.5 Output Format

The XnslDraw widget and the Draw widgets can be saved. Two formats are provided: a Draw format and an Fm format. The Fm format is readable by **XFaceMaker**. Use this format to generate the C code version of your drawing: **XFaceMaker** can generate C code from the Fm format.

The Draw format is:

```
DrawWidget      ::=  EMPTY
                  |  BEGINBLOCK <ResourceList> ENDBLOCK
WidgetList      ::=  EMPTY
                  |  <WidgetList> <WidgetRec>
WidgetRec       ::=  STRING STRING BEGINBLOCK <ResourceList>
                  ENDBLOCK
ResourceList     ::=  EMPTY
                  |  <ResourceList> STRING IS STRING SEP
                  |  <ResourceList> <WidgetRec>
                  |  <ResourceList> STRING SEP
```


2.3 XDrawMaker

XDrawMaker is a drawing editor. It makes full use of the features available in the **XnsIDraw** widget and the **Draw** widgets. The features are implemented as follows:

- m **Objects**: The widgets **Arc**, **Label**, **Ellipse**, **Segment**, **Spline**, **Line**, etc. are available in the tools icon bar. A **Selection To Path** item in the **Arrange** menu is used to form a **Path** widget with the current selection. Similarly, a **Selection To Group** item in the **Arrange** menu is used to form a **Group** widget with the current selection. The **Object** box has been subdivided into more specific boxes, each dedicated to a set of resources.
- m **Tree**: A tree display has been added to list the contents of the drawing. Selection and name editing are possible in the Tree box.
- m **Scales**: The **Scale** item lets you specify two scale factors: one for prescale, one for postscale.
- m **Zoom**: Items are available in the **View** menu to zoom and unzoom a part of the drawing.
- m **Slots**: Items are available in the **Tools** icon set to place, move or remove slots on the edit widget.
- m **Links**: Links can be placed between slots. An icon lets you switch to *Link Mode*. In this mode, only polygonal widgets can be used.
- m **CCITT Group 4**: A CCITT Group 4 output format is available.
- m **Script Editing**: As in **XFaceMaker**, a mechanism has been implemented to edit a widget or widget scripts with a custom editor.
- m **Usage with XFaceMaker**: It is possible to call **XDrawMaker** directly from **XFaceMaker**.

CHAPTER 3: Building Applications

This chapter describes what you need to do to build applications that use the XnsIDrawLibrary.

You can build an application without using any development tool, just making calls to the DrawLibrary functions, the Xt functions and the Xm functions. Applications built this way are described in the section “Draw Library and C code”.

You can build applications that use a drawing made with XDrawMaker but without using XFaceMaker to build the interface file. In this case, you will instantiate the widgets of your interface using Xt or Xm functions, and set the resources with Xt functions. Building this type of application is described in the section “Draw Library and XDrawMaker”.

You can build applications using the full tool suite, i.e. XDrawMaker to make your drawing, and XFaceMaker to build your interface and generate the C code of the interface. Building this type of application is described in the section “Draw Library XDrawMaker and XFaceMaker”. Note that such applications can run the Face scripts defined in a drawing with XDrawMaker in compiled rather than interpreted mode. The drawing has to be exported from XDrawMaker in XFaceMaker format, and the corresponding C code has to be generated by XFaceMaker.

A particular kind of application is a customized version of XFaceMaker. If, using the full tool suite, you have created new Xt widget classes that are subclassed from the XnsIDraw widget class, you might want to build an XFaceMaker executable that includes your new class. For instructions on adding XFaceMaker extensions, refer to the *XFaceMaker User Guide*.

The Draw Library distribution includes 2 copies of the XnsIDraw library: a *development* library that requests a license in order to instantiate the widgets, and a *run-time* library that is not protected by a license server. The development library is the only one available during product evaluation. The run-time library can be obtained only after purchase of the product. Final applications should be linked with the run-time library, as this library does not require that an NSL license server be running.

The following table lists the libraries that are part of the Draw Library distribution:

Table 3.1: Draw Library components

Library	Link flag	Description
XnslDraw	-lXnslDraw	The Draw widget and gadgets, the draw convenience functions: 2 versions: development and run-time
XnslTiff	-lXnslTiff	Tiff library for the XnslDraw widget
XnslDf	-lXnslDf	FACE interpreter for the XnslDraw widget

3.1 Draw Library and C code

To build an application directly with the Draw Library, you have to

1. Initialize the toolkit: `toplevel = XtInitialize( ... )`
2. Initialize the widget converters and attach the Draw Library functions to the Face language: `XnslDrawInit (toplevel)`
3. Write the application and interface code. The interface code will include calls to the Draw Library functions to create the drawings in the XnslDraw widget. Similarly, any behavior associated to the objects in the drawing will have to be written through calls to the convenience functions of the Draw Library.
4. If you are using the development library, write an exit function in which you free the XnslDrawWidget license: `XnslDrawComplete()` This function does nothing when a non-protected run-time library is used so that it is good practice to always call `XnslDrawComplete` when exiting an application linked with the Draw Library. Your *Close Handler* too should call `XnslDrawComplete`.
5. Compile the code and link the object code with any application connected libraries, and the following libraries:

XnslDraw, XnslTiff, XnslDf, Xm, Xt, X11

You may need to add the math library for floating point arithmetic, and some system libraries such as PW, gen, malloc, etc., depending on your environment. You may also have to specify compilation and link flags depending on your platform, your compiler, your linker. If you are using X11R6, you will need to link with the ICE and MRM libraries.

Your command line will look something like this:

```
cc -o myprog myprog.c -lXnsldraw -lXnsLTiff -lXnsldf -lXm \
-lXt -lX11 -lm
```

You must place the XnsL libraries before any libraries other than the application specific libraries; and the math library must be after the X11 library on the command line.

3.2 Draw Library and XDrawMaker

You can use the XDrawMaker drawing editor to layout your drawings and also to specify the behavior of the objects within your drawing when they receive user events. To build applications that use drawings made with XDrawMaker, proceed as follows:

1. With XDrawMaker, make your drawing, animate it, save the drawing as a **.draw** file, the default format used by XDrawMaker.
2. Initialize the toolkit: `toplevel = XtInitialize( ... )`
3. Initialize the widget converters and attach the Draw Library functions to the Face language: `XnsldrawInit (toplevel)`
4. Write the application and interface code. The interface code will need to set the `XnlsNdrawFile` resource of the `Xnsldraw` widget to the name of the **.draw** file that you have created in step 1.
5. If you are using the development library, write an exit function in which you free the `XnsldrawWidget` license: `XnsldrawComplete()` This function does nothing when a non-protected runtime library is used so that it is good practice to always call `XnsldrawComplete` when exiting an application linked with the Draw Library. Your *Close Handler* too should call `XnsldrawComplete`.
6. Compile the code and link the object code with any application connected libraries, and the following libraries:

XnslDraw, XnslTiff, XnslDf, Xm, Xt, X11

You may need to add the math library for floating point arithmetic, and some system libraries such as PW, gen, malloc, etc., depending on your environment. You may also have to specify compilation and link flags depending on your platform, your compiler, your linker. If you are using X11R6, you will need to link with the ICE and MRM libraries.

Your command line will look something like this:

```
cc -o myprog myprog.c -lXnslDraw -lXnslTiff -lXnslDf -lXm \
-lXt -lX11 -lm
```

You must place the Xnsl libraries before any libraries other than the application specific libraries; and the math library must be after the X11 library on the command line.

3.3 Draw Library XDrawMaker and XFaceMaker

When you use the full NSL tool suite to build your application, you can choose several execution modes for the interface and drawings of your application.

1. Execute both the interface and the drawing in interpreted mode. This is discussed in section “Interpreted Interface - Interpreted Drawing”.
2. Execute the interface in compiled mode and the drawing in interpreted mode. This is discussed in section “Compiled Interface - Interpreted Drawing”.
3. Execute both the interface and the drawing in compiled mode. This is discussed in section “Compiled Interface - Compiled Drawing”.

To build such applications you have to use an extended XFaceMaker executable. The XFaceMaker/WF distribution includes an executable called **xfmwf** that includes the Draw Library extensions. You can use that executable to build your interfaces and application files. You can also build your own executable if you need to include other extensions, for example one or several XRT widgets.

The Makefile generated by a given XFaceMaker executable refers to the include files and the library files of all entities included in the extensions *regardless of whether a particular interface makes any reference to the corresponding widgets or functions.*

For instructions on building a new extended XFaceMaker, refer to the section “Building an Interface Editor” in this document and to the XFaceMaker manuals.

Applications built with an extended XFaceMaker are linked with additional libraries:

Table 3.2: Basic Extended XFaceMaker Libraries

Library	Link flag	Description
XnslStand	-lXnslStand	Common widget support code for XFaceMaker
XnslCtrl	-lXnslCtrl	Control widgets: XnslBarGraph, XnslColorBox, XnslColorSB, XnslFontSelector, XnslIndicator, XnslJoystick, XnslMeter, XnslSlider, XnslStepper
XnslFm	-lXnslFm	XFaceMaker support

3.3.1 Interpreted Interface - Interpreted Drawing

Generally, you will use interpreted mode for both the interface and the drawing while you are developing your application: in interpreted mode, changes made in the interface using XFaceMaker are taken in account without your having to recompile the interface, or relink the application. This is because, in interpreted mode, the application just loads the .fm file. Similarly, the drawing will be loaded in the interface as a .draw file; if you modify the draw file using XDrawMaker, you need only re-launch the application to observe the changes you made. Note that you can modify the drawing directly within XFaceMaker, : select a draw widget and, activate the XDrawMaker editor through the XfaceMaker Edit menu.

To build an application that runs in interpreted mode for both the interface and the drawing, proceed as follows:

1. With XDrawMaker, make your drawing, animate it, save the drawing as a **.draw** file, the default format used by XDrawMaker.
2. With XFaceMaker, build your interface. Set the drawFile resource of the XnslDraw widgets of your interface to the appropriate file name: the drawing will be loaded into the draw widget whenever it is instantiated, i.e. in XFaceMaker and at run time in your application.

3. Make sure your interface includes a Quit or Exit button. In the button's `activateCallback`, call `XnslDrawComplete()` to free the `XnslDrawWidget` license. Also make sure you add a *Close Handler* that calls `XnslDrawComplete`.
4. Save your interface as a `.fm` file, generate the application files using `XFaceMaker` (Save Appli Files)
5. If your interface calls application functions, add the corresponding code in the `xxApp.c` file, or modify the Makefile generated (or the MakeFile pattern) as needed in order to link with the appropriate application code.
6. Compile with the command

```
make -f xxMake interpreted
```

where `xx` stands for the name of your interface file.

In interpreted mode, the application is linked with the `Fm` library of `XFaceMaker` and any libraries required by the extensions with which the extended `XFaceMaker` you are using was built. Among these, you will always have `XnslFm` and `XnslStand`. Note that the `XnslDf` library is not included in the link line as the Face interpreter is in the `Fm` library. Similarly, the `XnslCtrl` library is not included in the link line as the widgets are in the `Fm` library.

3.3.2 Compiled Interface - Interpreted Drawing

If you generate the C code for your interface and keep your drawing definition in a draw file, you will use this mode. Any changes you make in your interface will require that you rebuild your application. Since the interface, in this mode, continues to load the draw files in the `XnslDraw` widgets, changes in the drawings will be immediately visible when you relaunch the application, without having to rebuild anything. You can modify your drawings within `XFaceMaker` by launching `XDrawMaker` from `XFaceMaker`. The draw objects are not directly accessible in `XFaceMaker`, though.

To build an application that runs the interface in compiled mode and the drawings in interpreted mode, proceed as follows:

1. With `XDrawMaker`, make your drawing, animate it, save the drawing as a `.draw` file, the default format used by `XDrawMaker`.
2. With `XFaceMaker`, build your interface. Set the `drawFile` resource of the `XnslDraw` widgets of your interface to the appropriate file

name: the drawing will be loaded into the draw widget whenever it is instantiated, i.e. in XFaceMaker and at run time in your application.

3. Make sure your interface includes a Quit or Exit button. In the button's activateCallback , call XnsldDrawComplete() to free the XnsldDrawWidget license. Also make sure you add a *Close Handler* that calls XnsldDrawComplete.
4. If needed, set XFaceMaker's C code generation options.
5. Save your interface as C code, generate the application files using XFaceMaker (Save Appli Files)
6. If your interface calls application functions, add the corresponding code in the xxApp.c file, or modify the Makefile generated (or the MakeFile pattern) as needed in order to link with the appropriate application code.
7. Compile with the command

```
make -f xxMake compiled
```

where xx stands for the name of your interface file.

In compiled mode, the application is linked with the Fm_c library of XFaceMaker and any libraries required by the extensions with which the extended XFaceMaker you are using was built. Among these, you will always have XnslStand and XnslCtrl. Note that the XnslDf library is included in the link line as the Face interpreter is not in the Fm_c library. Likewise, the XnslCtrl has to be included, as it is not in the Fm_c library.

3.3.3 Compiled Interface - Compiled Drawing

You can build an application that will run both the interface and the drawing in compiled mode. This is achieved by exporting the drawing in XFaceMaker (fm) format, loading it in the interface and generating the C code with XFaceMaker.

To build an application that runs the interface and the drawings in compiled mode, proceed as follows:

1. With XDrawMaker, make your drawing, animate it, export the drawing in XFaceMaker format as a .fm file. It is good practice to also keep a copy of the .draw file.
2. With XFaceMaker, build your interface.
3. Read the .fm file(s) of your drawing(s) into XFaceMaker. Use the

widget cut and paste facility to place the drawing where it belongs in the widget hierarchy. Note that the drawing you read includes the `XnsIDraw` widget and its children so that you won't need to instantiate `XnsIDraw` widgets directly with `XFaceMaker`.

4. If you want to edit the drawing or its components, you can do so with `XFaceMaker` through `XDrawMaker` or directly with the `XFaceMaker` resource editor, as all the components of the drawing are directly accessible to `XFaceMaker`. Notice that the widget hierarchy of any paths or groups you have defined is flattened in the `XFaceMaker` widget tree representation. The groups or paths are still defined, though, and their behavior unchanged.
5. Make sure your interface includes a Quit or Exit button. In the button's `activateCallback`, call `XnsIDrawComplete()` to free the `XnsIDrawWidget` license. Also make sure you add a *Close Handler* that calls `XnsIDrawComplete`.
6. If needed, set `XFaceMaker`'s C code generation options.
7. Save your interface as C code, generate the application files using `XFaceMaker` (Save Appli Files)
8. If your interface calls application functions, add the corresponding code in the `xxApp.c` file, or modify the Makefile generated (or the MakeFile pattern) as needed in order to link with the appropriate application code.
9. Compile with the command

```
make -f xxMake compiled
```

where `xx` stands for the name of your interface file.

In compiled mode, the application is linked with the `Fm_c` library of `XFaceMaker` and any libraries required by the extensions with which the extended `XFaceMaker` you are using was built. Among these, you will always have `XnsIStand` and `XnsICtrl`. Note that the `XnsIDf` library is included in the link line even though the drawing is not interpreted. This is because the `DrawLibrary` functions have to be attached to the Face language. `h e XnsICtrl` has to be included, as it is not in the `Fm_c` library.

3.4 Building an Interface Editor

As shown below, it is easy to build a customized XFaceMaker that contains only the extensions you need.

The NSL product distribution includes one XFaceMaker executable:

- **xfm** - contains no extensions. This is XFaceMaker with the Motif widgets and the NSL Control widgets.

If you have installed XFaceMaker/WF, you have installed an extended XFaceMaker that contains the Draw Library extension. The executable is:

- **xfmwf** - contains the Draw extension. To build applications with the Makefiles generated by xfmwf, the Draw Library has to be installed.

You may need to include additional extensions in the XFaceMaker executable you use.

To build an extended XFaceMaker, proceed as follows:

1. Set any environment variables needed for the extensions you will include, e.g. XRTHOME if you include any XRT widget extension
2. Launch xfm (or xfmwf, or any XFaceMaker executable)
3. In the File menu, select *Load Extensions*
4. A dialog box is popped with a list of standard extensions. Select the ones you want. You may also browse any additional extensions you have defined.
5. Specify the name of the executable you are building in the dialog that is popped. XfaceMaker compiles the extensions you selected and the new executable. A dialog box displays the commands being executed.
6. When the executable has been built, XFaceMaker offers to launch it as a new design process. In most cases, launching your new executable as a design process will not suffice. Exit your XFaceMaker session, copy the executable to your bin directory (or add the current directory to your PATH environment variable).
7. Use your new executable to build interfaces and applications.

You can define your own extensions, e.g. to add widgets of your own, or third party widgets. Refer to the *XFaceMaker User Guide* for instructions on building new extensions.

CHAPTER 4: The XnsIDraw Class

4.1 Synopsis

`#include <XnsI/Draw.h>`

4.2 Classes

The **XnsIDraw** class inherits resources and behavior from the **Core**, **Composite**, **Constraint**, and **XmManager** classes.

The class pointer is **xnsIDrawWidgetClass**.

The class name is **XnsIDraw**.

4.3 Description

The **XnsIDraw** widget is a container widget (a subclass of the Motif Manager widget) able to contain any Composite or Primitive widget. It provides interactive editing capabilities. The **XnsIDraw** widget can operate in two modes:

- m Edit Mode: Selection is enabled; pointer events are not dispatched to children. Once a widget is created with an **XnsIDraw** widget as parent, the user can select, move and resize this widget. Selection is indicated by handles drawn on the selected objects.
- m Run Mode: Events are dispatched to children. Children have their normal behavior.

The **XnsIDraw** widget can also attach slots to its children. These slots are used as attachment points to dynamic links. Any polygonal object (line, polygon, spline, arc, segment) can be used as a dynamic link between objects. These links are called dynamic because they are updated automatically when a source or a destination object is moved or resized.

The **XnsIDraw** widget provides a set of resources called **Collect Reactors** which provide a general mechanism for interactive user input and the associated feedback. These resources handle point coordinate acquisition, the drawing of rectangular or line segment ghost images which follow mouse motion, the rotation of objects, etc.

The **XnsIDraw** widget also provides functions to modify the stacking order of its children. The **XnsIDraw** widget implements functions to save and load files containing Draw objects and non-Draw objects.

4.4 Resources

The **XnsIDraw** widget defines the following resource set:

XnsIDraw Resource Set

Name Class	Default Type	Access
XnsINdrawActivateCallback XnsICdrawActivateCallback	Null XtRCallback	C
XnsINdrawActivateCallbackList XnsICdrawActivateCallbackList	NULL XnsIRDrawActivateCallbackList	CSG
XnsINdrawAddCallback XnsICdrawAddCallback	NULL XtRCallback	C
XnsINdrawAlignment XnsICdrawAlignment	XnsIDRAW_MIDDLE XnsIRDrawAlignment	CSG
XnsINdrawAutoRedisplay XnsICdrawAutoRedisplay	True XnsIRBoolean	CSG
XnsINdrawAngleUnits XnsICdrawAngleUnits	XnsINO_ANGLE_UNITS XnsIRDrawAngleUnits	CSG
XnsINdrawBottomMargin XnsICdrawBottomMargin	10 XtRInt	CSG
XnsINdrawCollect XnsICdrawCollect	XnsIDRAW_SLEEP XnsIRDrawCollect	CSG
XnsINdrawCompleteButton XnsICdrawCompleteButton	Button3 XnsIRDrawButton	CSG
XnsINdrawDoneCallback XnsICdrawDoneCallback	NULL XtRCallback	C
XnsINdrawDoubleBuffer XnsICdrawDoubleBuffer	False XtRBoolean	CSG
XnsINdrawEdited XnsICdrawEdited	NULL XtRWidget	CSG
XnsINdrawEditedCallback XnsICdrawEditedCallback	NULL XtRCallback	C
XnsINdrawFile XnsICdrawFile	NULL XnsIRDrawFile	CSG
XnsINdrawFocusCursor XtCCursor	0 XtRCursor	CSG
XnsINdrawGlobalVariables XnsICdrawGlobalVariables	False Boolean	CSG

XnsIDraw Resource Set

Name Class	Default Type	Access
XnsIDrawGridAngle XnsICDrawGridAngle	dynamic XtRFloat	CSG
XnsIDrawGridCrossSize XnsICDrawGridCrossSize	3 XnsIRInt	CSG
XnsIDrawGridHorizontalSpace XnsICDrawGridHorizontalSpace	dynamic XtRFloat	CSG
XnsIDrawGridOneOf XnsICDrawGridOneOf	5 XnsIRInt	CSG
XnsIDrawGridScale XnsICDrawGridScale	dynamic XtRFloat	CSG
XnsIDrawGridSnapTo XnsICDrawGridSnapTo	False XnsIRBoolean	CSG
XnsIDrawGridSnapToAngle XnsICDrawGridSnapToAngle	False XnsIRBoolean	CSG
XnsIDrawGridSnapToScale XnsICDrawGridSnapToScale	False XnsIRBoolean	CSG
XnsIDrawGridVerticalSpace XnsICDrawGridVerticalSpace	20 XnsIRInt	CSG
XnsIDrawGridVisible XnsICDrawGridVisible	False XnsIRBoolean	CSG
XnsIDrawHandleCursor XtCCursor	crosshair XtRCursor	CSG
XnsIDrawHandleSize XnsICDrawHandleSize	7 XnsIRInt	CSG
XnsIDrawInputCallback XnsICDrawInputCallback	NULL XtRCallback	C
XnsIDrawInsertWidgetCallback XnsICDrawInsertWidgetCallback	NULL XtRCallback	C
XnsIDrawKeepCollecting XnsICDrawKeepCollecting	False XtRBoolean	CSG
XnsIDrawLeftMargin XnsICDrawLeftMargin	10 XtRInt	CSG
XnsIDrawLoadWidgetCallback XnsICDrawLoadWidgetCallback	NULL XtRCallback	C

XnsIDraw Resource Set

Name Class	Default Type	Access
XnsINdrawMode XnsICDrawMode	XnsIDRAW_RUN XnsIRDrawMode	CSG
XnsINdrawModifyCallback XnsICDrawModifyCallback	NULL XtRCallback	C
XnsINdrawModifyVerifyCallback XnsICDrawModifyVerifyCallback	NULL XtRCallback	C
XnsINdrawNumCollectPoints XnsICDrawNumCollectPoints	-1 XtRInt	CSG
XnsINdrawOutputFormat XnsICDrawOutputFormat	XnsIDRAW_DRAW XnsIRDrawOutputFormat	CSG
XnsINdrawPageHeight XnsICDrawPageHeight	dynamic XtRFloat	CSG
XnsINdrawPageNumColumns XnsICDrawPageNumColumns	1 XnsIRInt	CSG
XnsINdrawPageNumRows XnsICDrawPageNumRows	1 XnsIRInt	CSG
XnsINdrawPageOrientation XnsICDrawPageOrientation	XnsIDRAW_PORTRAIT XnsIRDrawPageOrientation	CSG
XnsINdrawPageSize XnsICDrawPageSize	NULL XnsIRDrawPageSize	CSG
XnsINdrawPageWidth XnsICDrawPageWidth	dynamic XtRFloat	CSG
XnsINdrawPlaceSlotsCallback XnsICDrawPlaceSlotsCallback	NULL XtRCallback	C
XnsINdrawReadInProgressCallback XnsICdrawReadInProgressCallback	NULL XtRCallback	C
XnsINdrawResizePolicy XnsICDrawResizePolicy	XnsIDRAW_NONE XnsIRDrawResizePolicy	CSG
XnsINdrawRightMargin XnsICDrawRightMargin	10 XtRInt	CSG
XnsINdrawSavePixmapCallback XnsICDrawSavePixmapCallback	NULL XtRCallback	C
XnsINdrawSaveWidgetCallback XnsICDrawSaveWidgetCallback	NULL XtRCallback	C

XnslDraw Resource Set

Name Class	Default Type	Access
XnslNdrawSaveWidgetHookCallback XnslCDrawSaveWidgetHookCallback	NULL XtRCallback	C
XnslNdrawScriptEnabled XnslCDrawScriptEnabled	True XtRBoolean	CSG
XnslNdrawSelectButton XnslCDrawSelectButton	Button1 XnslRDrawButton	CSG
XnslNdrawSelectCallback XnslCDrawSelectCallback	NULL XtRCallback	C
XnslNdrawSelectVerifyCallback XnslCDrawSelectVerifyCallback	NULL XtRCallback	C
XnslNdrawShowPageSize XnslCDrawShowPageSize	True XtRBoolean	CSG
XnslNdrawSlotCursor XtCCursor	target XtRCursor	CSG
XnslNdrawTolerance XtCWidth	4 XtRInt	CSG
XnslNdrawTopMargin XnslCDrawTopMargin	10 XtRInt	CSG
XnslNdrawTrackCursor XtCCursor	fleur XtRCursor	CSG
XnslNdrawUnits XnslCDrawUnits	XnslNO_UNITS XnslRDrawUnits	CSG
XnslNdrawUpdateLinks XnslCDrawUpdateLinks	True XnslRBoolean	CSG
XnslNdrawUseRgbCube XnslCDrawUseRgbCube	False XtRBoolean	CG
XnslNdrawZoomCursor XtCCursor	bottom_right_corner XtRCursor	CSG

XnsIDrawActivateCallback

This callback is triggered if a child is activated (i.e. if an incoming event matches the activate condition and activate location of the child. The callback is a duplication of the activate callback of the child and catches children activate callback in the Draw widget. This allows an application to supply a single callback routine for handling all children activate callbacks. Note that the children's activate callbacks are not disabled if this resource is set.

XnsIDrawActivateCallbackList

This resource is only usable with the **FACE** interpreter. It defines a list of scripts to be triggered when an **XnsIDrawObject** is activated by user input. This resource is active only in run mode.

XnsIDrawAddCallback

This callback is invoked when a bitmap, pixmap, color or font is loaded by an **XnsIDrawObject**. The structure whose address is passed to this callback is **XnsIDrawAddCallbackStruct**. The reason is **XnsICR_DRAW_ADD_PIXMAP** when a pixmap is added, **XnsICR_DRAW_ADD_BITMAP** when a bitmap is added, **XnsICR_DRAW_ADD_COLOR** is added.

XnsIDrawAlignment

Specifies the positioning of children when the **XnsIDraw** widget is resized. This resource is effective only if the **XnsIDrawResizePolicy** resource value is not **XnsIDRAW_ALIGN**. Possible values are:

- m **XnsIDRAW_NORTH_WEST**
- m **XnsIDRAW_WEST**
- m **XnsIDRAW_SOUTH_WEST**
- m **XnsIDRAW_SOUTH**
- m **XnsIDRAW_SOUTH_EAST**
- m **XnsIDRAW_EAST**
- m **XnsIDRAW_NORTH_EAST**
- m **XnsIDRAW_NORTH**
- m **XnsIDRAW_MIDDLE**

XnslNdrawAutoRedisplay

If this resource is set to **False** and **XnslNdrawDoubleBuffer** is **True**, the drawing is not automatically refreshed. The drawing is redisplayed by calling the **XnslDrawRedisplay** function.

XnslNdrawAngleUnits

This resource controls the **XnslNdrawAngleUnits** of children if these children have no angle unit specifications. Possible units are:

m **XnslNO_ANGLE_UNITS**

m **XnslRADIANS**

m **XnslDEGREES**

m **XnslREVOLUTIONS**

XnslNdrawBottomMargin

Specifies the bottom margin between the widget and any of its child widgets. Effective if **XnslNdrawResizePolicy** is different from **XnslDRAW_NONE**.

XnslNdrawCollect

Implements the **Collect Reactors** mechanism. Once this resource is set to a value other than **XnslDRAW_SLEEP**, the **XnslDraw** widget waits for user input and handles appropriate feedback. This resource is effective only in **XnslDRAW_EDIT** mode. When the operation is completed, the **XnslNdraw-DoneCallback** is called. Acceptable values are:

m **XnslDRAW_GET_ANGLE**: The widget waits for the movement of the mouse to define a new rotated position.

m **XnslDRAW_GET_HLINE**: The widget waits for the definition of two horizontally aligned points.

m **XnslDRAW_GET_LINE**: The widget waits for the definition of two points.

m **XnslDRAW_GET_LINK**: The widget waits for the definition of an arbitrary number of points. The number of points is specified by the **XnslNdrawNum-CollectPoints** resource. The first and the last point must be in a slot. The Complete button cancels the action.

m **XnslDRAW_GET_POINT_ARM**: The widget waits for a button press to define a point.

m **XnslDRAW_GET_POINT_DISARM**: The widget waits for a button release to define a point.

- m **XnsIDRAW_GET_POLY**: The widget waits for the definition of an arbitrary number of points. The number of points is specified by the **XnsINdrawNumCollectPoints** resource.
- m **XnsIDRAW_GET_POST_SCALE**: The widget waits for the movement of the mouse to define a new size to be applied after the eventual rotation is applied.
- m **XnsIDRAW_GET_PRE_SCALE**: The widget waits for the movement of the mouse to define a new size to be applied before the eventual rotation is applied.
- m **XnsIDRAW_GET_RECT**: The widget waits for the definition of a rectangle.
- m **XnsIDRAW_SLEEP**: The widget is in normal mode.
- m **XnsIDRAW_GET_VLINE**: The widget waits for the definition of two vertically aligned points.

XnsINdrawCompleteButton

Name of the button that, when pressed, signals that the definition of an arbitrary number of points is complete. Possible values are:

- m **Button1**
- m **Button2**
- m **Button3**

It must not be in conflict with the **XnsINdrawSelectButton** resource.

XnsINdrawDoneCallback

This callback is activated when the definition of points specified by the **XnsINdrawCollect** resource is complete. The associated callback structure includes the list of points and the number of points defined. The structure whose address is passed to this callback is **XnsIDrawDoneCallbackStruct**. The reason is **XnsICR_DRAW_DONE**.

XnslNdrawDoubleBuffer

When true, drawings are refreshed in a secondary pixmap, thus avoiding unpleasant flashing effects.

XnslNdrawEdited

Takes the value of the current **Edit** widget, i.e. the widget most recently selected.

XnslNdrawEditedCallback

This callback is activated when the **Edit** widget changes. The structure whose address is passed to this callback is **XnslDrawEditedCallbackStruct**. The reason is **XnslCR_DRAW_EDIT**.

XnslNdrawFile

If this resource is set to the name of a valid **Draw** file name, the objects contained in the file are inserted in the widget.

XnslNdrawFocusCursor

Specifies the **XnslDraw** widget cursor.

XnslNdrawGlobalVariables

This resource controls if a global FACE variable is to be created for each child of a DrawObject. For backward compatibility, the default value of the resource is True. Applications should set this variable to False to increase widget creation speed

XnslNdrawGridAngle

This resource indicates the minimum angular step when the user does interactive rotation. This resource is in effect if the **XnslNdrawGridSnapToAngle** is set to **True**.

XnslNdrawGridCrossSize

Specifies the size of crosses used to display the grid, in pixels.

XnslNdrawGridHorizontalSpace

Specifies the horizontal spacing of grid points, in pixels.

XnslNdrawGridOneOf

Specifies the number of grid points to be skipped when displaying the grid.

XnslNdrawGridScale

This resource indicates the minimum scale factor step when the user does interactive scaling. This resource is in effect if the **XnsINdrawGridSnapToScale** is set to **True**.

XnsINdrawGridSnapTo

If **True**, mouse pointer motions are constrained to the grid.

XnsINdrawGridSnapToAngle

This resources enables the **XnsINdrawGridAngle** resource.

XnsINdrawGridSnapToScale

This resource enables the **XnsINdrawGridScale** resource.

XnsINdrawGridVerticalSpace

Specifies the vertical spacing of grid points, in pixels.

XnsINdrawGridVisible

If **True**, the grid is displayed.

XnsINdrawHandleCursor

Specifies the cursor when the pointer is over a selection handle.

XnsINdrawHandleSize

Specifies the size of the selection handles, in pixels. Selection handles being square, you specify the size of a side.

XnsINdrawInputCallback

This callback is activated on a button up, button down, key press or key release. The associated callback structure elements contain the widget and the slot or selection handle under the cursor, if applicable. The structure whose address is passed to this callback is **XnsIDrawInputCallbackStruct**. The reason is **XnsICR_DRAW_INPUT**.

XnsINdrawInsertWidgetCallback

This callback is activated every time a child is created. The structure whose address is passed to this callback is **XnsIDrawInsertWidgetCallbackStruct**. The reason is **XnsICR_DRAW_INSERT_CHILD**.

XnslNdrawKeepCollecting

If **False**, the widget switches to selection mode after the points defining an object have been acquired. If **True**, the widget continues to ask for points after the points defining an object have been acquired.

XnslNdrawLeftMargin

Specifies the left margin between the widget and any of its child widgets. Effective if **XnslNdrawResizePolicy** is different from **XnslDRAW_NONE**.

XnslNdrawLoadWidgetCallback

This callback is activated when a non-**Draw** graphic object is loaded in the widget. The structure whose address is passed to this callback is **XnslDrawLoadWidgetCallbackStruct**. The reason is **XnslCR_DRAW_LOAD_WIDGET**.

XnslNdrawMode

Specifies the mode of the widget. Acceptable values are:

- m **XnslDRAW_RUN**: Child widgets have their normal behavior. Object editing and point definition are disabled.
- m **XnslDRAW_EDIT**: Object editing is enabled. Events are not passed on to children.

XnslNdrawModifyCallback

This callback is activated when a child or its handles are moved through mouse interaction. The associated callback structure contains elements for the widget and the type of modification. The structure whose address is passed to this callback is **XnslDrawModifyCallbackStruct**. The reason is **XnslCR_DRAW_MODIFY** when handles are moved, **XnslCR_DRAW_MOVE** when a child is moved.

XnslNdrawModifyVerifyCallback

This callback is activated when a child or its handles are about to be moved through mouse interaction. The application can accept or refuse the modification, or impose new values. The structure whose address is passed to this callback is **XnslDrawModifyVerifyCallbackStruct**. The reason is **XnslCR_DRAW_MODIFY** when handles are concerned, **XnslCR_DRAW_MOVE** when child is concerned.

XnslNdrawNumCollectPoints

Specifies the number of points to be collected. If this number is **-1**, the number of points is not limited.

XnsIDrawOutputFormat

Controls the format of the output of an **XnsIDraw** widget. This resource affects the functions **XnsIDrawSaveWidget** and **XnsIDrawWidgetToString** functions. Possible values are:

- m **XnsIDRAW_DRAW**: Output is in **Draw** format.
- m **XnsIDRAW_FM**: Output is in **XFaceMaker** format.
- m **XnsIDRAW_PS**: Output is in **PostScript** format.

XnsIDrawPageHeight

Controls the height of the widget's page in term of widget's unit. This resource only takes effect when the name of the page is custom.

XnsIDrawPageNumColumns

This resource sets the number of columns (i.e. number of pages) and resizes the **XnsIDraw** widget accordingly.

XnsIDrawPageNumRows

This resource sets the number of rows (i.e. number of pages) and resizes the **XnsIDraw** widget accordingly.

XnsIDrawPageOrientation

This resource controls the orientation of the page. Possible values are:

- m **XnsIDRAW_LANDSCAPE**
- m **XnsIDRAW_PORTRAIT**

XnsIDrawPageSize

Controls the size of the **XnsIDraw** widget in terms of pages. The value affected to this resource is a pointer to a structure containing the size, margin and name of the page style.

XnsIDrawPageWidth

Controls the width of the widget's page in terms of the widget's unit. This resource only takes effect when the name of the page is custom.

XnsIDrawPlaceSlotsCallback

This callback is activated when the widget asks the application to position the **Slots** attached to a child object. The structure whose address is passed to this callback is **XnsIDrawPlaceSlotsCallbackStruct**. The reason is **XnsICR_DRAW_UPDATE_SLOTS**.

XnsINdrawReadInProgressCallback

This callback is invoked while a draw file is being loaded. The structure whose address is passed to this callback is **XnsIDrawReadInProgressCallbackStruct**. The reason is **XnsIDRAW_RD_PROGRESS_START** when the load first starts, then **XnsICR_DRAW_RD_PROGRESS** with , in the *pct* field giving a measure of the progress, and finally **XnsIDRAW_RD_PROGRESS_END** when loading is completed. In the latter case, *pct* is set to 0.

XnsINdrawResizePolicy

Controls child placement policy when the **XnsIDraw** widget is resized. Acceptable values are:

- m **XnsIDRAW_NONE**: Children are neither aligned nor resized.
- m **XnsIDRAW_ALIGN**: The objects are aligned.
- m **XnsIDRAW_UNIFORM**: Resize the objects so as to maintain the relative sizes.
- m **XnsIDRAW_FULL**: Resize the objects so as to fill the **XnsIDraw** widget.

XnsINdrawRightMargin

Specifies the right margin between the widget and any of its child widgets. Effective if **XnsINdrawResizePolicy** is different from **XnsIDRAW_NONE**.

XnsINdrawSavePixmapCallback

This callback is activated when the pixmap of a child not subclassed from **XnsIDrawObject** is to be output to a Pixmap, e.g. for CCITT Group 4 output. The child Pixmap output must be handled by the application for non **XnsIDrawObject** widgets. The structure whose address is passed to this callback is **XnsIDrawSavePixmapCallbackStruct**. The reason is **XnsICR_DRAW_SAVE_PIXMAP**.

XnslNdrawSaveWidgetCallback

This callback is activated when a widget not subclassed from **XnslDrawObject** is saved. The structure whose address is passed to this callback is **XnslDrawSaveWidgetCallbackStruct**. The reason is **XnslCR_DRAW_SAVE_WIDGET**.

XnslNdrawSaveWidgetHookCallback

This callback is invoked when the **XnslDrawSaveWidget** function is called. The structure whose address is passed to this callback is **XnslDrawSaveWidgetCallbackStruct**. The reason is **XnslCR_DRAW_SAVE_WIDGET_HOOK**.

XnslNdrawScriptEnabled

If this resource is set to **False**, the scripts of the widget are disabled.

XnslNdrawSelectButton

Specifies the active mouse button name for widget selection and modification. Acceptable names are: **Button[1-5]**.

XnslNdrawSelectCallback

This callback is activated when a child is selected. The structure whose address is passed to this callback is **XnslDrawSelectCallbackStruct**. The reason is **XnslCR_DRAW_SELECT**.

XnslNdrawSelectVerifyCallback

This callback is activated when a child is about to be selected. The application can enable or disable the selection. The structure whose address is passed to this callback is **XnslDrawSelectVerifyCallbackStruct**. The reason is **XnslCR_DRAW_SELECT_VERIFY**.

XnslNdrawShowPageSize

Indicates if the page outline is to be displayed in the **XnslDraw** widget.

XnslNdrawSlotCursor

Specifies the cursor when the pointer is over a slot.

XnslNdrawTolerance

Specifies the tolerance in pixels for selecting or activating a child.

XnsINdrawTopMargin

Specifies the top margin between the widget and any of its child widgets. Effective if **XnsINdrawResizePolicy** is different from **XnsIDRAW_NONE**.

XnsINdrawTrackCursor

Specifies the cursor when the pointer moves a selection.

XnsINdrawUnits

This resource defines the units in which the coordinates of the widgets are given.

m **XnsIPIXELS**

m **XnsI100TH_POINTS**

m **XnsI1000TH_INCHES**

m **XnsI100TH_MILLIMETERS**

m **XnsI100TH_FONT_UNITS**

m **XnsI_POINTS**

m **XnsI_INCHES**

m **XnsI_MILLIMETERS**

m **XnsINO_UNITS** (same as **XnsIPIXELS**)

XnsINdrawUpdateLinks

If **False**, disable link updating.

XnsINdrawUseRgbCube

This resource can only be set at creation time. If **True**, the widget tries to allocate an RGB cube in its colormap. This cube is used to find colors for Draw pix-maps.

It first tries to allocate a 6x6x6 cube. If it's not possible, it tries with a 5x5x5, a 4x4x4, or a 3x3x3 cube.

XnsINdrawZoomCursor

Specifies the cursor when a zoom operation is required.

4.4.1 Backward Compatibility Resources

XnsIDraw Backward Compatibility resources

Name Class	Default Type	Access
XnsINautoResize XnsICAutoResize	False XtRBoolean	Access
XnsINbottomMargin XnsICBottomMargin	10 XtRInt	CSG
XnsINleftMargin XnsICLeftMargin	10 XtRInt	CSG
XnsINresizeImages XnsICResizeImages	False XtRBoolean	CSG
XnsINresizeLineWidths XnsICResizeLineWidths	False XtRBoolean	CSG
XnsINresizeTexts XnsICResizeTexts	NULL XnsIDrawNumberedPoint	CSG
XnsINrightMargin XnsICRightMargin	10 XtRInt	CSG
XnsINscriptDoneCallback XnsICScriptDoneCallback	NULL XtRCallback	C
XnsINscriptVerifyCallback XnsICScriptVerifyCallback	NULL XtRCallback	C
XnsINtopMargin XnsICTopMargin	10 XtRInt	CSG

XnsINautoResize

If set to True, the **XnsINdrawResizePolicy** of all the **XnsIDrawObjects** is set to **XnsIDRAW_UNIFORM**.

XnsINbottomMargin

Specifies the bottom margin between the widget and any of its child widgets. Effective if **XnsINdrawResizePolicy** is different from **XnsIDRAW_NONE**.

XnslNleftMargin

Specifies the left margin between the widget and any of its child widgets. Effective if **XnslNdrawResizePolicy** is different from **XnslDRAW_NONE**.

XnslNresizeImages

If set to **True**, all **XnslDrawImages** objects have their **XnslNdrawImageTransformImage** resource set to **True**.

XnslNresizeLineWidths

If set to **True**, all line widths of child objects are resizable.

XnslNresizeTexts

If set to **True**, all **XnslDrawLabel** objects have their **XnslNdrawLabelTransformString** resource set to **True**.

XnslNdrawRightMargin

Specifies the right margin between the widget and any of its child widgets. Effective if **XnslNdrawResizePolicy** is different from **XnslDRAW_NONE**.

XnslNscriptDoneCallback

This callback is invoked when a script has been launched and is completed. The structure whose address is passed to this callback is **XnslDrawScriptDoneCallbackStruct**. The reason is **XnslCR_DRAW_SCRIPT_DONE**.

XnslNscriptVerifyCallback

Dynamically controls the activation of a script. Before activating a script, this callback is called. The associated callback structure provides a field to enable or disable the execution of this script. The structure whose address is passed to this callback is **XnslDrawScriptVerifyCallbackStruct**. The reason is **XnslCR_DRAW_SCRIPT_VERIFY**.

XnslNdrawTopMargin

Specifies the top margin between the widget and any of its child widgets. Effective if **XnslNdrawResizePolicy** is different from **XnslDRAW_NONE**.

4.4.2 Constraint Resources

Constraint resources are placed on children of the **XnsIDraw** widget.

XnsIDraw Constraint Resource Set

Name Class	Default Type	Access
XnsINdrawAddSlot XnsICDrawAddSlot	NULL XnsIRDrawAddSlot	CSG
XnsINdrawClickCount XnsICDrawClickCount	0 XnsIRInt	CSG
XnsINdrawFromSlot XnsICDrawFromSlot	-1 XnsIRInt	CSG
XnsINdrawNumReplaceSlots XnsICDrawNumReplaceSlots	0 XtRInt	CSG
XnsINdrawNumSlots XnsICDrawNumSlots	0 XnsIRInt	CSG
XnsINdrawPlaceSlotsPolicy XnsICDrawPlaceSlotsPolicy	XnsIDRAW_PLACE_RUBBER XnsIRDrawPlaceSlotsPolicy	CSG
XnsINdrawRemoveSlot XnsICDrawRemoveSlot	-1 XnsIRInt	CSG
XnsINdrawReplaceSlots XnsICDrawReplaceSlots	0 XnsIRDrawReplaceSlots	CSG
XnsINdrawSelectSlotCallback XmCCallback	0 XtRCallback	C
XnsINdrawSelectable XnsICDrawSelectable	True XtRBoolean	CSG
XnsINdrawSelected XnsICDrawSelected	False XtRBoolean	CSG
XnsINdrawSlots XnsICDrawSlots	0 XnsIRDrawSlots	CSG
XnsINdrawSlotAttributes XnsICDrawSlotAttributes	0 XnsIRDrawSlotAttributes	CSG
XnsINdrawSlotsPixmap XnsICDrawSlotsPixmap	-1 XnsIRDrawSlotsPixmap	CSG
XnsINdrawSlotsSize XnsICDrawSlotsSize	10 XtRInt	CSG

XnslDraw Constraint Resource Set

Name Class	Default Type	Access
XnslNdrawSlotsVisible XnslCDrawSlotsVisible	True XtRBoolean	CSG
XnslNdrawToSlot XnslCDrawToSlot	-1 XnslRInt	CSG

XnslNdrawAddSlot

Points to a **XnslDrawNumberedPoint** structure which indicates where to add the slot, and specifies its coordinates.

XnslNdrawClickCount

Count of sequences of [button release, button pressed] within the time span specified by the display's multi-click time.

XnslNdrawFromSlot

For a link object, specifies the source **Slot** number to which it is attached. This resource is used along with the **XnslNdrawFromWidget** resource for the default placing of links.

XnslNdrawNumReplaceSlots

Specifies how many slots should be replaced in a slot replacement operation.

XnslNdrawNumSlots

Indicates the number of slots for the child.

XnslNdrawPlaceSlotsPolicy

Indicates the slot placing policy. Acceptable values are:

- m **XnslDRAW_PLACE_OFFSET**: Each point value in the list of slot definition points is interpreted as an offset from the center of the object.
- m **XnslDRAW_PLACE_RUBBER**: Each point value in the list of slot definition points is interpreted as a percentage of the half-width (for the x coordinate) and of the half-height (for the y coordinate) of the object.
- m **XnslDRAW_PLACE_CALLBACK**: The **XnslNdrawPlace-SlotsCallback** is invoked. It will specify slot placement.

XnslNdrawRemoveSlot

Indicates the index of a slot to be removed.

XnsIDrawReplaceSlots

A set of pairs [point coordinates, index] used to replace slot definition points.

XnsIDrawSelectSlotsCallback

Since slots can be selected individually, a callback is provided to notify the application that a slot has been selected or deselected

XnsIDrawSelectable

If **True**, enables selection of the child.

XnsIDrawSelected

if **True** the child is selected.

XnsIDrawSlots

A set of point coordinates specifying slot positions for the child. The slot coordinates are stated as a percentage of one-half the height and width of the surrounding rectangle, relative to the object's center.

XnsIDrawSlotAttributes

This resource is used to store individual slot information like size and pixmap

XnsIDrawSlotsPixmap

Index to a pixmap returned by **XnsIDrawGetPixmap**. The slots are filled with the specified pixmap. If the index value is -1 the slots are drawn as unfilled squares.

XnsIDrawSlotsSize

Specifies the size of slots in pixels.

XnsIDrawSlotsVisible

If **True**, slots are displayed.

XnsIDrawToSlot

For a link object, specifies the destination **Slot** number to which it is attached. This resource is used along with the **XnsIDrawToWidget** resource for the default placing of links.

4.4.3 Inherited Resources

XnsIDraw inherits the following resources from the named superclass. For a description of each resource, refer to the man page of the superclass.

XmManager Resource Set

Name Class	Default Type	Access
XmNbottomShadowColor XmCBottomShadowColor	dynamic Pixel	CSG
XmNbottomShadowPixmap XmCBottomShadowPixmap	XmUNSPECIFIED_PIXMAP Pixmap	CSG
XmNforeground XmCForeground	dynamic Pixel	CSG
XmNhelpCallback XmCCallback	NULL XtCallbackList	C
XmNhighlightColor XmCHighlightColor	dynamic Pixel	CSG
XmNhighlightPixmap XmCHighlightPixmap	dynamic Pixmap	CSG
XmNinitialFocus XmCInitialFocus	dynamic Widget	CSG
XmNnavigationType XmCNavigationType	XmTAB_GROUP XmNavigationType	CSG
XmNshadowThickness XmCShadowThickness	dynamic Dimension	CSG
XmNstringDirection XmCStringDirection	dynamic XmStringDirection	CG
XmNtopShadowColor XmCTopShadowColor	dynamic Pixel	CSG
XmNtopShadowPixmap XmCTopShadowPixmap	dynamic Pixmap	CSG
XmNtraversalOn XmCTraversalOn	True Boolean	CSG
XmNunitType XmCUnitType	dynamic unsigned char	CSG

XmManager Resource Set

Name Class	Default Type	Access
XmNuserData XmCUserData	NULL XtPointer	CSG

Composite Resource Set

Name Class	Default Type	Access
XmNchildren XmCReadOnly	NULL WidgetList	G
XmNinsertPosition XmCInsertPosition	NULL XtOrderProc	CSG
XmNnumChildren XmCReadOnly	0 Cardinal	G

Core Resource Set

Name Class	Default Type	Access
XmNaccelerators XmCAccelerators	dynamic XtAccelerators	N/A
XmNancestorSensitive XmCSensitive	dynamic Boolean	G
XmNbackground XmCBackground	dynamic Pixel	CSG
XmNbackgroundPixmap XmCPixmap	XmUNSPECIFIED_PIXMAP Pixmap	CSG
XmNborderColor XmCBorderColor	XtDefaultForeground Pixel	CSG

Core Resource Set

Name Class	Default Type	Access
XmNborderPixmap XmCPixmap	XmUNSPECIFIED_PIXMAP Pixmap	CSG
XmNborderWidth XmCBorderWidth	0 Dimension	CSG
XmNcolormap XmCColormap	dynamic Colormap	CG
XmNdepth XmCDepth	dynamic int	CG
XmNdestroyCallback XmCCallback	NULL XtCallbackList	C
XmNheight XmCHeight	dynamic Dimension	CSG
XmNinitialResourcesPersistent XmCInitialResourcesPersistent	True Boolean	C
XmNmappedWhenManaged XmCMappedWhenManaged	True Boolean	CSG
XmNscreen XmCScreen	dynamic Screen	CG
XmNsensitive XmCSensitive	True Boolean	CSG
XmNtranslations XmCTranslations	dynamic XtTranslations	CSG
XmNwidth XmCWidth	dynamic Dimension	CSG
XmNx XmCPosition	0 Position	CSG
XmNy XmCPosition	0 Position	CSG

4.5 Callback Information

4.5.1 XnsIDrawDoneCallback

A pointer to the following structure is passed to the **XnsIDrawDoneCallback**.

```
typedef struct {
    int                reason;
    XEvent             *event;
    XPoint              *x_points;
    XnsIDrawPoint       *draw_points;
    int                num_points;
    int                collected;
    int                from_slot;
    Widget             from_widget;
    int                to_slot;
    Widget             to_widget;
} XnsIDrawDoneCallbackStruct;
```

- *reason* Indicates why the callback was invoked. For this callback, the reason is **XnsICR_DRAW_DONE**.
- *event* Pointer to the XEvent that triggered the callback.
- *x_points* Pointer to the list of points that were defined in pixels.
- *draw_points* Pointer to the list of points that were defined in term of the widget's unit.
- *num_points* Number of points that were defined.
- *collected* Indicates the value of the **XnsIDrawCollect** resource which invoked the callback.
- *from_slot* Indicates the slot number in which the collecting started.
- *from_widget* Indicates the widget corresponding to the *from_slot*.
- *to_slot* Indicates the slot number in which the collecting ended.
- *to_widget* Indicates the widget corresponding to the *to_slot*.

4.5.2 XnsIldrawSelectVerifyCallback

A pointer to the following structure is passed to the **XnsIldrawSelectVerifyCallback**.

```
typedef struct {
    int                reason;
    XEvent             *event;
    Boolean            doit;
    Widget             widget;
    int                count;
} XnsIldrawSelectVerifyCallbackStruct;
```

- *reason* Indicates why the callback was invoked. For this callback, the reason is **XnsICR_DRAW_SELECT**.
- *event* Pointer to the XEvent that triggered the callback.
- *doit* Indicates whether the selection action is to be performed. If set to **True** the widget is selected; if set to **False**, the widget is not selected.
- *widget* Indicates the widget which is about to be selected.
- *count* Count of sequences of [button release, button pressed] within the time span specified by the display's multi-click time.

4.5.3 XnsIldrawSelectCallback

A pointer to the following structure is passed to the **XnsIldrawSelectCallback**.

```
typedef struct {
    int                reason;
    XEvent             *event;
    Widget             widget;
    int                selected;
    int                count;
} XnsIldrawSelectCallbackStruct;
```

- *reason* Indicates why the callback was invoked. For this callback, the reason is **XnsICR_DRAW_SELECT**.
- *event* Pointer to the XEvent that triggered the callback.
- *widget* Indicates the widget that has been selected or deselected.
- *selected* Indicates the select state of the widget.
- *count* Count of sequences of [button release, button pressed] within the time span specified by the display's multi-click time.

4.5.4 XnsldrawEditedCallback

A pointer to the following structure is passed to the **XnsldrawEditedCallback**.

```
typedef struct {
    int                reason;
    XEvent             *event;
    Widget             widget;
} XnsldrawEditedCallbackStruct;
```

- *reason* Indicates why the callback was invoked. For this callback, the reason is **XnsICR_DRAW_EDIT**
- *event* NULL for this callback.
- *widget* Widget that becomes the edited object.

4.5.5 XnsldrawModifyVerifyCallback

A pointer to the following structure is passed to the **XnsldrawModifyVerifyCallback**.

```
typedef struct {
    int                reason;
    XEvent             *event;
    XnsldrawDoit       doit;
    Widget             widget;
    int                mask;
    int                x, y;
    int                handle;
} XnsldrawModifyVerifyCallbackStruct;
```

- *reason* Indicates why the callback was invoked. For this callback, the reason can be:
 - **XnsICR_DRAW_MOVE**: The widget asks to be moved.
 - **XnsICR_DRAW_MODIFY**: The widget asks to be modified because of user action on a selection handle.
- *event* Pointer to the XEvent that triggered the callback.
- *doit* If set to **XnslYes**, the modification becomes effective. If set to **XnslNo**, the modification is ignored. If set to **XnslAlmost**, the modification is applied as specified by the fields *mask*, *x*, *y*, *width*, *height*, and *geometry*.
- *widget* Indicates the widget that will be modified.
- *mask* Indicates which fields contain valid information. The following values and their combination are acceptable:

- `XnslX`
- `XnslY`
- `x`, `y` New position suggested.
- `handle` modification handle index.

4.5.6 `XnslNdrawModifyCallback`

A pointer to the following structure is passed to the `XnslNdrawModifyCallback`.

```
typedef struct {
    int                reason;
    XEvent             *event;
    Widget             widget;
    int                x, y;
    int                width, height;
} XnslDrawModifyCallbackStruct;
```

- *reason* Indicates why the callback was invoked. For this callback, the reason can be:
 - `XnslCR_DRAW_MOVE`: The widget has been moved.
 - `XnslCR_DRAW_MODIFY`: The widget has been modified because of user action on a selection handle.
- *event* Pointer to the XEvent that triggered the callback.
- *widget* Indicates the widget that has been modified.
- `x`, `y` New position.
- *width* New width.
- *height* New height.

4.5.7 XnsIDrawInputCallback

A pointer to the following structure is passed to the **XnsIDrawInputCallback**.

```
typedef struct {
    int                reason;
    XEvent             *event;
    Widget             widget;
    Widget             slot_widget;
    int                slot;
    Widget             handle_widget;
    int                handle;
} XnsIDrawInputCallbackStruct;
```

- *reason* Indicates why the callback was invoked. For this callback, the reason is **XnsICR_DRAW_ARM**.
- *event* Pointer to the XEvent that triggered the callback.
- *widget* Indicates the widget in which the event occurred or 0.
- *slot_widget* Indicates the widget whose slot was under the event or 0.
- *slot* Indicates the slot number over which the event occurred or -1.
- *handle_widget* Indicates the widget whose selection handle was under the event or 0.
- *handle* Indicates the selection handle number over which the event occurred or -1.

4.5.8 XnsIDrawPlaceSlotCallback

A pointer to the following structure is passed to the **XnsIDrawPlaceSlotCallback**.

```
typedef struct {
    int                reason;
    XEvent             *event;
    Widget             widget;
    XnsIDrawXPointList *x_point_list;
    XnsIDrawPointList *slots_point_list;
} XnsIDrawPlaceSlotsCallbackStruct;
```


- *reason* Indicates why the callback was invoked. For this callback, the reason is **XnsICR_DRAW_UPDATE_SLOTS**.
- *event* **NULL** for this callback.
- *widget* Indicates the widget whose slots are to be placed.
- *x_point_list* List of **XPoint** used to display slots.
- *slots_point_list* List of **DrawPoint** that define the slots before transformation; these should be modified by the application to define the slots after transformation.

4.5.9 XnsINdrawSavePixmapCallback

A pointer to the following structure is passed to the **XnsINdrawSavePixmapCallback**. This callback is activated when the pixmap of a child not subclassed from **XnsIDrawObject** is to be output to a Pixmap, e.g. for CCITT Group 4 output. The child Pixmap output must be handled by the application for non-**XnsIDrawObject** widgets.

```
typedef struct {
    int                reason;
    XEvent             *event;
    Widget             widget;
    Pixmap             pixmap;
} XnsIDrawSavePixmapCallbackStruct;
```

- *reason* Indicates why the callback was invoked. For this callback, the reason is **XnsICR_DRAW_SAVE_PIXMAP**.
- *event* **NULL** for this callback.
- *widget* Indicates the widget that is to be drawn in the destination pixmap.
- *pixmap* Indicates the destination pixmap.

4.5.10 XnsIDrawSaveWidgetCallback

A pointer to the following structure is passed to the **XnsIDrawSaveWidgetCallback**.

```
typedef struct {
    int                reason;
    XEvent             *event;
    Widget             widget;
    char               *string;
} XnsIDrawSaveWidgetCallbackStruct;
```

- *reason* Indicates why the callback was invoked. For this callback, the reason is **XnsICR_DRAW_SAVE_WIDGET**
- *event* **NULL** for this callback.
- *widget* Indicates the widget that is to be saved.
- *string* Pointer to a string that must be provided by the application. This string is a sequence of **resource_name**, **value** which describes the resources the application wants to save.

4.5.11 XnsIDrawLoadWidgetCallback

A pointer to the following structure is passed to the **<XnsIDrawLoadWidgetCallback**.

```
typedef struct {
    int                reason;
    XEvent             *event;
    char               *class_name;
    WidgetClass        widget_class;
} XnsIDrawLoadWidgetCallbackStruct;
```

- *reason* Indicates why the callback was invoked. For this callback, the reason is **XnsICR_DRAW_LOAD_WIDGET**
- *event* **NULL** for this callback.
- *class_name* Indicates the name of the widget class to be loaded.
- *widget_class* Pointer to the widget class to be provided by the application.

4.5.12 XnsINdrawInsertWidgetCallback

A pointer to the following structure is passed to the **XnsINdrawInsertWidgetCallback**.

```
typedef struct {
    int                reason;
    XEvent             *event;
    Widget             widget;
} XnsIDrawInsertWidgetCallbackStruct;
```

- *reason* Indicates why the callback was invoked. For this callback, the reason is **XnsICR_DRAW_INSERT_CHILD**.
- *event* **NULL** for this callback.
- *widget* The widget that has been created.

4.5.13 XnsINdrawAddCallback

A pointer to the following structure is passed to the **XnsINdrawAddCallback**.

```
typedef struct {
    int                reason;
    XEvent             *event;
    int                index;
    String             name;
} XnsIDrawAddCallbackStruct;
```

- *reason* Indicates why the callback was invoked. Possible values are:
 - **XnsICR_DRAW_ADD_PIXMAP**: A call to **XnsIDrawGetPixmap** has loaded a new pixmap.
 - **XnsICR_DRAW_ADD_BITMAP**: A call to **XnsIDrawGetBitmap** has loaded a new bitmap.
 - **XnsICR_DRAW_ADD_FONT**: A call to **XnsIDrawGetFont** has loaded a new font.
 - **XnsICR_DRAW_ADD_COLOR**: A call to **XnsIDrawGetColor** has loaded a new color.
- *event* **NULL** for this callback.
- *index* The index of the loaded resource.
- *name* The name of the loaded resource.

4.5.14 XnsINscriptVerifyCallback

A pointer to the following structure is passed to the **XnsINscriptVerifyCallback**

```
typedef struct {  
    int                reason;  
    XEvent             *event;  
    int                doit;  
    XnsIDrawObject     *object;  
    int                where;  
    char               *action;  
} XnsIDrawScriptVerifyCallbackStruct;
```

- *reason* Indicates why the callback was invoked. For this callback, the reason is **XnsICR_DRAW_SCRIPT_VERIFY**.
- *event* Indicates the event that triggered the callback.
- *doit* If set to **False** by the application, the script is not executed.
- *object* For a script associated with a **XnsIDraw** widget, this field is set to **NULL**. Otherwise, this field contain the id of the object for which the script is to be activated.
- *where* Indicates the location associated with the script.
- *action* Indicates the action associated with the script.

4.5.15 XnsINscriptDoneCallback

A pointer to the following structure is passed to the **XnsINscriptDoneCallback**

```
typedef struct {  
    int                reason;  
    XEvent             *event;  
    int                failed;  
    XnsIDrawObject     *object;  
    int                where;  
    char               *action;  
} XnsIDrawScriptDoneCallbackStruct;
```

- *reason* Indicates why the callback was invoked. For this callback, the reason is **XnsICR_DRAW_SCRIPT_DONE**.
- *event* Indicates the event that triggered the callback.
- *failed* Indicates the result returned by the script execution.
- *object* For a script associated with a **XnsIDraw** widget, this field is set to **NULL**. Otherwise, this field contain the id of the object for which the script is to be activated.
- *where* Indicates the location associated with the script.

- *action* Indicates the action associated with the script.

4.5.16 XnsIDrawReadInProgressCallback

A pointer to the following structure is passed to the **XnsIDrawReadInProgressCallback**

```
typedef struct {
    int                reason;
    XEvent             *event;
    int                status;
    int                pct;
} XnsIDrawReadInProgressCallbackStruct;
```

- *reason* Indicates why the callback was invoked. For this callback, the reason is **XnsICR_DRAW_RD_PROGRESS**.
- *event* NULL for this callback.
- *status* Values can be **XnsIDRAW_RD_PROGRESS_START** when starting loading a file, **XnsIDRAW_RD_PROGRESS** during the read and **XnsIDRAW_RD_PROGRESS_END** at the end of the read
- *pct* Indicates the percent of the file that has been read

4.5.17 XnsIDrawSelectSlotCallback

A pointer to the following structure is passed to the **XnsIDrawSelectSlotCallback**

```
typedef struct {
    int                reason;
    XEvent             *event;
    Widget             widget;
    int                slot;
    int                selected;
    int                pixmap;
    int                size;}
XnsIDrawSelectSlotCallbackStruct;
```

- *reason* Indicates why the callback was invoked. For this callback, the reason is **XnsICR_DRAW_SELECT_SLOT**.
- *event* NULL for this callback.
- *widget* The widget to which the slot belongs
- *slot* Indicates the slot index that has been selected or deselected

- *selected* Indicates if the slot has been selected or deselected
- *pixmap* Indicates the pixmap index of the slot. If the slot has no pixmap, this field is set to -1
- *size* Indicates the size of the slot. This value is zero if the slot has no size(i.e. if the slot takes its size from its widget)

CHAPTER 5: The XnsIDrawArc Class

4.6 Synopsis

```
#include <XnsI/DArc.h>
```

4.7 Classes

The **XnsIDrawArc** class inherits resources and behavior from the **Object**, **RectObj**, **XmGadget**, **XnsIDrawObject** and **XnsIDrawPoly** classes.

The class pointer is **xnsIDrawArcClass**.

The class name is **XnsIDrawArc**.

4.8 Description

The arc object is defined by three points. These three points are used to build a quarter of an ellipse.

4.9 Resources

The **XnsIDrawArc** widget defines the following resources.

XnsIDrawArc Resource Set

Name Class	Default Type	Access
XnsNdrawArcStrictRegion XnsCDrawArcStrictRegion	True XtRBoolean	CSG

XnsNdrawArcStrictRegion

If **True**, the object frame is the smallest rectangle enclosing the arc. If **False**, the frame is the region occupied by the object's defining points.

4.9.1 Inherited Resources

XnsIDrawArc inherits the following resources from the named superclass. For a complete description of each resource, refer to the man page for that superclass.

XnslDrawPoly Resource Set

Name Class	Default Type	Access
XnslNdrawResizeArrows XnslCDrawResizeArrows	True XtRBoolean	CSG
XnslNdrawArrowDir XnslCDrawArrowDir	XnslDRAW_NO_ARROW XnslRDrawArrowDir	CSG
XnslNdrawArrowStyle XnslCDrawArrowStyle	XnslDRAW_ARROW_FILLED XnslRDrawArrowStyle	CSG
XnslNdrawArrowWidth XnslCDrawArrowWidth	5 XnslRInt	CSG
XnslNdrawArrowHeight XnslCDrawArrowHeight	15 XnslRInt	CSG
XnslNdrawFromWidget XnslCDrawFromWidget	NULL XnslRDrawLinkWidget	CSG
XnslNdrawXFrom XnslCDrawXFrom	NULL XtRFloat	CSG
XnslNdrawYFrom XnslCDrawYFrom	NULL XtRFloat	CSG
XnslNdrawUpdateFrom XnslCDrawUpdateFrom	XnslDRAW_AUTO_UPDATE XnslRDrawUpdateLinkPolicy	CSG
XnslNdrawToWidget XnslCDrawToWidget	NULL XnslRDrawLinkWidget	CSG
XnslNdrawXTo XnslCDrawXTo	NULL XtRFloat	CSG
XnslNdrawYTo XnslCDrawYTo	NULL XtRFloat	CSG
XnslNdrawUpdateTo XnslCDrawUpdateTo	XnslDRAW_AUTO_UPDATE XnslRDrawUpdateLinkPolicy	CSG
XnslNdrawUpdateLinkCallback XnslCDrawUpdateLinkCallback	NULL XtRCallback	C

XnslDrawObject Resource Set

Name Class	Default Type	Access
XnslNdrawActivateCallback XnslCDrawActivateCallback	NULL XtRCallback	C
XnslNdrawActivateCallbackList XnslCDrawActivateCallbackList	NULL XnslRDrawActivateCallbackList	CSG
XnslNdrawActivateCondition XnslCDrawActivateCondition	XnslDRAW_INSIDE XnslRDrawSelectCondition	CSG
XnslNdrawAddPoint XnslCDrawAddPoint	NULL XnslRDrawAddPoint	CSG
XnslNdrawAlignment XnslCDrawAlignment	XnslDRAW_MIDDLE XnslRDrawAlignment	CSG
XnslNdrawAngle XnslCDrawAngle	NULL XtRFloat	CSG
XnslNdrawAngleUnits XnslCDrawAngleUnits	XnslRADIANS XnslRDrawAngleUnits	CSG
XnslNdrawBackground XnslCDrawPixel	-1 XnslRDrawPixel	CSG
XnslNdrawBackgroundPixmap XnslCDrawBackgroundPixmap	-1 XnslRDrawImage	CSG
XnslNdrawBlinkTime XnslCDrawBlinkTime	0 XtRInt	CSG
XnslNdrawBottomMargin XnslCDrawBottomMargin	5 XtRInt	CSG
XnslNdrawClipContents XnslCDrawClipContents	False XtRBoolean	CSG
XnslNdrawCloseMode XnslCDrawCloseMode	XnslDRAW_OPEN XnslRDrawCloseMode	CSG
XnslNdrawDashes XnslCDrawDashes	0 XnslRDrawDashes	CSG
XnslNdrawDefaultFontName XnslCdrawDefaultFontName	“Fixed*” String	CSG
XnslNdrawEventMask XnslCDrawEventMask	BUTTON1DOWN XnslRDrawEventMask	CSG

XnsIDrawObject Resource Set

Name Class	Default Type	Access
XnsINdrawFillDir XnsICDrawFillDir	XnsIDRAW_FILL_BOTTOM XnsIRDrawFillDir	CSG
XnsINdrawFillMode XnsICDrawFillMode	XnsIDRAW_NON_FILLED XnsIRDrawFillMode	CSG
XnsINdrawFont XnsICDrawFont	-1 XnsIRDrawFont	CSG
XnsINdrawForeground XnsICDrawPixel	-1 XnsIRDrawPixel	CSG
XnsINdrawFrameBackground XnsICDrawPixel	-1 XnsIRDrawPixel	CSG
XnsINdrawFrameDashes XnsICDrawDashes	0 XnsIRDrawDashes	CSG
XnsINdrawFrameForeground XnsICDrawPixel	-1 XnsIRDrawPixel	CSG
XnsINdrawFrameFunction XnsICDrawFunction	GXcopy XnsIRDrawFunction	CSG
XnsINdrawFrameLineWidth XnsICDrawLineWidth	0 XnsIRInt	CSG
XnsINdrawFrameStipple XnsICDrawStipple	-1 XnsIRDrawStipple	CSG
XnsINdrawFrameXtSOrigin XnsICDrawXTSOrigin	0 XnsIRInt	CSG
XnsINdrawFrameYtSOrigin XnsICDrawYTSOrigin	0 XnsIRInt	CSG
XnsINdrawFunction XnsICDrawFunction	GXcopy XnsIRDrawFunction	CSG
XnsINdrawGeoAttributes XnsICDrawGeoAttributes	NULL XnsIRDrawGeoAttributes	G
XnsINdrawGlobalVariables XnsICdrawGlobalVariables	True Boolean	CSG
XnsINdrawGraphAttributes XnsICDrawGraphAttributes	NULL XnsIRDrawGraphAttributes	G
XnsINdrawHeight XtCHeight	MAX_DIM XtRDimension	CSG

XnslDrawObject Resource Set

Name Class	Default Type	Access
XnslNdrawInheritAttributes XnslCDrawInheritAttributes	XnslDRAW_INHERIT_ATTRIBUTES XnslRDrawInheritAttributes	CSG
XnslNdrawLeftMargin XnslCDrawLeftMargin	5 XtRInt	CSG
XnslNdrawLineWidth XnslCDrawLineWidth	0 XnslRInt	CSG
XnslNdrawModifyCallback XnslCDrawModifyCallback	NULL XtRCallback	C
XnslNdrawModifyVerifyCallback XnslCDrawModifyVerifyCallback	NULL XtRCallback	C
XnslNdrawName XnslCDrawName	NULL XtRString	CSG
XnslNdrawNthUserData XnslCDrawNthUserData	0 XnslRInt	CSG
XnslNdrawNumPoints XnslCDrawNumPoints	0 XnslRInt	CSG
XnslNdrawNumReplacePoints XnslCDrawNumReplacePoints	0 XnslRInt	CSG
XnslNdrawObjectTolerance XtCWidth	4 XtRInt	CSG
XnslNdrawOpaque XnslCDrawOpaque	False XtRBoolean	CSG
XnslNdrawParent XnslCDrawParent	NULL XnslRObject	CSG
XnslNdrawPctFill XnslCDrawPctFill	0 XnslRInt	CSG
XnslNdrawPoints XnslCDrawPoints	NULL XnslRDrawPoints	CSG
XnslNdrawRecomputeSize XnslCDrawRecomputeSize	True XtRBoolean	CSG
XnslNdrawRegular XnslCDrawRegular	False XtRBoolean	CSG
XnslNdrawRemovePoint XnslCDrawRemovePoint	-1 XnslRInt	CSG

XnsIDrawObject Resource Set

Name Class	Default Type	Access
XnsINdrawReplacePoints XnsICDrawReplacePoints	NULL XnsIRDrawReplacePoints	CSG
XnsINdrawResizeCallback XnsICDrawResizeCallback	NULL XtRCallback	C
XnsINdrawResizeLineWidth XnsICDrawResizeLineWidth	True XtRBoolean	CSG
XnsINdrawResizePolicy XnsICDrawResizePolicy	XnsIDRAW_ALIGN XnsIRDrawResizePolicy	CSG
XnsINdrawRightMargin XnsICDrawRightMargin	5 XtRInt	CSG
XnsINdrawScriptEnabled XnsICDrawScriptEnabled	True XtRBoolean	CSG
XnsINdrawSelectCondition XnsICDrawSelectCondition	XnsIDRAW_ON_BORDER XnsIRDrawSelectCondition	CSG
XnsINdrawSelectMode XnsICDrawSelectMode	XnsIDRAW_TRACK_POINTS XnsIRDrawSelectMode	CSG
XnsINdrawShowBorder XnsICDrawShowBorder	False XtRBoolean	CSG
XnsINdrawShowFrame XnsICDrawShowFrame	False XtRBoolean	CSG
XnsINdrawStipple XnsICDrawStipple	-1 XnsIRDrawStipple	CSG
XnsINdrawStrictFrame XnsICDrawStrictFrame	False XtRBoolean	CSG
XnsINdrawStringValue XnsICDrawStringValue	NULL XtRString	CSG
XnsINdrawTopMargin XnsICDrawTopMargin	5 XtRInt	CSG
XnsINdrawTrackMode XnsICDrawTrackMode	XnsIDRAW_SELECT_POINTS XnsIRDrawTrackMode	CSG
XnsINdrawUnits XnsICDrawUnits	XnsIPIXELS XnsIRDrawUnits	CSG
XnsINdrawUserData XnsICDrawUserData	NULL XnsIRDrawUserData	CSG

XnslDrawObject Resource Set

Name Class	Default Type	Access
XnslNdrawUseRgbCube XnslCDrawUseRgbCube	False XtRBoolean	CG
XnslNdrawWidth XtCWidth	MAX_DIM XtRDimension	CSG
XnslNdrawX XtCPosition	MAX_POS XtRPosition	CSG
XnslNdrawXCenter XnslCDrawXCenter	NULL XtRFloat	CSG
XnslNdrawXPostScale XnslCDrawXPostScale	NULL XtRFloat	CSG
XnslNdrawXPreScale XnslCDrawXPreScale	NULL XtRFloat	CSG
XnslNdrawXTranslation XnslCDrawXTranslation	NULL XtRFloat	CSG
XnslNdrawXtSOrigin XnslCDrawXTSOrigin	0 XnslRInt	CSG
XnslNdrawY XtCPosition	MAX_POS XtRPosition	CSG
XnslNdrawYCenter XnslCDrawYCenter	NULL XtRFloat	CSG
XnslNdrawYPostScale XnslCDrawYPostScale	NULL XtRFloat	CSG
XnslNdrawYPreScale XnslCDrawYPreScale	NULL XtRFloat	CSG
XnslNdrawYTranslation XnslCDrawYTranslation	NULL XtRFloat	CSG
XnslNdrawYtSOrigin XnslCDrawYTSOrigin	0 XnslRInt	CSG

XmGadget Resource Set

Name Class	Default Type	Access
XmNbottomShadowColor XmCBottomShadowColor	dynamic Pixel	G
XmNhelpCallback XmCCallback	NULL XtCallbackList	C
XmNhighlightColor XmCHighlightColor	dynamic Pixel	G
XmNnavigationType XmCNavigationType	XmNONE XmNavigationType	CSG
XmNshadowThickness XmCShadowThickness	2 Dimension	CSG
XmNtopShadowColor XmCTopShadowColor	dynamic Pixel	G
XmNtraversalOn XmCTraversalOn	True Boolean	CSG
XmNunitType XmCUnitType	dynamic unsigned char	CSG
XmNuserData XmCUserData	NULL XtPointer	CSG

RectObj Resource Set

Name Class	Default Type	Access
XmNancestorSensitive XmCSensitive	dynamic Boolean	G
XmNborderWidth XmCBorderWidth	0 Dimension	CSG
XmNheight XmCHeight	dynamic Dimension	CSG
XmNsensitive XmCSensitive	True Boolean	CSG
XmNwidth XmCWidth	dynamic Dimension	CSG

RectObj Resource Set

Name Class	Default Type	Access
XmNx XmCPosition	0 Position	CSG
XmNy XmCPosition	0 Position	CSG

Object Resource Set

Name Class	Default Type	Access
XmNdestroyCallback XmCCallback	NULL XtCallbackList	C

CHAPTER 6: The XnsIDrawEllipse Class

6.1 Synopsis

```
#include <XnsI/DEllipse.h>
```

6.2 Classes

The **XnsIDrawEllipse** class inherits resources and behavior from the **Object**, **RectObj**, **XmGadget**, **XnsIDrawObject** and **XnsIDrawRect** classes.

The class pointer is **xnsIDrawEllipseClass**.

The class name is **XnsIDrawEllipse**.

6.3 Description

An ellipse is defined by its enclosing rectangle. A start angle and an end angle are provided to draw only a part of an ellipse.

6.4 Resources

The **XnsIDrawEllipse** widget defines the following resources.

XnsIDrawEllipse Resource Set

Name Class	Default Type	Access
XnsINdrawEllipseStartAngle XnsICDrawEllipseStartAngle	0.0 XtRFloat	CSG
XnsINdrawEllipseStrictRegion XnsICDrawEllipseStrictRegion	False XtRBoolean	CSG
XnsINdrawEllipseEndAngle XnsICDrawEllipseEndAngle	2*M_PI XtRFloat	CSG

XnsINdrawEllipseStartAngle

Angle in widget's unit defining the start point of the ellipse.

XnsINdrawEllipseStrictRegion

If **True**, the object frame is the smallest rectangle enclosing the curve. If **False**, the frame is the region occupied by the object's defining rectangle.

XnsINdrawEllipseEndAngle

Angle in widget’s unit defining the end point of the ellipse.

6.4.1 Inherited Resources

XnsIDrawEllipse inherits the following resources from the named superclass. For a complete description of each resource, refer to the man page for that superclass.

XnsIDrawRect Resource Set

Name Class	Default Type	Access
XnsINdrawRadius XnsICDrawRadius	0.0 XtRFloat	CSG

XnsIDrawObject Resource Set

Name Class	Default Type	Access
XnsINdrawActivateCallback XnsICDrawActivateCallback	NULL XtRCallback	C
XnsINdrawActivateCallbackList XnsICDrawActivateCallbackList	NULL XnsIRDrawActivateCallbackList	CSG
XnsINdrawActivateCondition XnsICDrawActivateCondition	XnsIDRAW_INSIDE XnsIRDrawSelectCondition	CSG
XnsINdrawAddPoint XnsICDrawAddPoint	NULL XnsIRDrawAddPoint	CSG
XnsINdrawAlignment XnsICDrawAlignment	XnsIDRAW_MIDDLE XnsIRDrawAlignment	CSG
XnsINdrawAngle XnsICDrawAngle	NULL XtRFloat	CSG
XnsINdrawAngleUnits XnsICDrawAngleUnits	XnsIRADIANS XnsIRDrawAngleUnits	CSG
XnsINdrawBackground XnsICDrawPixel	-1 XnsIRDrawPixel	CSG
XnsINdrawBackgroundPixmap XnsICDrawBackgroundPixmap	-1 XnsIRDrawImage	CSG

XnsIDrawObject Resource Set

Name Class	Default Type	Access
XnsINdrawBlinkTime XnsICDrawBlinkTime	0 XtRInt	CSG
XnsINdrawBottomMargin XnsICDrawBottomMargin	5 XtRInt	CSG
XnsINdrawClipContents XnsICDrawClipContents	False XtRBoolean	CSG
XnsINdrawCloseMode XnsICDrawCloseMode	XnsIDRAW_OPEN XnsIRDrawCloseMode	CSG
XnsINdrawDashes XnsICDrawDashes	0 XnsIRDrawDashes	CSG
XnsINdrawDefaultFontName XnsICdrawDefaultFontName	“Fixed*” String	CSG
XnsINdrawEventMask XnsICDrawEventMask	BUTTON1DOWN XnsIRDrawEventMask	CSG
XnsINdrawFillDir XnsICDrawFillDir	XnsIDRAW_FILL_BOTTOM XnsIRDrawFillDir	CSG
XnsINdrawFillMode XnsICDrawFillMode	XnsIDRAW_NON_FILLED XnsIRDrawFillMode	CSG
XnsINdrawFont XnsICDrawFont	-1 XnsIRDrawFont	CSG
XnsINdrawForeground XnsICDrawPixel	-1 XnsIRDrawPixel	CSG
XnsINdrawFrameBackground XnsICDrawPixel	-1 XnsIRDrawPixel	CSG
XnsINdrawFrameDashes XnsICDrawDashes	0 XnsIRDrawDashes	CSG
XnsINdrawFrameForeground XnsICDrawPixel	-1 XnsIRDrawPixel	CSG
XnsINdrawFrameFunction XnsICDrawFunction	GXcopy XnsIRDrawFunction	CSG
XnsINdrawFrameLineWidth XnsICDrawLineWidth	0 XnsIRInt	CSG
XnsINdrawFrameStipple XnsICDrawStipple	-1 XnsIRDrawStipple	CSG

XnsIDrawObject Resource Set

Name Class	Default Type	Access
XnsINdrawFrameXtOrigin XnsICDrawXTSOrigin	0 XnsIRInt	CSG
XnsINdrawFrameYtOrigin XnsICDrawYTSOrigin	0 XnsIRInt	CSG
XnsINdrawFunction XnsICDrawFunction	GXcopy XnsIRDrawFunction	CSG
XnsINdrawGeoAttributes XnsICDrawGeoAttributes	NULL XnsIRDrawGeoAttributes	G
XnsINdrawGlobalVariables XnsICDrawGlobalVariables	True Boolean	CSG
XnsINdrawGraphAttributes XnsICDrawGraphAttributes	NULL XnsIRDrawGraphAttributes	G
XnsINdrawHeight XtCHeight	MAX_DIM XtRDimension	CSG
XnsINdrawInheritAttributes XnsICDrawInheritAttributes	XnsIDRAW_INHERIT_ATTRIBUTES XnsIRDrawInheritAttributes	CSG
XnsINdrawLeftMargin XnsICDrawLeftMargin	5 XtRInt	CSG
XnsINdrawLineWidth XnsICDrawLineWidth	0 XnsIRInt	CSG
XnsINdrawModifyCallback XnsICDrawModifyCallback	NULL XtRCallback	C
XnsINdrawModifyVerifyCallback XnsICDrawModifyVerifyCallback	NULL XtRCallback	C
XnsINdrawName XnsICDrawName	NULL XtRString	CSG
XnsINdrawNthUserData XnsICDrawNthUserData	0 XnsIRInt	CSG
XnsINdrawNumPoints XnsICDrawNumPoints	0 XnsIRInt	CSG
XnsINdrawNumReplacePoints XnsICDrawNumReplacePoints	0 XnsIRInt	CSG
XnsINdrawObjectTolerance XtCWidth	4 XtRInt	CSG

XnsIDrawObject Resource Set

Name Class	Default Type	Access
XnsINdrawOpaque XnsICDrawOpaque	False XtRBoolean	CSG
XnsINdrawParent XnsICDrawParent	NULL XnsIRObject	CSG
XnsINdrawPctFill XnsICDrawPctFill	0 XnsIRInt	CSG
XnsINdrawPoints XnsICDrawPoints	NULL XnsIRDrawPoints	CSG
XnsINdrawRecomputeSize XnsICDrawRecomputeSize	True XtRBoolean	CSG
XnsINdrawRegular XnsICDrawRegular	False XtRBoolean	CSG
XnsINdrawRemovePoint XnsICDrawRemovePoint	-1 XnsIRInt	CSG
XnsINdrawReplacePoints XnsICDrawReplacePoints	NULL XnsIRDrawReplacePoints	CSG
XnsINdrawResizeCallback XnsICDrawResizeCallback	NULL XtRCallback	C
XnsINdrawResizeLineWidth XnsICDrawResizeLineWidth	True XtRBoolean	CSG
XnsINdrawResizePolicy XnsICDrawResizePolicy	XnsIDRAW_ALIGN XnsIRDrawResizePolicy	CSG
XnsINdrawRightMargin XnsICDrawRightMargin	5 XtRInt	CSG
XnsINdrawScriptEnabled XnsICDrawScriptEnabled	True XtRBoolean	CSG
XnsINdrawSelectCondition XnsICDrawSelectCondition	XnsIDRAW_ON_BORDER XnsIRDrawSelectCondition	CSG
XnsINdrawSelectMode XnsICDrawSelectMode	XnsIDRAW_TRACK_POINTS XnsIRDrawSelectMode	CSG
XnsINdrawShowBorder XnsICDrawShowBorder	False XtRBoolean	CSG
XnsINdrawShowFrame XnsICDrawShowFrame	False XtRBoolean	CSG

XnsIDrawObject Resource Set

Name Class	Default Type	Access
XnsINdrawStipple XnsICDrawStipple	-1 XnsIRDrawStipple	CSG
XnsINdrawStrictFrame XnsICDrawStrictFrame	False XtrBoolean	CSG
XnsINdrawStringValue XnsICDrawStringValue	NULL XtrString	CSG
XnsINdrawTopMargin XnsICDrawTopMargin	5 XtrInt	CSG
XnsINdrawTrackMode XnsICDrawTrackMode	XnsIDRAW_SELECT_POINTS XnsIRDrawTrackMode	CSG
XnsINdrawUnits XnsICDrawUnits	XnsIPIXELS XnsIRDrawUnits	CSG
XnsINdrawUserData XnsICDrawUserData	NULL XnsIRDrawUserData	CSG
XnsINdrawUseRgbCube XnsICDrawUseRgbCube	False XtrBoolean	CG
XnsINdrawWidth XtCWidth	MAX_DIM XtrDimension	CSG
XnsINdrawX XtCPosition	MAX_POS XtrPosition	CSG
XnsINdrawXCenter XnsICDrawXCenter	NULL XtrFloat	CSG
XnsINdrawXPostScale XnsICDrawXPostScale	NULL XtrFloat	CSG
XnsINdrawXPreScale XnsICDrawXPreScale	NULL XtrFloat	CSG
XnsINdrawXTranslation XnsICDrawXTranslation	NULL XtrFloat	CSG
XnsINdrawXtSOrigin XnsICDrawXTSOrigin	0 XnsIRInt	CSG
XnsINdrawY XtCPosition	MAX_POS XtrPosition	CSG
XnsINdrawYCenter XnsICDrawYCenter	NULL XtrFloat	CSG

XnsIDrawObject Resource Set

Name Class	Default Type	Access
XnsINdrawYPostScale XnsICDrawYPostScale	NULL XtRFloat	CSG
XnsINdrawYPreScale XnsICDrawYPreScale	NULL XtRFloat	CSG
XnsINdrawYTranslation XnsICDrawYTranslation	NULL XtRFloat	CSG
XnsINdrawYtSOrigin XnsICDrawYtSOrigin	0 XnsIRInt	CSG

XmGadget Resource Set

Name Class	Default Type	Access
XmNbottomShadowColor XmCBottomShadowColor	dynamic Pixel	G
XmNhelpCallback XmCCallback	NULL XtCallbackList	C
XmNhighlightColor XmCHighlightColor	dynamic Pixel	G
XmNnavigationType XmCNavigationType	XmNONE XmNavigationType	CSG
XmNshadowThickness XmCShadowThickness	2 Dimension	CSG
XmNtopShadowColor XmCTopShadowColor	dynamic Pixel	G
XmNtraversalOn XmCTraversalOn	True Boolean	CSG
XmNunitType XmCUnitType	dynamic unsigned char	CSG
XmNuserData XmCUserData	NULL XtPointer	CSG

RectObj Resource Set

Name Class	Default Type	Access
XmNancestorSensitive XmCSensitive	dynamic Boolean	G
XmNborderWidth XmCBorderWidth	0 Dimension	CSG
XmNheight XmCHeight	dynamic Dimension	CSG
XmNsensitive XmCSensitive	True Boolean	CSG
XmNwidth XmCWidth	dynamic Dimension	CSG
XmNx XmCPosition	0 Position	CSG
XmNy XmCPosition	0 Position	CSG

Object Resource Set

Name Class	Default Type	Access
XmNdestroyCallback XmCCallback	NULL XtCallbackList	C

CHAPTER 7: The XnsIDrawGroup Class

7.1 Synopsis

```
#include <XnsI/DGroup.h>
```

7.2 Classes

The **XnsIDrawGroup** class inherits resources and behavior from the **Object**, **RectObj**, **XmGadget**, **XnsIDrawObject** and **XnsIDrawRect** classes.

The class pointer is **xnsIDrawGroupClass**.

The class name is **XnsIDrawGroup**.

7.3 Description

The **XnsIDrawGroup** object is used to group **XnsIDrawObjects**. When grouped, a set of objects becomes a new composite object. Geometric transformations specified for the group are applied to the children.

7.4 Resources

The **XnsIDrawGroup** widget defines the following resources.

XnsIDrawGroup Resource Set

Name Class	Default Type	Access
XnsINdrawChildren XnsICDrawChildren	NULL XnsIRDrawWidgetList	CSG
XnsINdrawGroupBottomMargin † XnsICDrawGroupBottomMargin	0 XtRInt	CG
XnsINdrawGroupDestroyChildren XnsICDrawGroupDestroyChildren	True XtRBoolean	CSG
XnsINdrawGroupLeftMargin † XnsICDrawGroupLeftMargin	0 XtRInt	CG
XnsINdrawGroupModifyPolicy XnsICDrawGroupModifyPolicy	XnsIDRAW_GROUP_CENTER XnsIRDrawGroupModifyPolicy	CSG

XnslDrawGroup Resource Set

Name Class	Default Type	Access
XnslNdrawGroupPassAttributes XnslCDrawGroupPassAttributes	True XtRBoolean	CSG
XnslNdrawGroupRightMargin † XnslCDrawGroupRightMargin	0 XtRInt	CG
XnslNdrawGroupTopMargin † XnslCDrawGroupTopMargin	0 XtRInt	CG
XnslNdrawNumChildren XnslCDrawNumChildren	0 XnslRCardinal	CSG

XnslNdrawChildren

Specifies a pointer to an array of **XnslDrawObject** objects to be grouped.

XnslNdrawGroupBottomMargin †

Reserved for future use.

XnslNdrawGroupDestroyChildren

Specifies whether a group's children are to be destroyed along with the group.

XnslNdrawGroupLeftMargin †

Reserved for future use.

XnslNdrawGroupModifyPolicy

Specifies the behavior of the children when the **XnslDrawGroup** is resized. Valid values are:

- m **XnslDRAW_GROUP_CENTER**: The children remain centered in the group.
- m **XnslDRAW_GROUP_DISTRIBUTE**: The children are distributed in the group with rubber positioning.
- m **XnslDRAW_GROUP_SCALE**: The children are resized with the group.

XnslNdrawGroupPassAttributes

If set to **True**, the graphic attributes are not used by the **XnslDrawGroup** but are available for children inheritance. If set to **False**, graphic attributes are used by the group.

XnslNdrawGroupRightMargin †

Reserved for future use.

XnslNdrawGroupTopMargin

Reserved for future use.

XnslNdrawNumChildren

Specifies the size of the **XnslNdrawChildren** array.

7.4.1 Inherited Resources

XnslDrawGroup inherits the following resources from the named superclass. For a complete description of each resource, refer to the man page for that superclass.

XnslDrawRect Resource Set

Name Class	Default Type	Access
XnslNdrawRadius XnslCDrawRadius	0.0 XtRFloat	CSG

XnslDrawObject Resource Set

Name Class	Default Type	Access
XnslNdrawActivateCallback XnslCDrawActivateCallback	NULL XtRCallback	C
XnslNdrawActivateCallbackList XnslCDrawActivateCallbackList	NULL XnslRDrawActivateCallbackList	CSG
XnslNdrawActivateCondition XnslCDrawActivateCondition	XnslDRAW_INSIDE XnslRDrawSelectCondition	CSG
XnslNdrawAddPoint XnslCDrawAddPoint	NULL XnslRDrawAddPoint	CSG
XnslNdrawAlignment XnslCDrawAlignment	XnslDRAW_MIDDLE XnslRDrawAlignment	CSG
XnslNdrawAngle XnslCDrawAngle	NULL XtRFloat	CSG
XnslNdrawAngleUnits XnslCDrawAngleUnits	XnslRADIANS XnslRDrawAngleUnits	CSG

XnsIDrawObject Resource Set

Name Class	Default Type	Access
XnsINdrawBackground XnsICDrawPixel	-1 XnsIRDrawPixel	CSG
XnsINdrawBackgroundPixmap XnsICDrawBackgroundPixmap	-1 XnsIRDrawImage	CSG
XnsINdrawBlinkTime XnsICDrawBlinkTime	0 XtRInt	CSG
XnsINdrawBottomMargin XnsICDrawBottomMargin	5 XtRInt	CSG
XnsINdrawClipContents XnsICDrawClipContents	False XtRBoolean	CSG
XnsINdrawCloseMode XnsICDrawCloseMode	XnsIDRAW_OPEN XnsIRDrawCloseMode	CSG
XnsINdrawDashes XnsICDrawDashes	0 XnsIRDrawDashes	CSG
XnsINdrawDefaultFontName XnsICdrawDefaultFontName	“Fixed*” String	CSG
XnsINdrawEventMask XnsICDrawEventMask	BUTTON1DOWN XnsIRDrawEventMask	CSG
XnsINdrawFillDir XnsICDrawFillDir	XnsIDRAW_FILL_BOTTOM XnsIRDrawFillDir	CSG
XnsINdrawFillMode XnsICDrawFillMode	XnsIDRAW_NON_FILLED XnsIRDrawFillMode	CSG
XnsINdrawFont XnsICDrawFont	-1 XnsIRDrawFont	CSG
XnsINdrawForeground XnsICDrawPixel	-1 XnsIRDrawPixel	CSG
XnsINdrawFrameBackground XnsICDrawPixel	-1 XnsIRDrawPixel	CSG
XnsINdrawFrameDashes XnsICDrawDashes	0 XnsIRDrawDashes	CSG
XnsINdrawFrameForeground XnsICDrawPixel	-1 XnsIRDrawPixel	CSG
XnsINdrawFrameFunction XnsICDrawFunction	GXcopy XnsIRDrawFunction	CSG

XnslDrawObject Resource Set

Name Class	Default Type	Access
XnslNdrawFrameLineWidth XnslCDrawLineWidth	0 XnslRInt	CSG
XnslNdrawFrameStipple XnslCDrawStipple	-1 XnslRDrawStipple	CSG
XnslNdrawFrameXtSOrigin XnslCDrawXTSOrigin	0 XnslRInt	CSG
XnslNdrawFrameYtSOrigin XnslCDrawYTSOrigin	0 XnslRInt	CSG
XnslNdrawFunction XnslCDrawFunction	GXcopy XnslRDrawFunction	CSG
XnslNdrawGeoAttributes XnslCDrawGeoAttributes	NULL XnslRDrawGeoAttributes	G
XnslNdrawGlobalVariables XnslCdrawGlobalVariables	True Boolean	CSG
XnslNdrawGraphAttributes XnslCDrawGraphAttributes	NULL XnslRDrawGraphAttributes	G
XnslNdrawHeight XtCHeight	MAX_DIM XtRDimension	CSG
XnslNdrawInheritAttributes XnslCDrawInheritAttributes	XnslDRAW_INHERIT_ATTRIBUTES XnslRDrawInheritAttributes	CSG
XnslNdrawLeftMargin XnslCDrawLeftMargin	5 XtRInt	CSG
XnslNdrawLineWidth XnslCDrawLineWidth	0 XnslRInt	CSG
XnslNdrawModifyCallback XnslCDrawModifyCallback	NULL XtRCallback	C
XnslNdrawModifyVerifyCallback XnslCDrawModifyVerifyCallback	NULL XtRCallback	C
XnslNdrawName XnslCDrawName	NULL XtRString	CSG
XnslNdrawNthUserData XnslCDrawNthUserData	0 XnslRInt	CSG
XnslNdrawNumPoints XnslCDrawNumPoints	0 XnslRInt	CSG

XnsIDrawObject Resource Set

Name Class	Default Type	Access
XnsINdrawNumReplacePoints XnsICDrawNumReplacePoints	0 XnsIRInt	CSG
XnsINdrawObjectTolerance XtCWidth	4 XtRInt	CSG
XnsINdrawOpaque XnsICDrawOpaque	False XtRBoolean	CSG
XnsINdrawParent XnsICDrawParent	NULL XnsIRObject	CSG
XnsINdrawPctFill XnsICDrawPctFill	0 XnsIRInt	CSG
XnsINdrawPoints XnsICDrawPoints	NULL XnsIRDrawPoints	CSG
XnsINdrawRecomputeSize XnsICDrawRecomputeSize	True XtRBoolean	CSG
XnsINdrawRegular XnsICDrawRegular	False XtRBoolean	CSG
XnsINdrawRemovePoint XnsICDrawRemovePoint	-1 XnsIRInt	CSG
XnsINdrawReplacePoints XnsICDrawReplacePoints	NULL XnsIRDrawReplacePoints	CSG
XnsINdrawResizeCallback XnsICDrawResizeCallback	NULL XtRCallback	C
XnsINdrawResizeLineWidth XnsICDrawResizeLineWidth	True XtRBoolean	CSG
XnsINdrawResizePolicy XnsICDrawResizePolicy	XnsIDRAW_ALIGN XnsIRDrawResizePolicy	CSG
XnsINdrawRightMargin XnsICDrawRightMargin	5 XtRInt	CSG
XnsINdrawScriptEnabled XnsICDrawScriptEnabled	True XtRBoolean	CSG
XnsINdrawSelectCondition XnsICDrawSelectCondition	XnsIDRAW_ON_BORDER XnsIRDrawSelectCondition	CSG
XnsINdrawSelectMode XnsICDrawSelectMode	XnsIDRAW_TRACK_POINTS XnsIRDrawSelectMode	CSG

XnslDrawObject Resource Set

Name Class	Default Type	Access
XnslNdrawShowBorder XnslCDrawShowBorder	False XtRBoolean	CSG
XnslNdrawShowFrame XnslCDrawShowFrame	False XtRBoolean	CSG
XnslNdrawStipple XnslCDrawStipple	-1 XnslRDrawStipple	CSG
XnslNdrawStrictFrame XnslCDrawStrictFrame	False XtRBoolean	CSG
XnslNdrawStringValue XnslCDrawStringValue	NULL XtRString	CSG
XnslNdrawTopMargin XnslCDrawTopMargin	5 XtRInt	CSG
XnslNdrawTrackMode XnslCDrawTrackMode	XnslDRAW_SELECT_POINTS XnslRDrawTrackMode	CSG
XnslNdrawUnits XnslCDrawUnits	XnslPIXELS XnslRDrawUnits	CSG
XnslNdrawUserData XnslCDrawUserData	NULL XnslRDrawUserData	CSG
XnslNdrawUseRgbCube XnslCDrawUseRgbCube	False XtRBoolean	CG
XnslNdrawWidth XtCWidth	MAX_DIM XtRDimension	CSG
XnslNdrawX XtCPosition	MAX_POS XtRPosition	CSG
XnslNdrawXCenter XnslCDrawXCenter	NULL XtRFloat	CSG
XnslNdrawXPostScale XnslCDrawXPostScale	NULL XtRFloat	CSG
XnslNdrawXPreScale XnslCDrawXPreScale	NULL XtRFloat	CSG
XnslNdrawXTranslation XnslCDrawXTranslation	NULL XtRFloat	CSG
XnslNdrawXtSOrigin XnslCDrawXTSOrigin	0 XnslRInt	CSG

XnsIDrawObject Resource Set

Name Class	Default Type	Access
XnsIDrawY XtCPosition	MAX_POS XtRPosition	CSG
XnsIDrawYCenter XnsICDrawYCenter	NULL XtRFloat	CSG
XnsIDrawYPostScale XnsICDrawYPostScale	NULL XtRFloat	CSG
XnsIDrawYPreScale XnsICDrawYPreScale	NULL XtRFloat	CSG
XnsIDrawYTranslation XnsICDrawYTranslation	NULL XtRFloat	CSG
XnsIDrawYtSOrigin XnsICDrawYTSOrigin	0 XnsIRInt	CSG

XmGadget Resource Set

Name Class	Default Type	Access
XmNbottomShadowColor XmCBottomShadowColor	dynamic Pixel	G
XmNhelpCallback XmCCallback	NULL XtCallbackList	C
XmNhighlightColor XmCHighlightColor	dynamic Pixel	G
XmNnavigationType XmCNavigationType	XmNONE XmNavigationType	CSG
XmNshadowThickness XmCShadowThickness	2 Dimension	CSG
XmNtopShadowColor XmCTopShadowColor	dynamic Pixel	G
XmNtraversalOn XmCTraversalOn	True Boolean	CSG
XmNunitType XmCUnitType	dynamic unsigned char	CSG

XmGadget Resource Set

Name Class	Default Type	Access
XmNuserData XmCUserData	NULL XtPointer	CSG

RectObj Resource Set

Name Class	Default Type	Access
XmNancestorSensitive XmCSensitive	dynamic Boolean	G
XmNborderWidth XmCBorderWidth	0 Dimension	CSG
XmNheight XmCHeight	dynamic Dimension	CSG
XmNsensitive XmCSensitive	True Boolean	CSG
XmNwidth XmCWidth	dynamic Dimension	CSG
XmNx XmCPosition	0 Position	CSG
XmNy XmCPosition	0 Position	CSG

Object Resource Set

Name Class	Default Type	Access
XmNdestroyCallback XmCCallback	NULL XtCallbackList	C

CHAPTER 8: The XnsldrawImage Class

8.1 Synopsis

#include <XnsldImage.h>

8.2 Classes

The **XnsldrawImage** class inherits behavior and resources from the **Object**, **RectObj**, **XmGadget**, **XnsldrawObject** and the **XnsldrawRect** classes.

The class pointer is **xnsldrawImageClass**.

The class name is **XnsldrawImage**.

8.3 Description

The **XnsldrawImage** widget displays images. Images can come from Gif¹, bitmap, xwd or xpm3 files. Colors used to display those images cannot be exactly the same as those defined in the files. The converter used to interpret the file sorts the colors according to their frequency of use and tries to allocate them. If some cannot be allocated, then the converter allocates the closest color in the colormap. Images can be scaled and/or rotated. The algorithm used for scaling and rotation does not allocate any new colors.

8.4 Resources

The **XnsldrawImage** defines the following resources:

XnsldrawImage Resource Set

Name Class	Default Type	Access
XnsldrawImage XnsldrawImage	-1 XnsldrawImage	CSG
XnsldrawImageFitPoints XnsldrawImageFitPoints	False XtRBoolean	CSG

1. The Graphics Interchange Format(c) is the Copyright property of CompuServe Incorporated. GIF(sm) is a Service Mark property of CompuServe Incorporated.

XnsIDrawImage Resource Set

Name Class	Default Type	Access
XnsINdrawImageMask XnsICDrawStipple	-1 XnsIRDrawStipple	CSG
XnsINdrawImageTransformImage XnsICDrawImageTransformImage	True XtRBoolean	CSG

The **XnsIDrawImage** also overrides the following resources:

XnsIDrawImage Resource Set

Name Class	Default Type	Access
XnsINdrawShowBorder XnsICDrawShowBorder	True XtRBoolean	CSG
XnsINdrawSelectCondition XnsICDrawSelectCondition	XnsIDRAW_ON_BORDER XnsIDRAW_INSIDE XnsIRDrawSelectCondition	CSG

XnsINdrawImage

Specifies the index of the image to display. This index is obtained by the function **XnsIDrawGetPixmap**.

XnsINdrawImageFitPoints

If set to **True**, the size of the widget is adjusted to the size of the image, regardless of dimension specifications applied to the widget.

XnsINdrawImageMask

Specifies the index of the mask used to display the image. This index is obtained by the function **XnsIDrawGetBitmap**. This resource is ignored when the widget is scaled or rotated.

XnsINdrawImageTransformImage

Indicates whether image rotation and scaling is enabled. If it is set to **False**, the widget does not scale or rotate the image.

8.4.1 Inherited Resources

XnsIDrawImage inherits the following resources from the named superclass. For a complete description of each resource, refer to the man page for that superclass.

XnsIDrawObject Resource Set

Name Class	Default Type	Access
XnsINdrawActivateCallback XnsICDrawActivateCallback	NULL XtRCallback	C
XnsINdrawActivateCallbackList XnsICDrawActivateCallbackList	NULL XnsIRDrawActivateCallbackList	CSG
XnsINdrawActivateCondition XnsICDrawActivateCondition	XnsIDRAW_INSIDE XnsIRDrawSelectCondition	CSG
XnsINdrawAddPoint XnsICDrawAddPoint	NULL XnsIRDrawAddPoint	CSG
XnsINdrawAlignment XnsICDrawAlignment	XnsIDRAW_MIDDLE XnsIRDrawAlignment	CSG
XnsINdrawAngle XnsICDrawAngle	NULL XtRFloat	CSG
XnsINdrawAngleUnits XnsICDrawAngleUnits	XnsIRADIANS XnsIRDrawAngleUnits	CSG
XnsINdrawBackground XnsICDrawPixel	-1 XnsIRDrawPixel	CSG
XnsINdrawBackgroundPixmap XnsICDrawBackgroundPixmap	-1 XnsIRDrawImage	CSG
XnsINdrawBlinkTime XnsICDrawBlinkTime	0 XtRInt	CSG
XnsINdrawBottomMargin XnsICDrawBottomMargin	5 XtRInt	CSG
XnsINdrawClipContents XnsICDrawClipContents	False XtRBoolean	CSG
XnsINdrawCloseMode XnsICDrawCloseMode	XnsIDRAW_OPEN XnsIRDrawCloseMode	CSG
XnsINdrawDashes XnsICDrawDashes	0 XnsIRDrawDashes	CSG
XnsINdrawDefaultFontName XnsICdrawDefaultFontName	“Fixed*” String	CSG

XnsIDrawObject Resource Set

Name Class	Default Type	Access
XnsINdrawEventMask XnsICDrawEventMask	BUTTONIDOWN XnsIRDrawEventMask	CSG
XnsINdrawFillDir XnsICDrawFillDir	XnsIDRAW_FILL_BOTTOM XnsIRDrawFillDir	CSG
XnsINdrawFillMode XnsICDrawFillMode	XnsIDRAW_NON_FILLED XnsIRDrawFillMode	CSG
XnsINdrawFont XnsICDrawFont	-1 XnsIRDrawFont	CSG
XnsINdrawForeground XnsICDrawPixel	-1 XnsIRDrawPixel	CSG
XnsINdrawFrameBackground XnsICDrawPixel	-1 XnsIRDrawPixel	CSG
XnsINdrawFrameDashes XnsICDrawDashes	0 XnsIRDrawDashes	CSG
XnsINdrawFrameForeground XnsICDrawPixel	-1 XnsIRDrawPixel	CSG
XnsINdrawFrameFunction XnsICDrawFunction	GXcopy XnsIRDrawFunction	CSG
XnsINdrawFrameLineWidth XnsICDrawLineWidth	0 XnsIRInt	CSG
XnsINdrawFrameStipple XnsICDrawStipple	-1 XnsIRDrawStipple	CSG
XnsINdrawFrameXtSOrigin XnsICDrawXTSOrigin	0 XnsIRInt	CSG
XnsINdrawFrameYtSOrigin XnsICDrawYTSOrigin	0 XnsIRInt	CSG
XnsINdrawFunction XnsICDrawFunction	GXcopy XnsIRDrawFunction	CSG
XnsINdrawGeoAttributes XnsICDrawGeoAttributes	NULL XnsIRDrawGeoAttributes	G
XnsINdrawGlobalVariables XnsICdrawGlobalVariables	True Boolean	CSG
XnsINdrawGraphAttributes XnsICDrawGraphAttributes	NULL XnsIRDrawGraphAttributes	G

XnslDrawObject Resource Set

Name Class	Default Type	Access
XnslNdrawHeight XtCHeight	MAX_DIM XtRDimension	CSG
XnslNdrawInheritAttributes XnslCdrawInheritAttributes	XnslDRAW_INHERIT_ATTRIBUTES XnslRDrawInheritAttributes	CSG
XnslNdrawLeftMargin XnslCdrawLeftMargin	5 XtRInt	CSG
XnslNdrawLineWidth XnslCdrawLineWidth	0 XnslRInt	CSG
XnslNdrawModifyCallback XnslCdrawModifyCallback	NULL XtRCallback	C
XnslNdrawModifyVerifyCallback XnslCdrawModifyVerifyCallback	NULL XtRCallback	C
XnslNdrawName XnslCdrawName	NULL XtRString	CSG
XnslNdrawNthUserData XnslCdrawNthUserData	0 XnslRInt	CSG
XnslNdrawNumPoints XnslCdrawNumPoints	0 XnslRInt	CSG
XnslNdrawNumReplacePoints XnslCdrawNumReplacePoints	0 XnslRInt	CSG
XnslNdrawObjectTolerance XtCWidth	4 XtRInt	CSG
XnslNdrawOpaque XnslCdrawOpaque	False XtRBoolean	CSG
XnslNdrawParent XnslCdrawParent	NULL XnslRObject	CSG
XnslNdrawPctFill XnslCdrawPctFill	0 XnslRInt	CSG
XnslNdrawPoints XnslCdrawPoints	NULL XnslRDrawPoints	CSG
XnslNdrawRecomputeSize XnslCdrawRecomputeSize	True XtRBoolean	CSG
XnslNdrawRegular XnslCdrawRegular	False XtRBoolean	CSG

XnsIDrawObject Resource Set

Name Class	Default Type	Access
XnsINdrawRemovePoint XnsICDrawRemovePoint	-1 XnsIRInt	CSG
XnsINdrawReplacePoints XnsICDrawReplacePoints	NULL XnsIRDrawReplacePoints	CSG
XnsINdrawResizeCallback XnsICDrawResizeCallback	NULL XtRCallback	C
XnsINdrawResizeLineWidth XnsICDrawResizeLineWidth	True XtRBoolean	CSG
XnsINdrawResizePolicy XnsICDrawResizePolicy	XnsIDRAW_ALIGN XnsIRDrawResizePolicy	CSG
XnsINdrawRightMargin XnsICDrawRightMargin	5 XtRInt	CSG
XnsINdrawScriptEnabled XnsICDrawScriptEnabled	True XtRBoolean	CSG
XnsINdrawSelectCondition XnsICDrawSelectCondition	XnsIDRAW_ON_BORDER XnsIRDrawSelectCondition	CSG
XnsINdrawSelectMode XnsICDrawSelectMode	XnsIDRAW_TRACK_POINTS XnsIRDrawSelectMode	CSG
XnsINdrawShowBorder XnsICDrawShowBorder	False XtRBoolean	CSG
XnsINdrawShowFrame XnsICDrawShowFrame	False XtRBoolean	CSG
XnsINdrawStipple XnsICDrawStipple	-1 XnsIRDrawStipple	CSG
XnsINdrawStrictFrame XnsICDrawStrictFrame	False XtRBoolean	CSG
XnsINdrawStringValue XnsICDrawStringValue	NULL XtRString	CSG
XnsINdrawTopMargin XnsICDrawTopMargin	5 XtRInt	CSG
XnsINdrawTrackMode XnsICDrawTrackMode	XnsIDRAW_SELECT_POINTS XnsIRDrawTrackMode	CSG
XnsINdrawUnits XnsICDrawUnits	XnsIPIXELS XnsIRDrawUnits	CSG

XnslDrawObject Resource Set

Name Class	Default Type	Access
XnslNdrawUserData XnslCDrawUserData	NULL XnslRDrawUserData	CSG
XnslNdrawUseRgbCube XnslCDrawUseRgbCube	False XtRBoolean	CG
XnslNdrawWidth XtCWidth	MAX_DIM XtRDimension	CSG
XnslNdrawX XtCPosition	MAX_POS XtRPosition	CSG
XnslNdrawXCenter XnslCDrawXCenter	NULL XtRFloat	CSG
XnslNdrawXPostScale XnslCDrawXPostScale	NULL XtRFloat	CSG
XnslNdrawXPreScale XnslCDrawXPreScale	NULL XtRFloat	CSG
XnslNdrawXTranslation XnslCDrawXTranslation	NULL XtRFloat	CSG
XnslNdrawXtSOrigin XnslCDrawXTSOrigin	0 XnslRInt	CSG
XnslNdrawY XtCPosition	MAX_POS XtRPosition	CSG
XnslNdrawYCenter XnslCDrawYCenter	NULL XtRFloat	CSG
XnslNdrawYPostScale XnslCDrawYPostScale	NULL XtRFloat	CSG
XnslNdrawYPreScale XnslCDrawYPreScale	NULL XtRFloat	CSG
XnslNdrawYTranslation XnslCDrawYTranslation	NULL XtRFloat	CSG
XnslNdrawYtSOrigin XnslCDrawYTSOrigin	0 XnslRInt	CSG

XmGadget Resource Set

Name Class	Default Type	Access
XmNbottomShadowColor XmCBottomShadowColor	dynamic Pixel	G
XmNhelpCallback XmCCallback	NULL XtCallbackList	C
XmNhighlightColor XmCHighlightColor	dynamic Pixel	G
XmNnavigationType XmCNavigationType	XmNONE XmNavigationType	CSG
XmNshadowThickness XmCShadowThickness	2 Dimension	CSG
XmNtopShadowColor XmCTopShadowColor	dynamic Pixel	G
XmNtraversalOn XmCTraversalOn	True Boolean	CSG
XmNunitType XmCUnitType	dynamic unsigned char	CSG
XmNuserData XmCUserData	NULL XtPointer	CSG

RectObj Resource Set

Name Class	Default Type	Access
XmNancestorSensitive XmCSensitive	dynamic Boolean	G
XmNborderWidth XmCBorderWidth	0 Dimension	CSG
XmNheight XmCHeight	dynamic Dimension	CSG
XmNsensitive XmCSensitive	True Boolean	CSG

RectObj Resource Set

Name Class	Default Type	Access
XmNwidth XmCWidth	dynamic Dimension	CSG
XmNx XmCPosition	0 Position	CSG
XmNy XmCPosition	0 Position	CSG

Object Resource Set

Name Class	Default Type	Access
XmNdestroyCallback XmCCallback	NULL XtCallbackList	C

CHAPTER 9: The XnsIDrawLabelClass Class

9.1 Synopsis

```
#include <XnsI/DLabel.h>
```

9.2 Classes

The **XnsIDrawLabel** inherits behavior and resources from the **Object**, **RectObj**, **XmGadget**, **XnsIDrawObject** and **XnsIDrawRect** classes.

The class pointer is **xnsIDrawLabelClass**.

The class name is **XnsIDrawLabel**.

9.3 Description

The **XnsIDrawLabel** widget displays text. The text can be on several lines and is displayed using the **XnsIDrawGetFont** function. The text can be scaled and rotated; an antialiasing mechanism is used to draw the text when it is rotated or scaled. The bounding box of the text is always centered in the widget.

9.4 Resources

The **XnsIDrawLabel** defines the following resources:

XnsIDrawLabel Resource Set

Name Class	Default Type	Access
XnsINdrawLabelAliasingColors XnsICDrawLabelAliasingColors	0 XtRInt	CSG
XnsINdrawLabelAlignment XnsICDrawLabelAlignment	XnsIDRAW_LABEL_ALIGN_LEFT XnsIRDrawLabelAlignment	CGS
XnsINdrawLabelFitPoints XnsICDrawLabelFitPoints	False XtRBoolean	CSG
XnsINdrawLabelString XnsICDrawLabelString	NULL XnsIRDrawLabelString	CSG
XnsINdrawLabelTransformString XnsICDrawLabelTransformString	True XtRBoolean	CSG

The **XnsIDrawLabel** also overrides the following resources.

Name Class	Default Type	Access
XnsINdrawShowBorder XnsICDrawShowBorder	True XtRBoolean	CSG
XnsINdrawSelectCondition XnsICDrawSelectCondition	XnsIDRAW_ON_BORDER XnsIDRAW_INSIDE XnsIRDrawSelectCondition	CSG

XnsINdrawLabelAlignment

For text that contains several lines, specifies how the text should be aligned. Acceptable values are:

- **XnsIDRAW_LABEL_ALIGN_LEFT**
- **XnsIDRAW_LABEL_ALIGN_RIGHT**
- **XnsIDRAW_LABEL_ALIGN_CENTER**
- **XnsIDRAW_LABEL_ALIGN_NORTH_WEST**
- **XnsIDRAW_LABEL_ALIGN_WEST**
- **XnsIDRAW_LABEL_ALIGN_SOUTH_WEST**
- **XnsIDRAW_LABEL_ALIGN_NORTH**
- **XnsIDRAW_LABEL_ALIGN_MIDDLE**
- **XnsIDRAW_LABEL_ALIGN_SOUTH**
- **XnsIDRAW_LABEL_ALIGN_NORTH_EAST**
- **XnsIDRAW_LABEL_ALIGN_EAST**
- **XnsIDRAW_LABEL_ALIGN_SOUTH_EAST**

XnsINdrawLabelAliasingColors

If the widget is scaled or rotated, this resource specifies the number of colors to use for the antialiasing mechanism.

XnsINdrawLabelFitPoints

If set to **True**, the size of the widget is adjusted to the size of the text bounding box, regardless of dimension specifications applied to the widget.

XnsINdrawLabelString

Specifies the text to display.

XnsINdrawLabelTransformString

Indicates whether text rotation and scaling are enabled. If it is set to **False**, the widget does not scale or rotate the label.

9.4.1 Inherited Resources

XnslDrawLabel inherits the following resources from the named superclass. For a complete description of each resource, refer to the man page for that superclass.

XnslDrawRect Resource Set

Name Class	Default Type	Access
XnslNdrawRadius XnslCdrawRadius	0.0 XtRFloat	CSG

XnslDrawObject Resource Set

Name Class	Default Type	Access
XnslNdrawActivateCallback XnslCdrawActivateCallback	NULL XtRCallback	C
XnslNdrawActivateCallbackList XnslCdrawActivateCallbackList	NULL XnslRdrawActivateCallbackList	CSG
XnslNdrawActivateCondition XnslCdrawActivateCondition	XnslDRAW_INSIDE XnslRdrawSelectCondition	CSG
XnslNdrawAddPoint XnslCdrawAddPoint	NULL XnslRdrawAddPoint	CSG
XnslNdrawAlignment XnslCdrawAlignment	XnslDRAW_MIDDLE XnslRdrawAlignment	CSG
XnslNdrawAngle XnslCdrawAngle	NULL XtRFloat	CSG
XnslNdrawAngleUnits XnslCdrawAngleUnits	XnslRADIANS XnslRdrawAngleUnits	CSG
XnslNdrawBackground XnslCdrawPixel	-1 XnslRdrawPixel	CSG
XnslNdrawBackgroundPixmap XnslCdrawBackgroundPixmap	-1 XnslRdrawImage	CSG

XnslDrawObject Resource Set

Name Class	Default Type	Access
XnslNdrawBlinkTime XnslCDrawBlinkTime	0 XtRInt	CSG
XnslNdrawBottomMargin XnslCDrawBottomMargin	5 XtRInt	CSG
XnslNdrawClipContents XnslCDrawClipContents	False XtRBoolean	CSG
XnslNdrawCloseMode XnslCDrawCloseMode	XnslDRAW_OPEN XnslRDrawCloseMode	CSG
XnslNdrawDashes XnslCDrawDashes	0 XnslRDrawDashes	CSG
XnslNdrawDefaultFontName XnslCDrawDefaultFontName	“Fixed*” String	CSG
XnslNdrawEventMask XnslCDrawEventMask	BUTTON1DOWN XnslRDrawEventMask	CSG
XnslNdrawFillDir XnslCDrawFillDir	XnslDRAW_FILL_BOTTOM XnslRDrawFillDir	CSG
XnslNdrawFillMode XnslCDrawFillMode	XnslDRAW_NON_FILLED XnslRDrawFillMode	CSG
XnslNdrawFont XnslCDrawFont	-1 XnslRDrawFont	CSG
XnslNdrawForeground XnslCDrawPixel	-1 XnslRDrawPixel	CSG
XnslNdrawFrameBackground XnslCDrawPixel	-1 XnslRDrawPixel	CSG
XnslNdrawFrameDashes XnslCDrawDashes	0 XnslRDrawDashes	CSG
XnslNdrawFrameForeground XnslCDrawPixel	-1 XnslRDrawPixel	CSG
XnslNdrawFrameFunction XnslCDrawFunction	GXcopy XnslRDrawFunction	CSG
XnslNdrawFrameLineWidth XnslCDrawLineWidth	0 XnslRInt	CSG
XnslNdrawFrameStipple XnslCDrawStipple	-1 XnslRDrawStipple	CSG

XnslDrawObject Resource Set

Name Class	Default Type	Access
XnslNdrawFrameXtSOrigin XnslCDrawXTSOrigin	0 XnslRInt	CSG
XnslNdrawFrameYtSOrigin XnslCDrawYTSOrigin	0 XnslRInt	CSG
XnslNdrawFunction XnslCDrawFunction	GXcopy XnslRDrawFunction	CSG
XnslNdrawGeoAttributes XnslCDrawGeoAttributes	NULL XnslRDrawGeoAttributes	G
XnslNdrawGlobalVariables XnslCdrawGlobalVariables	True Boolean	CSG
XnslNdrawGraphAttributes XnslCDrawGraphAttributes	NULL XnslRDrawGraphAttributes	G
XnslNdrawHeight XtCHeight	MAX_DIM XtRDimension	CSG
XnslNdrawInheritAttributes XnslCDrawInheritAttributes	XnslDRAW_INHERIT_ATTRIBUTES XnslRDrawInheritAttributes	CSG
XnslNdrawLeftMargin XnslCDrawLeftMargin	5 XtRInt	CSG
XnslNdrawLineWidth XnslCDrawLineWidth	0 XnslRInt	CSG
XnslNdrawModifyCallback XnslCDrawModifyCallback	NULL XtRCallback	C
XnslNdrawModifyVerifyCallback XnslCDrawModifyVerifyCallback	NULL XtRCallback	C
XnslNdrawName XnslCDrawName	NULL XtRString	CSG
XnslNdrawNthUserData XnslCDrawNthUserData	0 XnslRInt	CSG
XnslNdrawNumPoints XnslCDrawNumPoints	0 XnslRInt	CSG
XnslNdrawNumReplacePoints XnslCDrawNumReplacePoints	0 XnslRInt	CSG
XnslNdrawObjectTolerance XtCWidth	4 XtRInt	CSG

XnslDrawObject Resource Set

Name Class	Default Type	Access
XnslNdrawOpaque XnslCDrawOpaque	False XtRBoolean	CSG
XnslNdrawParent XnslCDrawParent	NULL XnslRObject	CSG
XnslNdrawPctFill XnslCDrawPctFill	0 XnslRInt	CSG
XnslNdrawPoints XnslCDrawPoints	NULL XnslRDrawPoints	CSG
XnslNdrawRecomputeSize XnslCDrawRecomputeSize	True XtRBoolean	CSG
XnslNdrawRegular XnslCDrawRegular	False XtRBoolean	CSG
XnslNdrawRemovePoint XnslCDrawRemovePoint	-1 XnslRInt	CSG
XnslNdrawReplacePoints XnslCDrawReplacePoints	NULL XnslRDrawReplacePoints	CSG
XnslNdrawResizeCallback XnslCDrawResizeCallback	NULL XtRCallback	C
XnslNdrawResizeLineWidth XnslCDrawResizeLineWidth	True XtRBoolean	CSG
XnslNdrawResizePolicy XnslCDrawResizePolicy	XnslDRAW_ALIGN XnslRDrawResizePolicy	CSG
XnslNdrawRightMargin XnslCDrawRightMargin	5 XtRInt	CSG
XnslNdrawScriptEnabled XnslCDrawScriptEnabled	True XtRBoolean	CSG
XnslNdrawSelectCondition XnslCDrawSelectCondition	XnslDRAW_ON_BORDER XnslRDrawSelectCondition	CSG
XnslNdrawSelectMode XnslCDrawSelectMode	XnslDRAW_TRACK_POINTS XnslRDrawSelectMode	CSG
XnslNdrawShowBorder XnslCDrawShowBorder	False XtRBoolean	CSG
XnslNdrawShowFrame XnslCDrawShowFrame	False XtRBoolean	CSG

XnslDrawObject Resource Set

Name Class	Default Type	Access
XnslNdrawStipple XnslCDrawStipple	-1 XnslRDrawStipple	CSG
XnslNdrawStrictFrame XnslCDrawStrictFrame	False XtRBoolean	CSG
XnslNdrawStringValue XnslCDrawStringValue	NULL XtRString	CSG
XnslNdrawTopMargin XnslCDrawTopMargin	5 XtRInt	CSG
XnslNdrawTrackMode XnslCDrawTrackMode	XnslDRAW_SELECT_POINTS XnslRDrawTrackMode	CSG
XnslNdrawUnits XnslCDrawUnits	XnslPIXELS XnslRDrawUnits	CSG
XnslNdrawUserData XnslCDrawUserData	NULL XnslRDrawUserData	CSG
XnslNdrawUseRgbCube XnslCDrawUseRgbCube	False XtRBoolean	CG
XnslNdrawWidth XtCWidth	MAX_DIM XtRDimension	CSG
XnslNdrawX XtCPosition	MAX_POS XtRPosition	CSG
XnslNdrawXCenter XnslCDrawXCenter	NULL XtRFloat	CSG
XnslNdrawXPostScale XnslCDrawXPostScale	NULL XtRFloat	CSG
XnslNdrawXPreScale XnslCDrawXPreScale	NULL XtRFloat	CSG
XnslNdrawXTranslation XnslCDrawXTranslation	NULL XtRFloat	CSG
XnslNdrawXtSOrigin XnslCDrawXTSOrigin	0 XnslRInt	CSG
XnslNdrawY XtCPosition	MAX_POS XtRPosition	CSG
XnslNdrawYCenter XnslCDrawYCenter	NULL XtRFloat	CSG

XnslDrawObject Resource Set

Name Class	Default Type	Access
XnslNdrawYPostScale XnslCDrawYPostScale	NULL XtRFloat	CSG
XnslNdrawYPreScale XnslCDrawYPreScale	NULL XtRFloat	CSG
XnslNdrawYTranslation XnslCDrawYTranslation	NULL XtRFloat	CSG
XnslNdrawYtSOrigin XnslCDrawYTSOrigin	0 XnslRInt	CSG

XmGadget Resource Set

Name Class	Default Type	Access
XmNbottomShadowColor XmCBottomShadowColor	dynamic Pixel	G
XmNhelpCallback XmCCallback	NULL XtCallbackList	C
XmNhighlightColor XmCHighlightColor	dynamic Pixel	G
XmNnavigationType XmCNavigationType	XmNONE XmNavigationType	CSG
XmNshadowThickness XmCShadowThickness	2 Dimension	CSG
XmNtopShadowColor XmCTopShadowColor	dynamic Pixel	G
XmNtraversalOn XmCTraversalOn	True Boolean	CSG
XmNunitType XmCUnitType	dynamic unsigned char	CSG
XmNuserData XmCUserData	NULL XtPointer	CSG

RectObj Resource Set

Name Class	Default Type	Access
XmNancestorSensitive XmCSensitive	dynamic Boolean	G
XmNborderWidth XmCBorderWidth	0 Dimension	CSG
XmNheight XmCHeight	dynamic Dimension	CSG
XmNsensitive XmCSensitive	True Boolean	CSG
XmNwidth XmCWidth	dynamic Dimension	CSG
XmNx XmCPosition	0 Position	CSG
XmNy XmCPosition	0 Position	CSG

Object Resource Set

Name Class	Default Type	Access
XmNdestroyCallback XmCCallback	NULL XtCallbackList	C

CHAPTER 10: The XnsIDrawLine Class

10.1 Synopsis

```
#include <Xnsl/DLine.h>
```

10.2 Classes

The **XnsIDrawLine** class inherits resources and behavior from the **Object**, **RectObj**, **XmGadget**, **XnsIDrawObject** and **XnsIDrawPoly** classes.

The class pointer is **xnsIDrawLineClass**.

The class name is **XnsIDrawLine**.

10.3 Description

Widgets instantiated from the **xnsIDrawLineClass** are defined by two points.

10.4 Resources

The **XnsIDrawLine** widget defines no new resources.

10.4.1 Inherited Resources

XnsIDrawLine inherits the following resources from the named superclass. For a complete description of each resource, refer to the man page for that superclass,

XnsIDrawPoly Resource Set

Name Class	Default Type	Access
XnsINdrawResizeArrows XnsICDrawResizeArrows	True XtRBoolean	CSG
XnsINdrawArrowDir XnsICDrawArrowDir	XnsIDRAW_NO_ARROW XnsIRDrawArrowDir	CSG
XnsINdrawArrowStyle XnsICDrawArrowStyle	XnsIDRAW_ARROW_FILLED XnsIRDrawArrowStyle	CSG
XnsINdrawArrowWidth XnsICDrawArrowWidth	5 XnsIRInt	CSG
XnsINdrawArrowHeight XnsICDrawArrowHeight	15 XnsIRInt	CSG

XnslDrawPoly Resource Set

Name Class	Default Type	Access
XnslNdrawFromWidget XnslCDrawFromWidget	NULL XnslRDrawLinkWidget	CSG
XnslNdrawXFrom XnslCDrawXFrom	NULL XtrFloat	CSG
XnslNdrawYFrom XnslCDrawYFrom	NULL XtrFloat	CSG
XnslNdrawUpdateFrom XnslCDrawUpdateFrom	XnslDRAW_AUTO_UPDATE XnslRDrawUpdateLinkPolicy	CSG
XnslNdrawToWidget XnslCDrawToWidget	NULL XnslRDrawLinkWidget	CSG
XnslNdrawXTo XnslCDrawXTo	NULL XtrFloat	CSG
XnslNdrawYTo XnslCDrawYTo	NULL XtrFloat	CSG
XnslNdrawUpdateTo XnslCDrawUpdateTo	XnslDRAW_AUTO_UPDATE XnslRDrawUpdateLinkPolicy	CSG
XnslNdrawUpdateLinkCallback XnslCDrawUpdateLinkCallback	NULL XtrCallback	C

XnslDrawObject Resource Set

Name Class	Default Type	Access
XnslNdrawActivateCallback XnslCDrawActivateCallback	NULL XtrCallback	C
XnslNdrawActivateCallbackList XnslCDrawActivateCallbackList	NULL XnslRDrawActivateCallbackList	CSG
XnslNdrawActivateCondition XnslCDrawActivateCondition	XnslDRAW_INSIDE XnslRDrawSelectCondition	CSG
XnslNdrawAddPoint XnslCDrawAddPoint	NULL XnslRDrawAddPoint	CSG
XnslNdrawAlignment XnslCDrawAlignment	XnslDRAW_MIDDLE XnslRDrawAlignment	CSG

XnsIDrawObject Resource Set

Name Class	Default Type	Access
XnsINdrawAngle XnsICDrawAngle	NULL XtRFloat	CSG
XnsINdrawAngleUnits XnsICDrawAngleUnits	XnsIRADIANS XnsIRDrawAngleUnits	CSG
XnsINdrawBackground XnsICDrawPixel	-1 XnsIRDrawPixel	CSG
XnsINdrawBackgroundPixmap XnsICDrawBackgroundPixmap	-1 XnsIRDrawImage	CSG
XnsINdrawBlinkTime XnsICDrawBlinkTime	0 XtRInt	CSG
XnsINdrawBottomMargin XnsICDrawBottomMargin	5 XtRInt	CSG
XnsINdrawClipContents XnsICDrawClipContents	False XtRBoolean	CSG
XnsINdrawCloseMode XnsICDrawCloseMode	XnsIDRAW_OPEN XnsIRDrawCloseMode	CSG
XnsINdrawDashes XnsICDrawDashes	0 XnsIRDrawDashes	CSG
XnsINdrawDefaultFontName XnsICdrawDefaultFontName	“Fixed*” String	CSG
XnsINdrawEventMask XnsICDrawEventMask	BUTTON1DOWN XnsIRDrawEventMask	CSG
XnsINdrawFillDir XnsICDrawFillDir	XnsIDRAW_FILL_BOTTOM XnsIRDrawFillDir	CSG
XnsINdrawFillMode XnsICDrawFillMode	XnsIDRAW_NON_FILLED XnsIRDrawFillMode	CSG
XnsINdrawFont XnsICDrawFont	-1 XnsIRDrawFont	CSG
XnsINdrawForeground XnsICDrawPixel	-1 XnsIRDrawPixel	CSG
XnsINdrawFrameBackground XnsICDrawPixel	-1 XnsIRDrawPixel	CSG
XnsINdrawFrameDashes XnsICDrawDashes	0 XnsIRDrawDashes	CSG

XnsIDrawObject Resource Set

Name Class	Default Type	Access
XnsINdrawFrameForeground XnsICDrawPixel	-1 XnsIRDrawPixel	CSG
XnsINdrawFrameFunction XnsICDrawFunction	GXcopy XnsIRDrawFunction	CSG
XnsINdrawFrameLineWidth XnsICDrawLineWidth	0 XnsIRInt	CSG
XnsINdrawFrameStipple XnsICDrawStipple	-1 XnsIRDrawStipple	CSG
XnsINdrawFrameXtSOrigin XnsICDrawXTSOrigin	0 XnsIRInt	CSG
XnsINdrawFrameYtSOrigin XnsICDrawYTSOrigin	0 XnsIRInt	CSG
XnsINdrawFunction XnsICDrawFunction	GXcopy XnsIRDrawFunction	CSG
XnsINdrawGeoAttributes XnsICDrawGeoAttributes	NULL XnsIRDrawGeo.Attributes	G
XnsINdrawGlobalVariables XnsICdrawGlobalVariables	True Boolean	CSG
XnsINdrawGraphAttributes XnsICDrawGraphAttributes	NULL XnsIRDrawGraphAttributes	G
XnsINdrawHeight XtCHeight	MAX_DIM XtRDimension	CSG
XnsINdrawInheritAttributes XnsICDrawInheritAttributes	XnsIDRAW_INHERIT_ATTRIBUTES XnsIRDrawInheritAttributes	CSG
XnsINdrawLeftMargin XnsICDrawLeftMargin	5 XtRInt	CSG
XnsINdrawLineWidth XnsICDrawLineWidth	0 XnsIRInt	CSG
XnsINdrawModifyCallback XnsICDrawModifyCallback	NULL XtRCallback	C
XnsINdrawModifyVerifyCallback XnsICDrawModifyVerifyCallback	NULL XtRCallback	C
XnsINdrawName XnsICDrawName	NULL XtRString	CSG

XnsIDrawObject Resource Set

Name Class	Default Type	Access
XnsINdrawNthUserData XnsICDrawNthUserData	0 XnsIRInt	CSG
XnsINdrawNumPoints XnsICDrawNumPoints	0 XnsIRInt	CSG
XnsINdrawNumReplacePoints XnsICDrawNumReplacePoints	0 XnsIRInt	CSG
XnsINdrawObjectTolerance XtCWidth	4 XtRInt	CSG
XnsINdrawOpaque XnsICDrawOpaque	False XtRBoolean	CSG
XnsINdrawParent XnsICDrawParent	NULL XnsIRObject	CSG
XnsINdrawPctFill XnsICDrawPctFill	0 XnsIRInt	CSG
XnsINdrawPoints XnsICDrawPoints	NULL XnsIRDrawPoints	CSG
XnsINdrawRecomputeSize XnsICDrawRecomputeSize	True XtRBoolean	CSG
XnsINdrawRegular XnsICDrawRegular	False XtRBoolean	CSG
XnsINdrawRemovePoint XnsICDrawRemovePoint	-1 XnsIRInt	CSG
XnsINdrawReplacePoints XnsICDrawReplacePoints	NULL XnsIRDrawReplacePoints	CSG
XnsINdrawResizeCallback XnsICDrawResizeCallback	NULL XtRCallback	C
XnsINdrawResizeLineWidth XnsICDrawResizeLineWidth	True XtRBoolean	CSG
XnsINdrawResizePolicy XnsICDrawResizePolicy	XnsIDRAW_ALIGN XnsIRDrawResizePolicy	CSG
XnsINdrawRightMargin XnsICDrawRightMargin	5 XtRInt	CSG
XnsINdrawScriptEnabled XnsICDrawScriptEnabled	True XtRBoolean	CSG

XnslDrawObject Resource Set

Name Class	Default Type	Access
XnslNdrawSelectCondition XnslCDrawSelectCondition	XnslDRAW_ON_BORDER XnslRDrawSelectCondition	CSG
XnslNdrawSelectMode XnslCDrawSelectMode	XnslDRAW_TRACK_POINTS XnslRDrawSelectMode	CSG
XnslNdrawShowBorder XnslCDrawShowBorder	False XtRBoolean	CSG
XnslNdrawShowFrame XnslCDrawShowFrame	False XtRBoolean	CSG
XnslNdrawStipple XnslCDrawStipple	-1 XnslRDrawStipple	CSG
XnslNdrawStrictFrame XnslCDrawStrictFrame	False XtRBoolean	CSG
XnslNdrawStringValue XnslCDrawStringValue	NULL XtRString	CSG
XnslNdrawTopMargin XnslCDrawTopMargin	5 XtRInt	CSG
XnslNdrawTrackMode XnslCDrawTrackMode	XnslDRAW_SELECT_POINTS XnslRDrawTrackMode	CSG
XnslNdrawUnits XnslCDrawUnits	XnslPIXELS XnslRDrawUnits	CSG
XnslNdrawUserData XnslCDrawUserData	NULL XnslRDrawUserData	CSG
XnslNdrawUseRgbCube XnslCDrawUseRgbCube	False XtRBoolean	CG
XnslNdrawWidth XtCWidth	MAX_DIM XtRDimension	CSG
XnslNdrawX XtCPosition	MAX_POS XtRPosition	CSG
XnslNdrawXCenter XnslCDrawXCenter	NULL XtRFloat	CSG
XnslNdrawXPostScale XnslCDrawXPostScale	NULL XtRFloat	CSG
XnslNdrawXPreScale XnslCDrawXPreScale	NULL XtRFloat	CSG

XnsIDrawObject Resource Set

Name Class	Default Type	Access
XnsINdrawXTranslation XnsICDrawXTranslation	NULL XtRFloat	CSG
XnsINdrawXtSOrigin XnsICDrawXTSOrigin	0 XnsIRInt	CSG
XnsINdrawY XtCPosition	MAX_POS XtRPosition	CSG
XnsINdrawYCenter XnsICDrawYCenter	NULL XtRFloat	CSG
XnsINdrawYPostScale XnsICDrawYPostScale	NULL XtRFloat	CSG
XnsINdrawYPreScale XnsICDrawYPreScale	NULL XtRFloat	CSG
XnsINdrawYTranslation XnsICDrawYTranslation	NULL XtRFloat	CSG
XnsINdrawYtSOrigin XnsICDrawYTSOrigin	0 XnsIRInt	CSG

XmGadget Resource Set

Name Class	Default Type	Access
XmNbottomShadowColor XmCBottomShadowColor	dynamic Pixel	G
XmNhelpCallback XmCCallback	NULL XtCallbackList	C
XmNhighlightColor XmCHighlightColor	dynamic Pixel	G
XmNnavigationType XmCNavigationType	XmNONE XmNavigationType	CSG
XmNshadowThickness XmCShadowThickness	2 Dimension	CSG
XmNtopShadowColor XmCTopShadowColor	dynamic Pixel	G

XmGadget Resource Set

Name Class	Default Type	Access
XmNtraversalOn XmCTraversalOn	True Boolean	CSG
XmNunitType XmCUnitType	dynamic unsigned char	CSG
XmNuserData XmCUserData	NULL XtPointer	CSG

RectObj Resource Set

Name Class	Default Type	Access
XmNancestorSensitive XmCSensitive	dynamic Boolean	G
XmNborderWidth XmCBorderWidth	0 Dimension	CSG
XmNheight XmCHeight	dynamic Dimension	CSG
XmNsensitive XmCSensitive	True Boolean	CSG
XmNwidth XmCWidth	dynamic Dimension	CSG
XmNx XmCPosition	0 Position	CSG
XmNy XmCPosition	0 Position	CSG

Object Resource Set

Name Class	Default Type	Access
XmNdestroyCallback XmCCallback	NULL XtCallbackList	C

CHAPTER 11: The XnsldrawObject Class

11.1 Synopsis

`#include <XnsldObject.h>`

11.2 Classes

The **XnsldrawObject** inherits resources and behavior from **Object**, **RectObj** and **XmGadget** classes.

The class pointer is **xnsldrawObjectClass**.

The class name is **XnsldrawObject**.

11.3 Description

XnsldrawObject is the superclass for the graphic widgets found in the **Draw** library. This class should not be instantiated directly; it defines the resources and behavior that are common to all the widgets in the graphic widget library: defining point coordinates, graphic, and geometric attributes.

Point coordinate definition consists of a list of points whose meaning is interpreted differently for each type of widget whose dimensions and/or shape they specify. These points are transformed by a geometric transformation matrix before the widget is drawn. The transforming matrix is generated using resources which specify a rotation angle, a scale factor, a translation, and a reference point (also called center).

The visual aspect of widgets is also specified by means of the graphics attributes.

Graphic widgets which are descendants of the **XnsldrawPoly** class (itself a subclass of **XnsldrawObjectClass**) can be used as links between two widgets. Resources let you specify a source and a destination widget for such links. Other resources and callbacks let you specify how the link should be placed when the source or destination widgets are moved. These links may be used within any type of composite widget and can be used to interconnect widgets of any type. When a graphic widget is used as a link, its geometric resources are disabled.

11.4 Resources

The **XnsIDrawObject** class defines the following resources:

XnsIDrawObject Resource Set

Name Class	Default Type	Access
XnsINdrawActivateCallback XnsICDrawActivateCallback	NULL XtRCallback	C
XnsINdrawActivateCallbackList XnsICDrawActivateCallbackList	NULL XnsIRDrawActivateCallbackList	CSG
XnsINdrawActivateCondition XnsICDrawActivateCondition	XnsIDRAW_INSIDE XnsIRDrawSelectCondition	CSG
XnsINdrawAddPoint XnsICDrawAddPoint	NULL XnsIRDrawAddPoint	CSG
XnsINdrawAlignment XnsICDrawAlignment	XnsIDRAW_MIDDLE XnsIRDrawAlignment	CSG
XnsINdrawAngle XnsICDrawAngle	0.0 XtRFloat	CSG
XnsINdrawAngleUnits XnsICDrawAngleUnits	XnsINO_ANGLE_UNITS XnsIRDrawAngleUnits	CSG
XnsINdrawBackground XnsICDrawPixel	-1 XnsIRDrawPixel	CSG
XnsINdrawBackgroundPixmap XnsICDrawBackgroundPixmap	-1 XnsIRDrawImage	CSG
XnsINdrawBlinkTime XnsICDrawBlinkTime	0 XtRInt	CSG
XnsINdrawBottomMargin XnsICDrawBottomMargin	5 XtRInt	CSG
XnsINdrawClipContents XnsICDrawClipContents	False XtRBoolean	CSG
XnsINdrawCloseMode XnsICDrawCloseMode	XnsIDRAW_OPEN XnsIRDrawCloseMode	CSG
XnsINdrawDashes XnsICDrawDashes	0 XnsIRDrawDashes	CSG
XnsINdrawDefaultFontName XnsICdrawDefaultFontName	“Fixed*” String	CSG

XnslDrawObject Resource Set

Name Class	Default Type	Access
XnslNdrawEventMask XnslCDrawEventMask	DEFAULT_EVENTS XnslRDrawEventMask	CSG
XnslNdrawFillDir XnslCDrawFillDir	XnslDRAW_FILL_BOTTOM XnslRDrawFillDir	CSG
XnslNdrawFillMode XnslCDrawFillMode	XnslDRAW_NON_FILLED XnslRDrawFillMode	CSG
XnslNdrawFont XnslCDrawFont	-1 XnslRDrawFont	CSG
XnslNdrawForeground XnslCDrawPixel	-1 XnslRDrawPixel	CSG
XnslNdrawFrameBackground XnslCDrawPixel	-1 XnslRDrawPixel	CSG
XnslNdrawFrameDashes XnslCDrawDashes	0 XnslRDrawDashes	CSG
XnslNdrawFrameForeground XnslCDrawPixel	-1 XnslRDrawPixel	CSG
XnslNdrawFrameFunction XnslCDrawFunction	GXcopy XnslRDrawFunction	CSG
XnslNdrawFrameLineWidth XnslCDrawLineWidth	0 XnslRInt	CSG
XnslNdrawFrameStipple XnslCDrawStipple	-1 XnslRDrawStipple	CSG
XnslNdrawFrameXtSOrigin XnslCDrawXTSOrigin	0 XnslRInt	CSG
XnslNdrawFrameYtSOrigin XnslCDrawYTSOrigin	0 XnslRInt	CSG
XnslNdrawFunction XnslCDrawFunction	GXcopy XnslRDrawFunction	CSG
XnslNdrawGeoAttributes XnslCDrawGeoAttributes	NULL XnslRDrawGeoAttributes	G
XnslNdrawGlobalVariables XnslCdrawGlobalVariables	False Boolean	CSG
XnslNdrawGraphAttributes XnslCDrawGraphAttributes	NULL XnslRDrawGraphAttributes	G

XnslDrawObject Resource Set

Name Class	Default Type	Access
XnslNdrawHeight XtCHeight	MAX_DIM XtRDimension	CSG
XnslNdrawInheritAttributes XnslCDrawInheritAttributes	XnslDRAW_INHERIT_ATTRIBUTES XnslRDrawInheritAttributes	CSG
XnslNdrawLeftMargin XnslCDrawLeftMargin	5 XtRInt	CSG
XnslNdrawLineWidth XnslCDrawLineWidth	0 XnslRInt	CSG
XnslNdrawModifyCallback XnslCDrawModifyCallback	NULL XtRCallback	C
XnslNdrawModifyVerifyCallback XnslCDrawModifyVerifyCallback	NULL XtRCallback	C
XnslNdrawName XnslCDrawName	NULL XtRString	CSG
XnslNdrawNthUserData XnslCDrawNthUserData	0 XnslRInt	CSG
XnslNdrawNumPoints XnslCDrawNumPoints	0 XnslRInt	CSG
XnslNdrawNumReplacePoints XnslCDrawNumReplacePoints	0 XnslRInt	CSG
XnslNdrawObjectTolerance XtCWidth	4 XtRInt	CSG
XnslNdrawOpaque XnslCDrawOpaque	False XtRBoolean	CSG
XnslNdrawParent XnslCDrawParent	NULL XnslRObject	CSG
XnslNdrawPctFill XnslCDrawPctFill	0 XnslRInt	CSG
XnslNdrawPoints XnslCDrawPoints	NULL XnslRDrawPoints	CSG
XnslNdrawRecomputeSize XnslCDrawRecomputeSize	True XtRBoolean	CSG
XnslNdrawRegular XnslCDrawRegular	False XtRBoolean	CSG

XnslDrawObject Resource Set

Name Class	Default Type	Access
XnslNdrawRemovePoint XnslCDrawRemovePoint	-1 XnslRInt	CSG
XnslNdrawReplacePoints XnslCDrawReplacePoints	NULL XnslRDrawReplacePoints	CSG
XnslNdrawResizeCallback XnslCDrawResizeCallback	NULL XtRCallback	C
XnslNdrawResizeLineWidth XnslCDrawResizeLineWidth	True XtRBoolean	CSG
XnslNdrawResizePolicy XnslCDrawResizePolicy	XnslDRAW_FULL XnslRDrawResizePolicy	CSG
XnslNdrawRightMargin XnslCDrawRightMargin	5 XtRInt	CSG
XnslNdrawScriptEnabled XnslCDrawScriptEnabled	True XtRBoolean	CSG
XnslNdrawSelectCondition XnslCDrawSelectCondition	XnslDRAW_ON_BORDER XnslRDrawSelectCondition	CSG
XnslNdrawSelectMode XnslCDrawSelectMode	XnslDRAW_SELECT_POINTS XnslRDrawSelectMode	CSG
XnslNdrawShowBorder XnslCDrawShowBorder	True XtRBoolean	CSG
XnslNdrawShowFrame XnslCDrawShowFrame	False XtRBoolean	CSG
XnslNdrawStipple XnslCDrawStipple	-1 XnslRDrawStipple	CSG
XnslNdrawStrictFrame XnslCDrawStrictFrame	False XtRBoolean	CSG
XnslNdrawStringValue XnslCDrawStringValue	NULL XtRString	CSG
XnslNdrawTopMargin XnslCDrawTopMargin	5 XtRInt	CSG
XnslNdrawTrackMode XnslCDrawTrackMode	XnslDRAW_TRACK_POINTS XnslRDrawTrackMode	CSG
XnslNdrawUnits XnslCDrawUnits	XnslNO_UNITS XnslRDrawUnits	CSG

XnslDrawObject Resource Set

Name Class	Default Type	Access
XnslNdrawUserData XnslCDrawUserData	NULL XnslRDrawUserData	CSG
XnslNdrawUseRgbCube XnslCDrawUseRgbCube	False XtRBoolean	CG
XnslNdrawWidth XtCWidth	MAX_DIM XtRDimension	CSG
XnslNdrawX XtCPosition	MAX_POS XtRPosition	CSG
XnslNdrawXCenter XnslCDrawXCenter	NULL XtRFloat	CSG
XnslNdrawXPostScale XnslCDrawXPostScale	1.0 XtRFloat	CSG
XnslNdrawXPreScale XnslCDrawXPreScale	1.0 XtRFloat	CSG
XnslNdrawXTranslation XnslCDrawXTranslation	0.0 XtRFloat	CSG
XnslNdrawXtSOrigin XnslCDrawXTSOrigin	0 XnslRInt	CSG
XnslNdrawY XtCPosition	MAX_POS XtRPosition	CSG
XnslNdrawYCenter XnslCDrawYCenter	NULL XtRFloat	CSG
XnslNdrawYPostScale XnslCDrawYPostScale	1.0 XtRFloat	CSG
XnslNdrawYPreScale XnslCDrawYPreScale	1.0 XtRFloat	CSG
XnslNdrawYTranslation XnslCDrawYTranslation	0.0 XtRFloat	CSG
XnslNdrawYtSOrigin XnslCDrawYTSOrigin	0 XnslRInt	CSG

XnsINdrawActivateCallback

This callback is invoked when the user generates an event that matches the **XnsINdrawEventMask** and the **XnsINdrawActivateCondition**.

XnsINdrawActivateCallbackList

This resource can be used only in conjunction with the **Face** interpreter. It is a pointer to a list of structures which specify the event type, the conditions and the script that will be interpreted when an event meeting the conditions occurs. User must provide a linked list of **XnsIDrawActivateCallbackList** structures to use this resource.

XnsINdrawActivateCondition

Specifies the condition to trigger the **XnsINdrawActivateCallback**. Valid values are the values possible for the **XnsINdrawSelectCondition**.

XnsINdrawAddPoint

Specifies a point and point number that the application wishes to add to the widget's point list. This resource is a pointer to an **XnsIDrawNumberedPoint** structure which contains an **XnsIDrawPoint** and a position at which the point must be inserted. Points are numbered from 0.

XnsINdrawAlignment

Controls the behavior of an widget when it is resized (i.e. when the **XmNwidth** and/or **XmNheight** resource of the widget's frame is modified) and when its **XnsINdrawResizePolicy** is set to **XnsINdrawAlign**. Acceptable values for this resource are:

- **XnsIDRAW_NORTH_WEST**
- **XnsIDRAW_WEST**
- **XnsIDRAW_SOUTH_WEST**
- **XnsIDRAW_SOUTH**
- **XnsIDRAW_SOUTH_EAST**
- **XnsIDRAW_EAST**
- **XnsIDRAW_NORTH_EAST**
- **XnsIDRAW_NORTH**
- **XnsIDRAW_MIDDLE**

XnsINdrawAngle

Pointer to a double which indicates the value in **XnsINdrawAngleUnits** of the

rotation to be applied to the widget.

XnsINdrawAngleUnits

Controls the interpretation of the **XnsINdrawAngle** resource. Possible units are:

- **XnsIRADIANS**
- **XnsIDEGREES**
- **XnsIREVOLUTIONS**

XnsINdrawBackground

Specifies the widget's background color. Its value is the index returned by **XnsIDrawGetColor**.

XnsINdrawBackgroundPixmap

Specifies the widget's background pixmap. This resource is an index returned by **XnsIDrawGetPixmap**.

XnsINdrawBlinkTime

This resource is used to blink a widget. The widget is redisplayed **XnsINdrawBlinkTime** milliseconds in normal state and **XnsINdrawBlinkTime** milliseconds with foreground and background inverted.

XnsINdrawBottomMargin

Specifies the bottom margin between the drawing and the widget frame. Effective if **XnsINdrawResizePolicy** is different from **XnsIDRAW_NONE**.

XnsINdrawClipContents

If set to **True**, The contents of the widget is clipped.

XnsINdrawCloseMode

Specifies the close mode of the widget. Acceptable values are:

- **XnsIDRAW_OPEN**: Do not close the widget.
- **XnsIDRAW_CENTER_CLOSED**: Close through the widget's center (Close the widget by joining its two end points to its center).
- **XnsIDRAW_LINE_CLOSED**: Close the widget by joining its two end points.
- **XnsIDRAW_OBJECT_CLOSED**: Close mode depends on the widget's type.

XnsINdrawDashes

Specifies the dashes that should be used to draw lines. The dashes are stored in a **String** containing alternately the number of pixels to be drawn with the foreground color and the number of pixels to be drawn with the background color. This string must be **NULL** terminated.

XnsINdrawDefaultFontName

Specifies a default font name. This resource was implemented because the default font given by an X server is server-dependent. Thus, if the user does not specify a font for text or labels, the size of the string might be different on different displays.

XnsINdrawEventMask

A list of event names indicating which events are dispatched to the widget. Valid event names are:

- **BUTTON1DOWN**
- **BUTTON2DOWN**
- **BUTTON3DOWN**
- **BUTTON1DOWN2**
- **BUTTON1DOWN3**
- **BUTTON2DOWN2**
- **BUTTON2DOWN3**
- **BUTTON3DOWN2**
- **BUTTON3DOWN3**
- **BUTTON1MOTION**
- **BUTTON2MOTION**
- **BUTTON3MOTION**
- **BUTTON1UP**
- **BUTTON2UP**
- **BUTTON3UP**
- **KEYPRESS**
- **KEYRELEASE**
- **MOUSEMOVED**

XnsINdrawFillDir

Specifies the side from which filling is started for partially filled widgets. Acceptable values are:

- **XnsIDRAW_FILL_BOTTOM**
- **XnsIDRAW_FILL_TOP**
- **XnsIDRAW_FILL_LEFT**
- **XnsIDRAW_FILL_RIGHT**

XnsINdrawFillMode

Specifies how the widget should be filled. Acceptable values are:

- **XnsIDRAW_NON_FILLED**: The widget is not filled.
- **XnsIDRAW_FILLED**: The widget is entirely filled.
- **XnsIDRAW_PCT_FILLED**: The widget is partially filled.

XnsINdrawFont

Specifies the font used by the widget. It is the value returned by **XnsIDrawGetFont**. This resource is used by **XnsIDrawLabel** and **XnsIDrawText** widgets.

XnsINdrawForeground

Specifies the widget's foreground color. Its value is the index returned by **XnsIDrawGetColor**

XnsINdrawFrameBackground

Specifies the background color of the widget's frame.

XnsINdrawFrameDashes

Specifies the dashes used to draw the frame. The format is the same as for the **XnsINdrawDashes** resource.

XnsINdrawFrameForeground

Specifies the foreground color of the frame.

XnsIDrawFrameFunction

Specifies the drawing function used to draw the frame (see **XnsIDrawFunction**).

XnsIDrawFrameLineWidth

Specifies the width of the line used to draw the widget's frame.

XnsIDrawFrameStipple

Specifies the stipple used to draw the frame.

XnsIDrawFrameXtSOrigin

Specifies the x stipple origin of the frame.

XnsIDrawFrameYtSOrigin

Specifies the y stipple origin of the frame.

XnsIDrawFunction

Specifies the widget's drawing mode. Possible values are:

- **XnsIDRAW_CLEAR**
- **XnsIDRAW_AND**
- **XnsIDRAW_AND_REVERSE**
- **XnsIDRAW_COPY**
- **XnsIDRAW_AND_INVERTED**
- **XnsIDRAW_NOOP**
- **XnsIDRAW_XOR**
- **XnsIDRAW_OR**
- **XnsIDRAW_NOR**
- **XnsIDRAW_EQUIV**
- **XnsIDRAW_INVERT**
- **XnsIDRAW_OR_REVERSE**
- **XnsIDRAW_COPY_INVERTED**
- **XnsIDRAW_OR_INVERTED**
- **XnsIDRAW_NAND**
- **XnsIDRAW_SET**

XnsIDrawGeoAttributes

Contains a copy of the geometric attributes of the widget.

XnsIDrawGlobalVariables

This resource controls if a global FACE variable is to be created for each DrawObject child. For backward compatibility, this resource default value is True. Applications should set this variable to False to increase widget creation speed.

XnsIDrawGraphicAttributes

Contains a copy of the graphic attributes of the widget.

XnsIDrawHeight

Specifies the height of the widget's frame.

XnsIDrawInheritAttributes

Specifies the behavior of the widget when placed in a group. Possible values are:

- **XnsIDRAW_COPY_ATTRIBUTES**: Graphic attributes are copied from the group.
- **XnsIDRAW_INHERIT_ATTRIBUTES**: Graphic attributes are taken from the group if they are not specified for the widget.
- **XnsIDRAW_DONT_INHERIT**: The widget do not look for graphic attributes in the group.

XnsIDrawLeftMargin

Specifies the left margin between the drawing and the widget frame. Effective if **XnsIDrawResizePolicy** is different from **XnsIDRAW_NONE**.

XnsIDrawLineWidth

Specifies the line width used to draw the widget.

XnsIDrawModifyCallback

This callback is triggered when either a geometric attribute or a graphic attribute has been modified.

XnsIDrawModifyVerifyCallback

This callback is triggered when a geometric attribute or a graphic attribute is about to be modified. The application can accept, reject the modification, or suggest a compromise for the modification.

XnsIDrawName

Specifies the name of the widget.

XnsINdrawNthUserData

Specifies the index of the current XnsINdrawUserData resource.

XnsINdrawNumPoints

Specifies the number of points which define the widget.

XnsINdrawNumReplacePoints

Specifies the number of replacement points specified in the **XnsINdrawReplacePoints** resource.

XnsINdrawObjectTolerance

Specifies the acceptable error margin in pixels during widget selection or activation. (See **XnsINdrawSelectCondition**)

XnsINdrawOpaque

When **True**, the background of the widget is filled with its background color.

XnsINdrawParent

This read only resource is a pointer to the group to which the widget belongs, or zero if the widget is not a member of a group.

XnsINdrawPctFill

Indicates the percentage of filling when the **XnsINdrawFillMode** is **XnsIDRAW_PCT_FILLED**.

XnsINdrawPoints

The set of points defining the widget: a pointer to an array of XnsIDrawPoints

XnsINdrawRecomputeSize

If **True**, the widget's frame is resized to accommodate the widget, when the widget is modified. If **False**, the frame size remains unchanged when the widget is modified.

XnsINdrawRegular

Not implemented.

XnsINdrawRemovePoint

Specifies the point number of the point that is to be removed from the widget's point list. Points are numbered starting from 0.

XnsINdrawReplacePoints

Pointer to an array of **XnsIDrawNumberedPoint** structures. Each structure contains an **XnsIDrawPoint** and the position at which the new point is to be placed. The number of items in the array is specified by the **XnsINdrawNum-ReplacePoints** resource.

XnsINdrawResizeCallback

Callback called when the widget is resized.

XnsINdrawResizeLineWidth

If **True** the line width of a widget is scaled when the widget is scaled.

XnsINdrawResizePolicy

Indicates the action taken when a widget is resized. Valid values are:

- **XnsIDRAW_ALIGN**: The drawing is aligned in the frame as specified by the **XnsINdrawAlignment** and margin resources.
- **XnsIDRAW_FULL**: The drawing is fully resized to fit in the frame.
- **XnsIDRAW_UNIFORM**: The drawing is resized to fit in the frame, but the aspect ratio is preserved.
- **XnsIDRAW_NONE**: The drawing is not affected.

XnsINdrawRightMargin

Specifies the right margin between the drawing and the widget frame. Effective if **XnsINdrawResizePolicy** is different from **XnsIDRAW_NONE**.

XnsINdrawScriptEnabled

If set to **False**, no script defined for the widget are executed.

XnsINdrawSelectCondition

Specifies the condition for the widget's selection. Acceptable values are:

- **XnsIDRAW_NOWHERE**: The widget is never selected.
- **XnsIDRAW_ON_BORDER**: The widget is selected when the user clicks on its border within the tolerance specified by **XnsINdrawObjectTolerance**.
- **XnsIDRAW_INSIDE**: The widget is selected when the user clicks inside the widget within the tolerance specified by **XnsINdrawObjectTolerance**.
- **XnsIDRAW_OUTSIDE**: The widget is selected when the user clicks outside the widget outside the tolerance specified by **XnsINdrawObjectTolerance**.

- **XnsIDRAW Everywhere**: The widget is selected when the user clicks, wherever the mouse is.

XnsINdrawSelectMode

Controls the selection mode of an widget:

- **XnsIDRAW_SELECT_POINTS**: Handles are shown for each point of the widget.
- **XnsIDRAW_SELECT_RECT**: Handles are shown at the corners of the frame.

XnsINdrawShowBorder

Indicates if the border of the widget is shown. For some widgets, such as line and ellipse, hiding the border is equivalent to hiding the draw line. For widgets such as label and image, the border is an enclosing rectangle inscribed in the frame. For a segment, the lines interconnecting the segments of a multi-segment widget belong to the border.

XnsINdrawShowFrame

If **True**, indicates that the widget's frame is to be shown.

XnsINdrawStipple

Indicates the stipple used to draw and/or fill the widget.

XnsINdrawStrictFrame

If set to **True** the frame of the widget is exactly computed to fit the widget. If set to **False**, the frame is approximated with lighter computation.

XnsINdrawStringValue

This read only resource is the string description of the widget.

XnsINdrawTopMargin

Specifies the top margin between the drawing and the widget frame. Effective if **XnsINdrawResizePolicy** is not **XnsIDRAW_NONE**.

XnsINdrawTrackMode

Defines the shape of the shadow that is drawn when the user moves a widget. This resource applies only if the widget is a child of an **XnsIDraw** widget. Acceptable values are:

- **XnsIDRAW_TRACK_POINTS**: The shadow is widget dependent.
- **XnsIDRAW_TRACK_RECT**: The shadow is the frame of the widget.

XnsINdrawUnits

This resource defines the units in which the coordinates of the widgets are given.

- **XnslPIXELS**
- **Xnsl100TH_POINTS**
- **Xnsl1000TH_INCHES**
- **Xnsl100TH_MILLIMETERS**
- **Xnsl100TH_FONT_UNITS**
- **Xnsl_POINTS**
- **Xnsl_INCHES**
- **Xnsl_MILLIMETERS**
- **XnslNO_UNITS** (take parent's units)

XnslNdrawUserData

Allows the application to attach any number of specific data to the object. Contains the value of the current user data as indexed by the **XnslNdrawNthUserData** resource. When the widget is saved, the contents of this resource is not saved. It is the application's responsibility to save the user data, if needed.

XnslNdrawUseRgbCube

This resource can only be set at creation time. If **True**, the widget tries to allocate an RGB cube in its colormap. This cube is used to find colors for Draw pix-maps.

It first tries to allocate a 6x6x6 cube. If it's not possible, it tries with a 5x5x5, a 4x4x4, or a 3x3x3 cube.

XnslNdrawWidth

Specifies the width of the frame; same as **XmNwidth**.

XnsINdrawX

Specifies the top left position (in x) of the widget; same as **XmNx**.

XnsINdrawXCenter

Specifies the center (in x) of the widget for geometric transformations.

XnsINdrawXPostScale

Post scaling factor (in x) for geometric transformations.

XnsINdrawXPreScale

Pre scaling factor (in x) for geometric transformations.

XnsINdrawXTranslation

Translation (in x) for geometric transformations.

XnsINdrawXtOrigin

Specifies the x origin of the stipple.

XnsINdrawY

Specifies the top left position (in y) of the widget; same as **XmNy**.

XnsINdrawYCenter

Specifies the center (in y) of the widget for geometric transformations.

XnsINdrawYPostScale

Post scaling factor (in y) for geometric transformations.

XnsINdrawYPreScale

Pre scaling factor (in y) for geometric transformations.

XnsINdrawYTranslation

Translation (in y) for geometric transformation.

XnsINdrawYtOrigin

Specifies the y origin of the stipple.

11.4.1 Inherited Resources

XnsIDrawObject inherits the following resources from the named superclass. For a complete description of each resource, refer to the man page for that superclass.

XmGadget Resource Set

Name Class	Default Type	Access
XmNbottomShadowColor XmCBottomShadowColor	dynamic Pixel	G
XmNhelpCallback XmCCallback	NULL XtCallbackList	C
XmNhighlightColor XmCHighlightColor	dynamic Pixel	G
XmNnavigationType XmCNavigationType	XmNONE XmNavigationType	CSG
XmNshadowThickness XmCShadowThickness	2 Dimension	CSG
XmNtopShadowColor XmCTopShadowColor	dynamic Pixel	G
XmNtraversalOn XmCTraversalOn	True Boolean	CSG
XmNunitType XmCUnitType	dynamic unsigned char	CSG
XmNuserData XmCUserData	NULL XtPointer	CSG

RectObj Resource Set

Name Class	Default Type	Access
XmNancestorSensitive XmCSensitive	dynamic Boolean	G
XmNborderWidth XmCBorderWidth	0 Dimension	CSG
XmNheight XmCHeight	dynamic Dimension	CSG
XmNsensitive XmCSensitive	True Boolean	CSG

RectObj Resource Set

Name Class	Default Type	Access
XmNwidth XmCWidth	dynamic Dimension	CSG
XmNx XmCPosition	0 Position	CSG
XmNy XmCPosition	0 Position	CSG

Object Resource Set

Name Class	Default Type	Access
XmNdestroyCallback XmCCallback	NULL XtCallbackList	C

11.5 Callback Information

11.5.1 XnsINdrawActivateCallback

A pointer to the following structure is passed to the **XnsINdrawActivateCallback**

```
typedef struct {
    int                reason;
    XEvent             *event;
    int                type;
    Widget             widget;
} XnsIDrawActivateCallbackStruct;
```

- reason Indicates why the callback was invoked. For this callback the reason is **XnsICR_DRAW_ACTIVATE**.
- event Pointer to the XEvent that triggered the callback.
- type The type of the event.
- widget The activated widget..

11.5.2 XnsINdrawModifyVerifyCallback

A pointer to the following structure is passed to the **XnsINdrawModifyVerifyCallback**.

```
typedef struct {
```

```

        int                    reason;
        XEvent                 *event;
        XnsldrawDoit           doit;
        Widget                 widget;
        int                    mask;
        int                    x, y;
        int                    handle;
} XnsldrawModifyVerifyCallbackStruct;

```

- *reason* Indicates why the callback was invoked. For this callback, the reason can be:
 - **XnsICR_DRAW_MOVE**: The widget asks to be moved.
 - **XnsICR_DRAW_MODIFY**: The widget asks to be modified because of user action on a selection handle.
- *event* Pointer to the XEvent that triggered the callback.
- *doit* If set to **XnsldrawYes**, the modification becomes effective. If set to **XnsldrawNo**, the modification is ignored. If set to **XnsldrawAlmost**, the modification is applied as specified by the fields *mask*, *x*, *y*, *width*, *height*, and *geometry*.
- *widget* Indicates the widget that will be modified.
- *mask* Indicates which fields contain valid information. The following values and their combination are acceptable:
 - XnsIX
 - XnsIY
- *x, y* New position suggested.
- *handle* Modification handle index.

11.5.3 Xnsln drawModifyCallback

A pointer to the following structure is passed to the **Xnsln drawModifyCallback**.

```

typedef struct {
        int                    reason;
        XEvent                 *event;
        Widget                 widget;
        int                    x, y;
} XnsldrawModifyCallbackStruct;

```

- *reason* Indicates why the callback was invoked. For this callback, the reason can be:
 - **XnsICR_DRAW_MOVE**: The widget has been moved.

- **XnsICR_DRAW_MODIFY**: The widget has been modified because of user action on a selection handle.
- *event* Pointer to the XEvent that triggered the callback.
- *widget* Indicates the widget that has been modified.
- *x, y* New position.

11.5.4 XnsINdrawResizeCallback

A pointer to the following structure is passed to the **XnsINdrawResizeCallback**

```
typedef struct {
    int                reason;
    XEvent             *event;
} XnsIDrawResizeCallbackStruct;
```

- *reason* Indicates why the callback was invoked. For this callback, the reason is **XnsICR_DRAW_RESIZE**;
- *event* **NULL** for this callback.

CHAPTER 12: The XnsIDrawPath Class

12.1 Synopsis

```
#include <Xnsl/DPath.h>
```

12.2 Classes

The **XnsIDrawPath** class inherits resources and behavior from the **Object**, **RectObj**, **XmGadget**, **XnsIDrawObject**, **XnsIDrawRect** and **XnsIDrawGroup** classes.

The class pointer is **xnsIDrawPathClass**.

The class name is **XnsIDrawPath**.

12.3 Description

The **XnsIDrawPath** is a subclass of **XnsIDrawGroup**. Only polygonal widgets can be members of a path. The objects are interconnected with lines to form a continuous curve.

12.4 Resources

The **XnsIDrawPath** widget defines the following new resources.

XnsIDrawPath Resource Set

Name Class	Default Type	Access
XnsINdrawGroupPassAttributes XnsICDrawGroupPassAttributes	False XtRBoolean	CSG

XnsINdrawGroupPassAttributes

If set to **True**, the graphic attributes are not used by the **XnsIDrawPath** but are available for children inheritance. If set to **False**, graphic attributes are used by the path.

12.4.1 Inherited Resources

XnsIDrawPath inherits the following resources from the named superclass. For a complete description of each resource, refer to the man page for that superclass.

XnslDrawGroup Resource Set

Name Class	Default Type	Access
XnslNdrawChildren XnslCDrawChildren	NULL XnslRDrawWidgetList	CSG
XnslNdrawNumChildren XnslCDrawNumChildren	0 XnslRCardinal	CSG
XnslNdrawGroupModifyPolicy XnslCDrawGroupModifyPolicy	XnslDRAW_GROUP_CENTER XnslRDrawGroupModifyPolicy	CSG
XnslNdrawGroupLeftMargin XnslCDrawGroupLeftMargin	5 XtRInt	CG
XnslNdrawGroupRightMargin XnslCDrawGroupRightMargin	5 XtRInt	CG
XnslNdrawGroupTopMargin XnslCDrawGroupTopMargin	5 XtRInt	CG
XnslNdrawGroupBottomMargin XnslCDrawGroupBottomMargin	5 XtRInt	CG
XnslNdrawGroupPassAttributes XnslCDrawGroupPassAttributes	True XtRBoolean	CSG

XnslDrawRect Resource Set

Name Class	Default Type	Access
XnslNdrawRadius XnslCDrawRadius	0.0 XtRFloat	CSG

XnslDrawObject Resource Set

Name Class	Default Type	Access
XnslNdrawActivateCallback XnslCDrawActivateCallback	NULL XtRCallback	C

XnsIDrawObject Resource Set

Name Class	Default Type	Access
XnsINdrawActivateCallbackList XnsICDrawActivateCallbackList	NULL XnsIRDrawActivateCallbackList	CSG
XnsINdrawActivateCondition XnsICDrawActivateCondition	XnsIDRAW_INSIDE XnsIRDrawSelectCondition	CSG
XnsINdrawAddPoint XnsICDrawAddPoint	NULL XnsIRDrawAddPoint	CSG
XnsINdrawAlignment XnsICDrawAlignment	XnsIDRAW_MIDDLE XnsIRDrawAlignment	CSG
XnsINdrawAngle XnsICDrawAngle	NULL XtRFloat	CSG
XnsINdrawAngleUnits XnsICDrawAngleUnits	XnsIRADIANS XnsIRDrawAngleUnits	CSG
XnsINdrawBackground XnsICDrawPixel	-1 XnsIRDrawPixel	CSG
XnsINdrawBackgroundPixmap XnsICDrawBackgroundPixmap	-1 XnsIRDrawImage	CSG
XnsINdrawBlinkTime XnsICDrawBlinkTime	0 XtRInt	CSG
XnsINdrawBottomMargin XnsICDrawBottomMargin	5 XtRInt	CSG
XnsINdrawClipContents XnsICDrawClipContents	False XtRBoolean	CSG
XnsINdrawCloseMode XnsICDrawCloseMode	XnsIDRAW_OPEN XnsIRDrawCloseMode	CSG
XnsINdrawDashes XnsICDrawDashes	0 XnsIRDrawDashes	CSG
XnsINdrawEventMask XnsICDrawEventMask	BUTTON1DOWN XnsIRDrawEventMask	CSG
XnsINdrawFillDir XnsICDrawFillDir	XnsIDRAW_FILL_BOTTOM XnsIRDrawFillDir	CSG
XnsINdrawFillMode XnsICDrawFillMode	XnsIDRAW_NON_FILLED XnsIRDrawFillMode	CSG
XnsINdrawFont XnsICDrawFont	-1 XnsIRDrawFont	CSG

XnsIDrawObject Resource Set

Name Class	Default Type	Access
XnsINdrawForeground XnsICDrawPixel	-1 XnsIRDrawPixel	CSG
XnsINdrawFrameBackground XnsICDrawPixel	-1 XnsIRDrawPixel	CSG
XnsINdrawFrameDashes XnsICDrawDashes	0 XnsIRDrawDashes	CSG
XnsINdrawDefaultFontName XnsICdrawDefaultFontName	“Fixed*” String	CSG
XnsINdrawFrameForeground XnsICDrawPixel	-1 XnsIRDrawPixel	CSG
XnsINdrawFrameFunction XnsICDrawFunction	GXcopy XnsIRDrawFunction	CSG
XnsINdrawFrameLineWidth XnsICDrawLineWidth	0 XnsIRInt	CSG
XnsINdrawFrameStipple XnsICDrawStipple	-1 XnsIRDrawStipple	CSG
XnsINdrawFrameXtSOrigin XnsICDrawXTSOrigin	0 XnsIRInt	CSG
XnsINdrawFrameYtSOrigin XnsICDrawYTSOrigin	0 XnsIRInt	CSG
XnsINdrawFunction XnsICDrawFunction	GXcopy XnsIRDrawFunction	CSG
XnsINdrawGeoAttributes XnsICDrawGeoAttributes	NULL XnsIRDrawGeoAttributes	G
XnsINdrawGlobalVariables XnsICdrawGlobalVariables	True Boolean	CSG
XnsINdrawGraphAttributes XnsICDrawGraphAttributes	NULL XnsIRDrawGraphAttributes	G
XnsINdrawHeight XtCHeight	MAX_DIM XtRDimension	CSG
XnsINdrawInheritAttributes XnsICDrawInheritAttributes	XnsIDRAW_INHERIT_ATTRIBUTES XnsIRDrawInheritAttributes	CSG
XnsINdrawLeftMargin XnsICDrawLeftMargin	5 XtRInt	CSG

XnslDrawObject Resource Set

Name Class	Default Type	Access
XnslNdrawLineWidth XnslCDrawLineWidth	0 XnslRInt	CSG
XnslNdrawModifyCallback XnslCDrawModifyCallback	NULL XtRCallback	C
XnslNdrawModifyVerifyCallback XnslCDrawModifyVerifyCallback	NULL XtRCallback	C
XnslNdrawName XnslCDrawName	NULL XtRString	CSG
XnslNdrawNthUserData XnslCDrawNthUserData	0 XnslRInt	CSG
XnslNdrawNumPoints XnslCDrawNumPoints	0 XnslRInt	CSG
XnslNdrawNumReplacePoints XnslCDrawNumReplacePoints	0 XnslRInt	CSG
XnslNdrawObjectTolerance XtCWidth	4 XtRInt	CSG
XnslNdrawOpaque XnslCDrawOpaque	False XtRBoolean	CSG
XnslNdrawParent XnslCDrawParent	NULL XnslRObject	CSG
XnslNdrawPctFill XnslCDrawPctFill	0 XnslRInt	CSG
XnslNdrawPoints XnslCDrawPoints	NULL XnslRDrawPoints	CSG
XnslNdrawRecomputeSize XnslCDrawRecomputeSize	True XtRBoolean	CSG
XnslNdrawRegular XnslCDrawRegular	False XtRBoolean	CSG
XnslNdrawRemovePoint XnslCDrawRemovePoint	-1 XnslRInt	CSG
XnslNdrawReplacePoints XnslCDrawReplacePoints	NULL XnslRDrawReplacePoints	CSG
XnslNdrawResizeCallback XnslCDrawResizeCallback	NULL XtRCallback	C

XnsIDrawObject Resource Set

Name Class	Default Type	Access
XnsINdrawResizeLineWidth XnsICDrawResizeLineWidth	True XtRBoolean	CSG
XnsINdrawResizePolicy XnsICDrawResizePolicy	XnsIDRAW_ALIGN XnsIRDrawResizePolicy	CSG
XnsINdrawRightMargin XnsICDrawRightMargin	5 XtRInt	CSG
XnsINdrawScriptEnabled XnsICDrawScriptEnabled	True XtRBoolean	CSG
XnsINdrawSelectCondition XnsICDrawSelectCondition	XnsIDRAW_ON_BORDER XnsIRDrawSelectCondition	CSG
XnsINdrawSelectMode XnsICDrawSelectMode	XnsIDRAW_TRACK_POINTS XnsIRDrawSelectMode	CSG
XnsINdrawShowBorder XnsICDrawShowBorder	False XtRBoolean	CSG
XnsINdrawShowFrame XnsICDrawShowFrame	False XtRBoolean	CSG
XnsINdrawStipple XnsICDrawStipple	-1 XnsIRDrawStipple	CSG
XnsINdrawStrictFrame XnsICDrawStrictFrame	False XtRBoolean	CSG
XnsINdrawStringValue XnsICDrawStringValue	NULL XtRString	CSG
XnsINdrawTopMargin XnsICDrawTopMargin	5 XtRInt	CSG
XnsINdrawTrackMode XnsICDrawTrackMode	XnsIDRAW_SELECT_POINTS XnsIRDrawTrackMode	CSG
XnsINdrawUnits XnsICDrawUnits	XnsIPIXELS XnsIRDrawUnits	CSG
XnsINdrawUserData XnsICDrawUserData	NULL XnsIRDrawUserData	CSG
XnsINdrawUseRgbCube XnsICDrawUseRgbCube	False XtRBoolean	CG
XnsINdrawWidth XtCWidth	MAX_DIM XtRDimension	CSG

XnsIDrawObject Resource Set

Name Class	Default Type	Access
XnsINdrawX XtCPosition	MAX_POS XtRPosition	CSG
XnsINdrawXCenter XnsICDrawXCenter	NULL XtRFloat	CSG
XnsINdrawXPostScale XnsICDrawXPostScale	NULL XtRFloat	CSG
XnsINdrawXPreScale XnsICDrawXPreScale	NULL XtRFloat	CSG
XnsINdrawXTranslation XnsICDrawXTranslation	NULL XtRFloat	CSG
XnsINdrawXtSOrigin XnsICDrawXTSOrigin	0 XnsIRInt	CSG
XnsINdrawY XtCPosition	MAX_POS XtRPosition	CSG
XnsINdrawYCenter XnsICDrawYCenter	NULL XtRFloat	CSG
XnsINdrawYPostScale XnsICDrawYPostScale	NULL XtRFloat	CSG
XnsINdrawYPreScale XnsICDrawYPreScale	NULL XtRFloat	CSG
XnsINdrawYTranslation XnsICDrawYTranslation	NULL XtRFloat	CSG
XnsINdrawYtSOrigin XnsICDrawYTSOrigin	0 XnsIRInt	CSG

XmGadget Resource Set

Name Class	Default Type	Access
XmNbottomShadowColor XmCBottomShadowColor	dynamic Pixel	G
XmNhelpCallback XmCCallback	NULL XtCallbackList	C

XmGadget Resource Set

Name Class	Default Type	Access
XmNhighlightColor XmCHighlightColor	dynamic Pixel	G
XmNnavigationType XmCNavigationType	XmNONE XmNavigationType	CSG
XmNshadowThickness XmCShadowThickness	2 Dimension	CSG
XmNtopShadowColor XmCTopShadowColor	dynamic Pixel	G
XmNtraversalOn XmCTraversalOn	True Boolean	CSG
XmNunitType XmCUnitType	dynamic unsigned char	CSG
XmNuserData XmCUserData	NULL XtPointer	CSG

RectObj Resource Set

Name Class	Default Type	Access
XmNancestorSensitive XmCSensitive	dynamic Boolean	G
XmNborderWidth XmCBorderWidth	0 Dimension	CSG
XmNheight XmCHeight	dynamic Dimension	CSG
XmNsensitive XmCSensitive	True Boolean	CSG
XmNwidth XmCWidth	dynamic Dimension	CSG
XmNx XmCPosition	0 Position	CSG
XmNy XmCPosition	0 Position	CSG

Object Resource Set

Name Class	Default Type	Access
XmNdestroyCallback XmCCallback	NULL XtCallbackList	C

CHAPTER 13: The XnsIDrawPoly Class

13.1 Synopsis

```
#include <XnsI/DPoly.h>
```

13.2 Classes

The **XnsIDrawPoly** class inherits resources and behavior from the **Object**, **RectObj**, **XmGadget** and **XnsIDrawObject** classes.

The class pointer is **xnsIDrawPolyClass**.

The class name is **XnsIDrawPoly**.

13.3 Description

The **XnsIDrawPoly** class defines resources and functions for polygonal widgets. These objects are defined by an arbitrary number of points. The first and the last segments of **XnsIDrawPoly** objects can be terminated with arrows.

The **XnsIDrawPoly** implements the **Links** mechanism. Each widget built from the **XnsIDrawPoly** class is able to become a dynamic link between widgets. A link is defined by a source widget and a destination widget.

Within an **XnsIDraw** widget, links can be attached to widgets via **Slots** or with the widget center as reference. In other **XmManager** widgets, links can be attached to widgets only with the object center as reference.

Widgets of class **XnsIDrawPoly** should not be instantiated.

13.4 Resources

The **XnsIDrawPoly** widget defines the following resource set:

XnsIDrawPoly Resource Set

Name Class	Default Type	Access
XnsNdrawArrowDir XnsICDrawArrowDir	XnsIDRAW_NO_ARROW XnsIRDrawArrowDir	CSG
XnsNdrawArrowHeight XnsICDrawArrowHeight	15 XnsIRInt	CSG

XnsIDrawPoly Resource Set

Name Class	Default Type	Access
XnsINdrawArrowStyle XnsICDrawArrowStyle	XnsIDRAW_ARROW_FILLED XnsIRDrawArrowStyle	CSG
XnsINdrawArrowWidth XnsICDrawArrowWidth	5 XnsIRInt	CSG
XnsINdrawFromWidget XnsICDrawFromWidget	NULL XnsIRDrawLinkWidget	CSG
XnsINdrawResizeArrows XnsICDrawResizeArrows	True XtRBoolean	CSG
XnsINdrawToWidget XnsICDrawToWidget	NULL XnsIRDrawLinkWidget	CSG
XnsINdrawUpdateFrom XnsICDrawUpdateFrom	XnsIDRAW_AUTO_UPDATE XnsIRDrawUpdateLinkPolicy	CSG
XnsINdrawUpdateLinkCallback XnsICDrawUpdateLinkCallback	NULL XtRCallback	C
XnsINdrawUpdateTo XnsICDrawUpdateTo	XnsIDRAW_AUTO_UPDATE XnsIRDrawUpdateLinkPolicy	CSG
XnsINdrawXFrom XnsICDrawXFrom	NULL XtRFloat	CSG
XnsINdrawXTo XnsICDrawXTo	NULL XtRFloat	CSG
XnsINdrawYFrom XnsICDrawYFrom	NULL XtRFloat	CSG
XnsINdrawYTo XnsICDrawYTo	NULL XtRFloat	CSG

XnsINdrawArrowDir

Specifies the position of the arrow. All arrows lead away from the object. Valid values are:

- **XnsIDRAW_NO_ARROW**: No arrows are defined.
- **XnsIDRAW_FROM_ARROW**: An arrow is drawn on the last point.
- **XnsIDRAW_TO_ARROW**: An arrow is drawn on the first point.
- **XnsIDRAW_FROM_TO_ARROW**: An arrow is drawn on the last point and on the first point.

XnslNdrawArrowHeight

Specifies the height of the arrow.

XnslNdrawArrowStyle

Specifies the drawing style of the arrow. Valid values are:

- **XnslDRAW_ARROW_FILLED**: Arrows are drawn as a filled triangle.
- **XnslDRAW_ARROW_OPEN**: Arrows are drawn as a V.
- **XnslDRAW_ARROW_CLOSED**: Arrows are drawn as a non filled triangle.

XnslNdrawArrowWidth

Specifies the width of the arrow.

XnslNdrawFromWidget

Specifies the source widget for a link behavior.

XnslNdrawResizeArrows

If **True**, the arrow is resized if the object is resized.

XnslNdrawToWidget

Specifies the destination widget for a link behavior.

XnslNdrawUpdateFrom

See **XnslNdrawUpdateTo**

XnslNdrawUpdateLinkCallback

Invoked when the link is to be updated and **XnslNdrawUpdateTo** or **XnslNdrawUpdateFrom** are set to **XnslDRAW_CALLBACK_UPDATE**.

XnslNdrawUpdateTo

Defines the update policy for an object used as a link. Links are updated when the size or the position of a source or destination widget changes. The link can be updated automatically, or under application control. Valid values are:

- **XnslDRAW_NO_UPDATE**: The link is not updated.
- **XnslDRAW_AUTO_UPDATE**: The link is automatically updated. The first point of the object is placed on the source point. The last point is placed on the destination point in accordance with the offset specifications.
- **XnslDRAW_CALLBACK_UPDATE**: The link is updated by the application. A callback is invoked and points must be provided by the application.

XnslNdrawXFrom

See **XnslNdrawYFrom**

XnslNdrawXTo

See **XnslNdrawYTo**

XnslNdrawYFrom

Specifies offsets for the link placement in relation to the **XnslNdrawFromWidget**. This offset is the distance from the attachment point which can be a slot (if the object is the child of an **XnslDraw** widget and if the **XnslNdrawFromSlot** is defined) or the center of the object.

XnslNdrawYTo

Specifies offsets for the link placement in relation to the **XnslNdrawToWidget**. This offset is the distance from the attachment point which can be a slot (if the object is child of an **XnslDraw** widget and if the **XnslNdrawToSlot** is defined) or the center of the object.

13.4.1 Inherited Resources

XnslDrawPoly inherits the following resources from the named superclass. For a complete description of each resource, refer to the man page for that superclass. .

XnslDrawObject Resource Set

Name Class	Default Type	Access
XnslNdrawActivateCallback XnslCDrawActivateCallback	NULL XtRCallback	C
XnslNdrawActivateCallbackList XnslCDrawActivateCallbackList	NULL XnslRDrawActivateCallbackList	CSG
XnslNdrawActivateCondition XnslCDrawActivateCondition	XnslDRAW_INSIDE XnslRDrawSelectCondition	CSG
XnslNdrawAddPoint XnslCDrawAddPoint	NULL XnslRDrawAddPoint	CSG
XnslNdrawAlignment XnslCDrawAlignment	XnslDRAW_MIDDLE XnslRDrawAlignment	CSG
XnslNdrawAngle XnslCDrawAngle	NULL XtRFloat	CSG
XnslNdrawAngleUnits XnslCDrawAngleUnits	XnslRADIANs XnslRDrawAngleUnits	CSG

XnsIDrawObject Resource Set

Name Class	Default Type	Access
XnsINdrawBackground XnsICDrawPixel	-1 XnsIRDrawPixel	CSG
XnsINdrawBackgroundPixmap XnsICDrawBackgroundPixmap	-1 XnsIRDrawImage	CSG
XnsINdrawBlinkTime XnsICDrawBlinkTime	0 XtRInt	CSG
XnsINdrawBottomMargin XnsICDrawBottomMargin	5 XtRInt	CSG
XnsINdrawClipContents XnsICDrawClipContents	False XtRBoolean	CSG
XnsINdrawCloseMode XnsICDrawCloseMode	XnsIDRAW_OPEN XnsIRDrawCloseMode	CSG
XnsINdrawDashes XnsICDrawDashes	0 XnsIRDrawDashes	CSG
XnsINdrawDefaultFontName XnsICdrawDefaultFontName	“Fixed*” String	CSG
XnsINdrawEventMask XnsICDrawEventMask	BUTTON1DOWN XnsIRDrawEventMask	CSG
XnsINdrawFillDir XnsICDrawFillDir	XnsIDRAW_FILL_BOTTOM XnsIRDrawFillDir	CSG
XnsINdrawFillMode XnsICDrawFillMode	XnsIDRAW_NON_FILLED XnsIRDrawFillMode	CSG
XnsINdrawFont XnsICDrawFont	-1 XnsIRDrawFont	CSG
XnsINdrawForeground XnsICDrawPixel	-1 XnsIRDrawPixel	CSG
XnsINdrawFrameBackground XnsICDrawPixel	-1 XnsIRDrawPixel	CSG
XnsINdrawFrameDashes XnsICDrawDashes	0 XnsIRDrawDashes	CSG
XnsINdrawFrameForeground XnsICDrawPixel	-1 XnsIRDrawPixel	CSG
XnsINdrawFrameFunction XnsICDrawFunction	GXcopy XnsIRDrawFunction	CSG

XnslDrawObject Resource Set

Name Class	Default Type	Access
XnslNdrawFrameLineWidth XnslCDrawLineWidth	0 XnslRInt	CSG
XnslNdrawFrameStipple XnslCDrawStipple	-1 XnslRDrawStipple	CSG
XnslNdrawFrameXtSOrigin XnslCDrawXTSOrigin	0 XnslRInt	CSG
XnslNdrawFrameYtSOrigin XnslCDrawYTSOrigin	0 XnslRInt	CSG
XnslNdrawFunction XnslCDrawFunction	GXcopy XnslRDrawFunction	CSG
XnslNdrawGeoAttributes XnslCDrawGeoAttributes	NULL XnslRDrawGeoAttributes	G
XnslNdrawGlobalVariables XnslCdrawGlobalVariables	True Boolean	CSG
XnslNdrawGraphAttributes XnslCDrawGraphAttributes	NULL XnslRDrawGraphAttributes	G
XnslNdrawHeight XtCHeight	MAX_DIM XtRDimension	CSG
XnslNdrawInheritAttributes XnslCDrawInheritAttributes	XnslDRAW_INHERIT_ATTRIBUTES XnslRDrawInheritAttributes	CSG
XnslNdrawLeftMargin XnslCDrawLeftMargin	5 XtRInt	CSG
XnslNdrawLineWidth XnslCDrawLineWidth	0 XnslRInt	CSG
XnslNdrawModifyCallback XnslCDrawModifyCallback	NULL XtRCallback	C
XnslNdrawModifyVerifyCallback XnslCDrawModifyVerifyCallback	NULL XtRCallback	C
XnslNdrawName XnslCDrawName	NULL XtRString	CSG
XnslNdrawNthUserData XnslCDrawNthUserData	0 XnslRInt	CSG
XnslNdrawNumPoints XnslCDrawNumPoints	0 XnslRInt	CSG

XnslDrawObject Resource Set

Name Class	Default Type	Access
XnslNdrawNumReplacePoints XnslCDrawNumReplacePoints	0 XnslRInt	CSG
XnslNdrawObjectTolerance XtCWidth	4 XtRInt	CSG
XnslNdrawOpaque XnslCDrawOpaque	False XtRBoolean	CSG
XnslNdrawParent XnslCDrawParent	NULL XnslRObject	CSG
XnslNdrawPctFill XnslCDrawPctFill	0 XnslRInt	CSG
XnslNdrawPoints XnslCDrawPoints	NULL XnslRDrawPoints	CSG
XnslNdrawRecomputeSize XnslCDrawRecomputeSize	True XtRBoolean	CSG
XnslNdrawRegular XnslCDrawRegular	False XtRBoolean	CSG
XnslNdrawRemovePoint XnslCDrawRemovePoint	-1 XnslRInt	CSG
XnslNdrawReplacePoints XnslCDrawReplacePoints	NULL XnslRDrawReplacePoints	CSG
XnslNdrawResizeCallback XnslCDrawResizeCallback	NULL XtRCallback	C
XnslNdrawResizeLineWidth XnslCDrawResizeLineWidth	True XtRBoolean	CSG
XnslNdrawResizePolicy XnslCDrawResizePolicy	XnslDRAW_ALIGN XnslRDrawResizePolicy	CSG
XnslNdrawRightMargin XnslCDrawRightMargin	5 XtRInt	CSG
XnslNdrawScriptEnabled XnslCDrawScriptEnabled	True XtRBoolean	CSG
XnslNdrawSelectCondition XnslCDrawSelectCondition	XnslDRAW_ON_BORDER XnslRDrawSelectCondition	CSG
XnslNdrawSelectMode XnslCDrawSelectMode	XnslDRAW_TRACK_POINTS XnslRDrawSelectMode	CSG

XnslDrawObject Resource Set

Name Class	Default Type	Access
XnslNdrawShowBorder XnslCDrawShowBorder	False XtRBoolean	CSG
XnslNdrawShowFrame XnslCDrawShowFrame	False XtRBoolean	CSG
XnslNdrawStipple XnslCDrawStipple	-1 XnslRDrawStipple	CSG
XnslNdrawStrictFrame XnslCDrawStrictFrame	False XtRBoolean	CSG
XnslNdrawStringValue XnslCDrawStringValue	NULL XtRString	CSG
XnslNdrawTopMargin XnslCDrawTopMargin	5 XtRInt	CSG
XnslNdrawTrackMode XnslCDrawTrackMode	XnslDRAW_SELECT_POINTS XnslRDrawTrackMode	CSG
XnslNdrawUnits XnslCDrawUnits	XnslPIXELS XnslRDrawUnits	CSG
XnslNdrawUserData XnslCDrawUserData	NULL XnslRDrawUserData	CSG
XnslNdrawUseRgbCube XnslCDrawUseRgbCube	False XtRBoolean	CG
XnslNdrawWidth XtCWidth	MAX_DIM XtRDimension	CSG
XnslNdrawX XtCPosition	MAX_POS XtRPosition	CSG
XnslNdrawXCenter XnslCDrawXCenter	NULL XtRFloat	CSG
XnslNdrawXPostScale XnslCDrawXPostScale	NULL XtRFloat	CSG
XnslNdrawXPreScale XnslCDrawXPreScale	NULL XtRFloat	CSG
XnslNdrawXTranslation XnslCDrawXTranslation	NULL XtRFloat	CSG
XnslNdrawXtSOrigin XnslCDrawXTSOrigin	0 XnslRInt	CSG

XnsIDrawObject Resource Set

Name Class	Default Type	Access
XnsINdrawY XtCPosition	MAX_POS XtRPosition	CSG
XnsINdrawYCenter XnsICDrawYCenter	NULL XtRFloat	CSG
XnsINdrawYPostScale XnsICDrawYPostScale	NULL XtRFloat	CSG
XnsINdrawYPreScale XnsICDrawYPreScale	NULL XtRFloat	CSG
XnsINdrawYTranslation XnsICDrawYTranslation	NULL XtRFloat	CSG
XnsINdrawYtSOrigin XnsICDrawYTSOrigin	0 XnsIRInt	CSG

XmGadget Resource Set

Name Class	Default Type	Access
XmNbottomShadowColor XmCBottomShadowColor	dynamic Pixel	G
XmNhelpCallback XmCCallback	NULL XtCallbackList	C
XmNhighlightColor XmCHighlightColor	dynamic Pixel	G
XmNnavigationType XmCNavigationType	XmNONE XmNavigationType	CSG
XmNshadowThickness XmCShadowThickness	2 Dimension	CSG
XmNtopShadowColor XmCTopShadowColor	dynamic Pixel	G
XmNtraversalOn XmCTraversalOn	True Boolean	CSG
XmNunitType XmCUnitType	dynamic unsigned char	CSG

XmGadget Resource Set

Name Class	Default Type	Access
XmNuserData XmCUserData	NULL XtPointer	CSG

RectObj Resource Set

Name Class	Default Type	Access
XmNancestorSensitive XmCSensitive	dynamic Boolean	G
XmNborderWidth XmCBorderWidth	0 Dimension	CSG
XmNheight XmCHeight	dynamic Dimension	CSG
XmNsensitive XmCSensitive	True Boolean	CSG
XmNwidth XmCWidth	dynamic Dimension	CSG
XmNx XmCPosition	0 Position	CSG
XmNy XmCPosition	0 Position	CSG

Object Resource Set

Name Class	Default Type	Access
XmNdestroyCallback XmCCallback	NULL XtCallbackList	C

13.5 Callback Information

13.5.1 XnsIDrawUpdateLinkCallback

A pointer to the following structure is passed to the **XnsIDrawUpdateLinkCallback**.


```
typedef struct {
    int                reason;
    XEvent             *event;
    Widget             widget;
    Widget             from;
    Widget             to;
    int                to_ref_x;
    int                to_ref_y;
    int                from_ref_x;
    int                from_ref_y;
} XnsIDrawUpdateLinkCallbackStruct;
```

- *reason* Indicates why the callback was invoked. For this callback the reason is **XnsICR_DRAW_UPDATE_LINK**.
- *event* **NULL** for this callback.
- *widget* The link.
- *from* The source widget for the link.
- *to* The destination widget for the link.
- *to_ref_x* The x coordinate source reference point.
- *to_ref_y* The y coordinate source reference point.
- *from_ref_x* The x coordinate destination reference point.
- *from_ref_y* The y coordinate destination reference point.

CHAPTER 14: The XnsIDrawPolygon Class

14.1 Synopsis

#include <XnsI/DPolygon.h>

14.2 Classes

The **XnsIDrawPolygon** class inherits resources and behavior from the **Object**, **RectObj**, **XmGadget**, **XnsIDrawObject** and **XnsIDrawPoly** classes.

The class pointer is **xnsIDrawPolygonClass**.

The class name is **XnsIDrawPolygon**.

14.3 Description

The **XnsIDrawPolygon** class inherits its behavior from the **XnsIDrawPoly** class.

14.4 Resources

The **XnsIDrawPolygon** widget defines no new resources.

14.4.1 Inherited Resources

XnsIDrawPolygon inherits the following resources from the named superclass. For a complete description of each resource, refer to the man page for that superclass.

XnsIDrawPoly Resource Set

Name Class	Default Type	Access
XnsINdrawResizeArrows XnsICDrawResizeArrows	True XtRBoolean	CSG
XnsINdrawArrowDir XnsICDrawArrowDir	XnsIDRAW_NO_ARROW XnsIRDrawArrowDir	CSG
XnsINdrawArrowStyle XnsICDrawArrowStyle	XnsIDRAW_ARROW_FILLED XnsIRDrawArrowStyle	CSG
XnsINdrawArrowWidth XnsICDrawArrowWidth	5 XnsIRInt	CSG
XnsINdrawArrowHeight XnsICDrawArrowHeight	15 XnsIRInt	CSG

XnslDrawPoly Resource Set

Name Class	Default Type	Access
XnslNdrawFromWidget XnslCDrawFromWidget	NULL XnslRDrawLinkWidget	CSG
XnslNdrawXFrom XnslCDrawXFrom	NULL XtRFloat	CSG
XnslNdrawYFrom XnslCDrawYFrom	NULL XtRFloat	CSG
XnslNdrawUpdateFrom XnslCDrawUpdateFrom	XnslDRAW_AUTO_UPDATE XnslRDrawUpdateLinkPolicy	CSG
XnslNdrawToWidget XnslCDrawToWidget	NULL XnslRDrawLinkWidget	CSG
XnslNdrawXTo XnslCDrawXTo	NULL XtRFloat	CSG
XnslNdrawYTo XnslCDrawYTo	NULL XtRFloat	CSG
XnslNdrawUpdateTo XnslCDrawUpdateTo	XnslDRAW_AUTO_UPDATE XnslRDrawUpdateLinkPolicy	CSG
XnslNdrawUpdateLinkCallback XnslCDrawUpdateLinkCallback	NULL XtRCallback	C

XnslDrawObject Resource Set

Name Class	Default Type	Access
XnslNdrawActivateCallback XnslCDrawActivateCallback	NULL XtRCallback	C
XnslNdrawActivateCallbackList XnslCDrawActivateCallbackList	NULL XnslRDrawActivateCallbackList	CSG
XnslNdrawActivateCondition XnslCDrawActivateCondition	XnslDRAW_INSIDE XnslRDrawSelectCondition	CSG
XnslNdrawAddPoint XnslCDrawAddPoint	NULL XnslRDrawAddPoint	CSG
XnslNdrawAlignment XnslCDrawAlignment	XnslDRAW_MIDDLE XnslRDrawAlignment	CSG

XnsIDrawObject Resource Set

Name Class	Default Type	Access
XnsINdrawAngle XnsICDrawAngle	NULL XtRFloat	CSG
XnsINdrawAngleUnits XnsICDrawAngleUnits	XnsIRADIANS XnsIRDrawAngleUnits	CSG
XnsINdrawBackground XnsICDrawPixel	-1 XnsIRDrawPixel	CSG
XnsINdrawBackgroundPixmap XnsICDrawBackgroundPixmap	-1 XnsIRDrawImage	CSG
XnsINdrawBlinkTime XnsICDrawBlinkTime	0 XtRInt	CSG
XnsINdrawBottomMargin XnsICDrawBottomMargin	5 XtRInt	CSG
XnsINdrawClipContents XnsICDrawClipContents	False XtRBoolean	CSG
XnsINdrawCloseMode XnsICDrawCloseMode	XnsIDRAW_OPEN XnsIRDrawCloseMode	CSG
XnsINdrawDashes XnsICDrawDashes	0 XnsIRDrawDashes	CSG
XnsINdrawDefaultFontName XnsICdrawDefaultFontName	“Fixed*” String	CSG
XnsINdrawEventMask XnsICDrawEventMask	BUTTON1DOWN XnsIRDrawEventMask	CSG
XnsINdrawFillDir XnsICDrawFillDir	XnsIDRAW_FILL_BOTTOM XnsIRDrawFillDir	CSG
XnsINdrawFillMode XnsICDrawFillMode	XnsIDRAW_NON_FILLED XnsIRDrawFillMode	CSG
XnsINdrawFont XnsICDrawFont	-1 XnsIRDrawFont	CSG
XnsINdrawForeground XnsICDrawPixel	-1 XnsIRDrawPixel	CSG
XnsINdrawFrameBackground XnsICDrawPixel	-1 XnsIRDrawPixel	CSG
XnsINdrawFrameDashes XnsICDrawDashes	0 XnsIRDrawDashes	CSG

XnsIDrawObject Resource Set

Name Class	Default Type	Access
XnsINdrawFrameForeground XnsICDrawPixel	-1 XnsIRDrawPixel	CSG
XnsINdrawFrameFunction XnsICDrawFunction	GXcopy XnsIRDrawFunction	CSG
XnsINdrawFrameLineWidth XnsICDrawLineWidth	0 XnsIRInt	CSG
XnsINdrawFrameStipple XnsICDrawStipple	-1 XnsIRDrawStipple	CSG
XnsINdrawFrameXtSOrigin XnsICDrawXTSOrigin	0 XnsIRInt	CSG
XnsINdrawFrameYtSOrigin XnsICDrawYTSOrigin	0 XnsIRInt	CSG
XnsINdrawFunction XnsICDrawFunction	GXcopy XnsIRDrawFunction	CSG
XnsINdrawGeoAttributes XnsICDrawGeoAttributes	NULL XnsIRDrawGeo.Attributes	G
XnsINdrawGlobalVariables XnsICdrawGlobalVariables	True Boolean	CSG
XnsINdrawGraphAttributes XnsICDrawGraphAttributes	NULL XnsIRDrawGraphAttributes	G
XnsINdrawHeight XtCHeight	MAX_DIM XtRDimension	CSG
XnsINdrawInheritAttributes XnsICDrawInheritAttributes	XnsIDRAW_INHERIT_ATTRIBUTES XnsIRDrawInheritAttributes	CSG
XnsINdrawLeftMargin XnsICDrawLeftMargin	5 XtRInt	CSG
XnsINdrawLineWidth XnsICDrawLineWidth	0 XnsIRInt	CSG
XnsINdrawModifyCallback XnsICDrawModifyCallback	NULL XtRCallback	C
XnsINdrawModifyVerifyCallback XnsICDrawModifyVerifyCallback	NULL XtRCallback	C
XnsINdrawName XnsICDrawName	NULL XtRString	CSG

XnslDrawObject Resource Set

Name Class	Default Type	Access
XnslNdrawNthUserData XnslCDrawNthUserData	0 XnslRInt	CSG
XnslNdrawNumPoints XnslCDrawNumPoints	0 XnslRInt	CSG
XnslNdrawNumReplacePoints XnslCDrawNumReplacePoints	0 XnslRInt	CSG
XnslNdrawObjectTolerance XtCWidth	4 XtRInt	CSG
XnslNdrawOpaque XnslCDrawOpaque	False XtRBoolean	CSG
XnslNdrawParent XnslCDrawParent	NULL XnslRObject	CSG
XnslNdrawPctFill XnslCDrawPctFill	0 XnslRInt	CSG
XnslNdrawPoints XnslCDrawPoints	NULL XnslRDrawPoints	CSG
XnslNdrawRecomputeSize XnslCDrawRecomputeSize	True XtRBoolean	CSG
XnslNdrawRegular XnslCDrawRegular	False XtRBoolean	CSG
XnslNdrawRemovePoint XnslCDrawRemovePoint	-1 XnslRInt	CSG
XnslNdrawReplacePoints XnslCDrawReplacePoints	NULL XnslRDrawReplacePoints	CSG
XnslNdrawResizeCallback XnslCDrawResizeCallback	NULL XtRCallback	C
XnslNdrawResizeLineWidth XnslCDrawResizeLineWidth	True XtRBoolean	CSG
XnslNdrawResizePolicy XnslCDrawResizePolicy	XnslDRAW_ALIGN XnslRDrawResizePolicy	CSG
XnslNdrawRightMargin XnslCDrawRightMargin	5 XtRInt	CSG
XnslNdrawScriptEnabled XnslCDrawScriptEnabled	True XtRBoolean	CSG

XnslDrawObject Resource Set

Name Class	Default Type	Access
XnslNdrawSelectCondition XnslCDrawSelectCondition	XnslDRAW_ON_BORDER XnslRDrawSelectCondition	CSG
XnslNdrawSelectMode XnslCDrawSelectMode	XnslDRAW_TRACK_POINTS XnslRDrawSelectMode	CSG
XnslNdrawShowBorder XnslCDrawShowBorder	False XtRBoolean	CSG
XnslNdrawShowFrame XnslCDrawShowFrame	False XtRBoolean	CSG
XnslNdrawStipple XnslCDrawStipple	-1 XnslRDrawStipple	CSG
XnslNdrawStrictFrame XnslCDrawStrictFrame	False XtRBoolean	CSG
XnslNdrawStringValue XnslCDrawStringValue	NULL XtRString	CSG
XnslNdrawTopMargin XnslCDrawTopMargin	5 XtRInt	CSG
XnslNdrawTrackMode XnslCDrawTrackMode	XnslDRAW_SELECT_POINTS XnslRDrawTrackMode	CSG
XnslNdrawUnits XnslCDrawUnits	XnslPIXELS XnslRDrawUnits	CSG
XnslNdrawUserData XnslCDrawUserData	NULL XnslRDrawUserData	CSG
XnslNdrawUseRgbCube XnslCDrawUseRgbCube	False XtRBoolean	CG
XnslNdrawWidth XtCWidth	MAX_DIM XtRDimension	CSG
XnslNdrawX XtCPosition	MAX_POS XtRPosition	CSG
XnslNdrawXCenter XnslCDrawXCenter	NULL XtRFloat	CSG
XnslNdrawXPostScale XnslCDrawXPostScale	NULL XtRFloat	CSG
XnslNdrawXPreScale XnslCDrawXPreScale	NULL XtRFloat	CSG

XnslDrawObject Resource Set

Name Class	Default Type	Access
XnslNdrawXTranslation XnslCDrawXTranslation	NULL XtRFloat	CSG
XnslNdrawXtSOrigin XnslCDrawXTSOrigin	0 XnslRInt	CSG
XnslNdrawY XtCPosition	MAX_POS XtRPosition	CSG
XnslNdrawYCenter XnslCDrawYCenter	NULL XtRFloat	CSG
XnslNdrawYPostScale XnslCDrawYPostScale	NULL XtRFloat	CSG
XnslNdrawYPreScale XnslCDrawYPreScale	NULL XtRFloat	CSG
XnslNdrawYTranslation XnslCDrawYTranslation	NULL XtRFloat	CSG
XnslNdrawYtSOrigin XnslCDrawYTSOrigin	0 XnslRInt	CSG

RectObj Resource Set

Name Class	Default Type	Access
XmNancestorSensitive XmCSensitive	dynamic Boolean	G
XmNborderWidth XmCBorderWidth	0 Dimension	CSG
XmNheight XmCHeight	dynamic Dimension	CSG
XmNsensitive XmCSensitive	True Boolean	CSG
XmNwidth XmCWidth	dynamic Dimension	CSG
XmNx XmCPosition	0 Position	CSG
XmNy XmCPosition	0 Position	CSG

Object Resource Set

Name Class	Default Type	Access
XmNdestroyCallback XmCCallback	NULL XtCallbackList	C

CHAPTER 15: The XnsIDrawRect Class

15.1 Synopsis

```
#include <XnsI/DRect.h>
```

15.2 Classes

The **XnsIDrawRect** class inherits resources and behavior from the **Object**, **RectObj**, **XmGadget**, and **XnsIDrawObject** classes.

The class pointer is **xnsIDrawRectClass**.

The class name is **XnsIDrawRect**.

15.3 Description

The **XnsIDrawRectClass** is the base class for rectangular objects. Two points are needed to define a rectangle. The first point specifies the coordinates of the upper left corner, the second point specifies the width and the height of the object.

15.4 Resources

The **XnsIDrawRect** widget defines the following new resources.

XnsIDrawRect Resource Set

Name Class	Default Type	Access
XnsINdrawRadius XnsICDrawRadius	0.0 XtRFloat	CSG

XnsINdrawRadius

If not 0.0, specifies the radius of the quarter circle used to round the rectangle's corners. If set to 0.0, the corners of the rectangle are not rounded.

15.4.1 Inherited Resources

XnsIDrawRect inherits the following resources from the named superclass. For a complete description of each resource, refer to the man page for that superclass.

XnsIDrawObject Resource Set

Name Class	Default Type	Access
XnsINdrawActivateCallback XnsICDrawActivateCallback	NULL XtRCallback	C
XnsINdrawActivateCallbackList XnsICDrawActivateCallbackList	NULL XnsIRDrawActivateCallbackList	CSG
XnsINdrawActivateCondition XnsICDrawActivateCondition	XnsIDRAW_INSIDE XnsIRDrawSelectCondition	CSG
XnsINdrawAddPoint XnsICDrawAddPoint	NULL XnsIRDrawAddPoint	CSG
XnsINdrawAlignment XnsICDrawAlignment	XnsIDRAW_MIDDLE XnsIRDrawAlignment	CSG
XnsINdrawAngle XnsICDrawAngle	NULL XtRFloat	CSG
XnsINdrawAngleUnits XnsICDrawAngleUnits	XnsIRADIANS XnsIRDrawAngleUnits	CSG
XnsINdrawBackground XnsICDrawPixel	-1 XnsIRDrawPixel	CSG
XnsINdrawBackgroundPixmap XnsICDrawBackgroundPixmap	-1 XnsIRDrawImage	CSG
XnsINdrawBlinkTime XnsICDrawBlinkTime	0 XtRInt	CSG
XnsINdrawBottomMargin XnsICDrawBottomMargin	5 XtRInt	CSG
XnsINdrawClipContents XnsICDrawClipContents	False XtRBoolean	CSG
XnsINdrawCloseMode XnsICDrawCloseMode	XnsIDRAW_OPEN XnsIRDrawCloseMode	CSG
XnsINdrawDashes XnsICDrawDashes	0 XnsIRDrawDashes	CSG
XnsINdrawDefaultFontName XnsICdrawDefaultFontName	“Fixed*” String	CSG

XnslDrawObject Resource Set

Name Class	Default Type	Access
XnslNdrawEventMask XnslCDrawEventMask	BUTTON1DOWN XnslRDrawEventMask	CSG
XnslNdrawFillDir XnslCDrawFillDir	XnslDRAW_FILL_BOTTOM XnslRDrawFillDir	CSG
XnslNdrawFillMode XnslCDrawFillMode	XnslDRAW_NON_FILLED XnslRDrawFillMode	CSG
XnslNdrawFont XnslCDrawFont	-1 XnslRDrawFont	CSG
XnslNdrawForeground XnslCDrawPixel	-1 XnslRDrawPixel	CSG
XnslNdrawFrameBackground XnslCDrawPixel	-1 XnslRDrawPixel	CSG
XnslNdrawFrameDashes XnslCDrawDashes	0 XnslRDrawDashes	CSG
XnslNdrawFrameForeground XnslCDrawPixel	-1 XnslRDrawPixel	CSG
XnslNdrawFrameFunction XnslCDrawFunction	GXcopy XnslRDrawFunction	CSG
XnslNdrawFrameLineWidth XnslCDrawLineWidth	0 XnslRInt	CSG
XnslNdrawFrameStipple XnslCDrawStipple	-1 XnslRDrawStipple	CSG
XnslNdrawFrameXtSOrigin XnslCDrawXTSOrigin	0 XnslRInt	CSG
XnslNdrawFrameYtSOrigin XnslCDrawYTSOrigin	0 XnslRInt	CSG
XnslNdrawFunction XnslCDrawFunction	GXcopy XnslRDrawFunction	CSG
XnslNdrawGeoAttributes XnslCDrawGeoAttributes	NULL XnslRDrawGeoAttributes	G
XnslNdrawGlobalVariables XnslCdrawGlobalVariables	True Boolean	CSG
XnslNdrawGraphAttributes XnslCDrawGraphAttributes	NULL XnslRDrawGraphAttributes	G

XnslDrawObject Resource Set

Name Class	Default Type	Access
XnslNdrawHeight XtCHeight	MAX_DIM XtRDimension	CSG
XnslNdrawInheritAttributes XnslCDrawInheritAttributes	XnslDRAW_INHERIT_ATTRIBUTES XnslRDrawInheritAttributes	CSG
XnslNdrawLeftMargin XnslCDrawLeftMargin	5 XtRInt	CSG
XnslNdrawLineWidth XnslCDrawLineWidth	0 XnslRInt	CSG
XnslNdrawModifyCallback XnslCDrawModifyCallback	NULL XtRCallback	C
XnslNdrawModifyVerifyCallback XnslCDrawModifyVerifyCallback	NULL XtRCallback	C
XnslNdrawName XnslCDrawName	NULL XtRString	CSG
XnslNdrawNthUserData XnslCDrawNthUserData	0 XnslRInt	CSG
XnslNdrawNumPoints XnslCDrawNumPoints	0 XnslRInt	CSG
XnslNdrawNumReplacePoints XnslCDrawNumReplacePoints	0 XnslRInt	CSG
XnslNdrawObjectTolerance XtCWidth	4 XtRInt	CSG
XnslNdrawOpaque XnslCDrawOpaque	False XtRBoolean	CSG
XnslNdrawParent XnslCDrawParent	NULL XnslRObject	CSG
XnslNdrawPctFill XnslCDrawPctFill	0 XnslRInt	CSG
XnslNdrawPoints XnslCDrawPoints	NULL XnslRDrawPoints	CSG
XnslNdrawRecomputeSize XnslCDrawRecomputeSize	True XtRBoolean	CSG
XnslNdrawRegular XnslCDrawRegular	False XtRBoolean	CSG

XnslDrawObject Resource Set

Name Class	Default Type	Access
XnslNdrawRemovePoint XnslCDrawRemovePoint	-1 XnslRInt	CSG
XnslNdrawReplacePoints XnslCDrawReplacePoints	NULL XnslRDrawReplacePoints	CSG
XnslNdrawResizeCallback XnslCDrawResizeCallback	NULL XtRCallback	C
XnslNdrawResizeLineWidth XnslCDrawResizeLineWidth	True XtRBoolean	CSG
XnslNdrawResizePolicy XnslCDrawResizePolicy	XnslDRAW_ALIGN XnslRDrawResizePolicy	CSG
XnslNdrawRightMargin XnslCDrawRightMargin	5 XtRInt	CSG
XnslNdrawScriptEnabled XnslCDrawScriptEnabled	True XtRBoolean	CSG
XnslNdrawSelectCondition XnslCDrawSelectCondition	XnslDRAW_ON_BORDER XnslRDrawSelectCondition	CSG
XnslNdrawSelectMode XnslCDrawSelectMode	XnslDRAW_TRACK_POINTS XnslRDrawSelectMode	CSG
XnslNdrawShowBorder XnslCDrawShowBorder	False XtRBoolean	CSG
XnslNdrawShowFrame XnslCDrawShowFrame	False XtRBoolean	CSG
XnslNdrawStipple XnslCDrawStipple	-1 XnslRDrawStipple	CSG
XnslNdrawStrictFrame XnslCDrawStrictFrame	False XtRBoolean	CSG
XnslNdrawStringValue XnslCDrawStringValue	NULL XtRString	CSG
XnslNdrawTopMargin XnslCDrawTopMargin	5 XtRInt	CSG
XnslNdrawTrackMode XnslCDrawTrackMode	XnslDRAW_SELECT_POINTS XnslRDrawTrackMode	CSG
XnslNdrawUnits XnslCDrawUnits	XnslPIXELS XnslRDrawUnits	CSG

XnslDrawObject Resource Set

Name Class	Default Type	Access
XnslNdrawUserData XnslCDrawUserData	NULL XnslRDrawUserData	CSG
XnslNdrawUseRgbCube XnslCDrawUseRgbCube	False XtrBoolean	CG
XnslNdrawWidth XtCWidth	MAX_DIM XtrDimension	CSG
XnslNdrawX XtCPosition	MAX_POS XtrPosition	CSG
XnslNdrawXCenter XnslCDrawXCenter	NULL XtrFloat	CSG
XnslNdrawXPostScale XnslCDrawXPostScale	NULL XtrFloat	CSG
XnslNdrawXPreScale XnslCDrawXPreScale	NULL XtrFloat	CSG
XnslNdrawXTranslation XnslCDrawXTranslation	NULL XtrFloat	CSG
XnslNdrawXtSOrigin XnslCDrawXTSOrigin	0 XnslRInt	CSG
XnslNdrawY XtCPosition	MAX_POS XtrPosition	CSG
XnslNdrawYCenter XnslCDrawYCenter	NULL XtrFloat	CSG
XnslNdrawYPostScale XnslCDrawYPostScale	NULL XtrFloat	CSG
XnslNdrawYPreScale XnslCDrawYPreScale	NULL XtrFloat	CSG
XnslNdrawYTranslation XnslCDrawYTranslation	NULL XtrFloat	CSG
XnslNdrawYtSOrigin XnslCDrawYTSOrigin	0 XnslRInt	CSG

RectObj Resource Set

Name Class	Default Type	Access
XmNancestorSensitive XmCSensitive	dynamic Boolean	G
XmNborderWidth XmCBorderWidth	0 Dimension	CSG
XmNheight XmCHeight	dynamic Dimension	CSG
XmNsensitive XmCSensitive	True Boolean	CSG
XmNwidth XmCWidth	dynamic Dimension	CSG
XmNx XmCPosition	0 Position	CSG
XmNy XmCPosition	0 Position	CSG

Object Resource Set

Name Class	Default Type	Access
XmNdestroyCallback XmCCallback	NULL XtCallbackList	C

CHAPTER 16: The XnsIDrawRectangle Class

16.1 Synopsis

#include <XnsI/DRectangle.h>

16.2 Classes

The **XnsIDrawRectangle** class inherits resources and behavior from the **Object**, **RectObj**, **XmGadget**, **XnsIDrawObject** and **XnsIDrawRect** classes.

The class pointer is **xnsIDrawRectangleClass**.

The class name is **XnsIDrawRectangle**.

16.3 Description

Two points are needed to define a rectangle. The first point specifies the coordinates of the upper left corner, the second point specifies the width and the height of the object.

16.4 Resources

The **XnsIDrawRectangle** widget defines no new resources.

16.4.1 Inherited Resources

XnsIDrawRectangle inherits the following resources from the named superclass. For a complete description of each resource, refer to the man page for that superclass.

XnsIDrawRect Resource Set

Name Class	Default Type	Access
XnsINdrawRadius XnsICDrawRadius	0.0 XtRFloat	CSG

XnslDrawObject Resource Set

Name Class	Default Type	Access
XnslNdrawActivateCallback XnslCDrawActivateCallback	NULL XtRCallback	C
XnslNdrawActivateCallbackList XnslCDrawActivateCallbackList	NULL XnslRDrawActivateCallbackList	CSG
XnslNdrawActivateCondition XnslCDrawActivateCondition	XnslDRAW_INSIDE XnslRDrawSelectCondition	CSG
XnslNdrawAddPoint XnslCDrawAddPoint	NULL XnslRDrawAddPoint	CSG
XnslNdrawAlignment XnslCDrawAlignment	XnslDRAW_MIDDLE XnslRDrawAlignment	CSG
XnslNdrawAngle XnslCDrawAngle	NULL XtRFloat	CSG
XnslNdrawAngleUnits XnslCDrawAngleUnits	XnslRADIANS XnslRDrawAngleUnits	CSG
XnslNdrawBackground XnslCDrawPixel	-1 XnslRDrawPixel	CSG
XnslNdrawBackgroundPixmap XnslCDrawBackgroundPixmap	-1 XnslRDrawImage	CSG
XnslNdrawBlinkTime XnslCDrawBlinkTime	0 XtRInt	CSG
XnslNdrawBottomMargin XnslCDrawBottomMargin	5 XtRInt	CSG
XnslNdrawClipContents XnslCDrawClipContents	False XtRBoolean	CSG
XnslNdrawCloseMode XnslCDrawCloseMode	XnslDRAW_OPEN XnslRDrawCloseMode	CSG
XnslNdrawDashes XnslCDrawDashes	0 XnslRDrawDashes	CSG
XnslNdrawDefaultFontName XnslCDrawDefaultFontName	“Fixed*” String	CSG
XnslNdrawEventMask XnslCDrawEventMask	BUTTON1DOWN XnslRDrawEventMask	CSG

XnslDrawObject Resource Set

Name Class	Default Type	Access
XnslNdrawFillDir XnslCDrawFillDir	XnslDRAW_FILL_BOTTOM XnslRDrawFillDir	CSG
XnslNdrawFillMode XnslCDrawFillMode	XnslDRAW_NON_FILLED XnslRDrawFillMode	CSG
XnslNdrawFont XnslCDrawFont	-1 XnslRDrawFont	CSG
XnslNdrawForeground XnslCDrawPixel	-1 XnslRDrawPixel	CSG
XnslNdrawFrameBackground XnslCDrawPixel	-1 XnslRDrawPixel	CSG
XnslNdrawFrameDashes XnslCDrawDashes	0 XnslRDrawDashes	CSG
XnslNdrawFrameForeground XnslCDrawPixel	-1 XnslRDrawPixel	CSG
XnslNdrawFrameFunction XnslCDrawFunction	GXcopy XnslRDrawFunction	CSG
XnslNdrawFrameLineWidth XnslCDrawLineWidth	0 XnslRInt	CSG
XnslNdrawFrameStipple XnslCDrawStipple	-1 XnslRDrawStipple	CSG
XnslNdrawFrameXtSOrigin XnslCDrawXTSOrigin	0 XnslRInt	CSG
XnslNdrawFrameYtSOrigin XnslCDrawYTSOrigin	0 XnslRInt	CSG
XnslNdrawFunction XnslCDrawFunction	GXcopy XnslRDrawFunction	CSG
XnslNdrawGeoAttributes XnslCDrawGeoAttributes	NULL XnslRDrawGeoAttributes	G
XnslNdrawGlobalVariables XnslCdrawGlobalVariables	True Boolean	CSG
XnslNdrawGraphAttributes XnslCDrawGraphAttributes	NULL XnslRDrawGraphAttributes	G
XnslNdrawHeight XtCHeight	MAX_DIM XtRDimension	CSG

XnsIDrawObject Resource Set

Name Class	Default Type	Access
XnsINdrawInheritAttributes XnsICDrawInheritAttributes	XnsIDRAW_INHERIT_ATTRIBUTES XnsIRDrawInheritAttributes	CSG
XnsINdrawLeftMargin XnsICDrawLeftMargin	5 XtRInt	CSG
XnsINdrawLineWidth XnsICDrawLineWidth	0 XnsIRInt	CSG
XnsINdrawModifyCallback XnsICDrawModifyCallback	NULL XtRCallback	C
XnsINdrawModifyVerifyCallback XnsICDrawModifyVerifyCallback	NULL XtRCallback	C
XnsINdrawName XnsICDrawName	NULL XtRString	CSG
XnsINdrawNthUserData XnsICDrawNthUserData	0 XnsIRInt	CSG
XnsINdrawNumPoints XnsICDrawNumPoints	0 XnsIRInt	CSG
XnsINdrawNumReplacePoints XnsICDrawNumReplacePoints	0 XnsIRInt	CSG
XnsINdrawObjectTolerance XtCWidth	4 XtRInt	CSG
XnsINdrawOpaque XnsICDrawOpaque	False XtRBoolean	CSG
XnsINdrawParent XnsICDrawParent	NULL XnsIRObject	CSG
XnsINdrawPctFill XnsICDrawPctFill	0 XnsIRInt	CSG
XnsINdrawPoints XnsICDrawPoints	NULL XnsIRDrawPoints	CSG
XnsINdrawRecomputeSize XnsICDrawRecomputeSize	True XtRBoolean	CSG
XnsINdrawRegular XnsICDrawRegular	False XtRBoolean	CSG
XnsINdrawRemovePoint XnsICDrawRemovePoint	-1 XnsIRInt	CSG

XnsIDrawObject Resource Set

Name Class	Default Type	Access
XnsINdrawReplacePoints XnsICDrawReplacePoints	NULL XnsIRDrawReplacePoints	CSG
XnsINdrawResizeCallback XnsICDrawResizeCallback	NULL XtRCallback	C
XnsINdrawResizeLineWidth XnsICDrawResizeLineWidth	True XtRBoolean	CSG
XnsINdrawResizePolicy XnsICDrawResizePolicy	XnsIDRAW_ALIGN XnsIRDrawResizePolicy	CSG
XnsINdrawRightMargin XnsICDrawRightMargin	5 XtRInt	CSG
XnsINdrawScriptEnabled XnsICDrawScriptEnabled	True XtRBoolean	CSG
XnsINdrawSelectCondition XnsICDrawSelectCondition	XnsIDRAW_ON_BORDER XnsIRDrawSelectCondition	CSG
XnsINdrawSelectMode XnsICDrawSelectMode	XnsIDRAW_TRACK_POINTS XnsIRDrawSelectMode	CSG
XnsINdrawShowBorder XnsICDrawShowBorder	False XtRBoolean	CSG
XnsINdrawShowFrame XnsICDrawShowFrame	False XtRBoolean	CSG
XnsINdrawStipple XnsICDrawStipple	-1 XnsIRDrawStipple	CSG
XnsINdrawStrictFrame XnsICDrawStrictFrame	False XtRBoolean	CSG
XnsINdrawStringValue XnsICDrawStringValue	NULL XtRString	CSG
XnsINdrawTopMargin XnsICDrawTopMargin	5 XtRInt	CSG
XnsINdrawTrackMode XnsICDrawTrackMode	XnsIDRAW_SELECT_POINTS XnsIRDrawTrackMode	CSG
XnsINdrawUnits XnsICDrawUnits	XnsIPIXELS XnsIRDrawUnits	CSG
XnsINdrawUserData XnsICDrawUserData	NULL XnsIRDrawUserData	CSG

XnslDrawObject Resource Set

Name Class	Default Type	Access
XnslNdrawUseRgbCube XnslCDrawUseRgbCube	False XtRBoolean	CG
XnslNdrawWidth XtCWidth	MAX_DIM XtRDimension	CSG
XnslNdrawX XtCPosition	MAX_POS XtRPosition	CSG
XnslNdrawXCenter XnslCDrawXCenter	NULL XtRFloat	CSG
XnslNdrawXPostScale XnslCDrawXPostScale	NULL XtRFloat	CSG
XnslNdrawXPreScale XnslCDrawXPreScale	NULL XtRFloat	CSG
XnslNdrawXTranslation XnslCDrawXTranslation	NULL XtRFloat	CSG
XnslNdrawXtSOrigin XnslCDrawXTSOrigin	0 XnslRInt	CSG
XnslNdrawY XtCPosition	MAX_POS XtRPosition	CSG
XnslNdrawYCenter XnslCDrawYCenter	NULL XtRFloat	CSG
XnslNdrawYPostScale XnslCDrawYPostScale	NULL XtRFloat	CSG
XnslNdrawYPreScale XnslCDrawYPreScale	NULL XtRFloat	CSG
XnslNdrawYTranslation XnslCDrawYTranslation	NULL XtRFloat	CSG
XnslNdrawYtSOrigin XnslCDrawYTSOrigin	0 XnslRInt	CSG

RectObj Resource Set

Name Class	Default Type	Access
XmNancestorSensitive XmCSensitive	dynamic Boolean	G
XmNborderWidth XmCBorderWidth	0 Dimension	CSG
XmNheight XmCHeight	dynamic Dimension	CSG
XmNsensitive XmCSensitive	True Boolean	CSG
XmNwidth XmCWidth	dynamic Dimension	CSG
XmNx XmCPosition	0 Position	CSG
XmNy XmCPosition	0 Position	CSG

Object Resource Set

Name Class	Default Type	Access
XmNdestroyCallback XmCCallback	NULL XtCallbackList	C

CHAPTER 17: The XnsIDrawSegment Class

17.1 Synopsis

```
#include <Xnsl/DSegment.h>
```

17.2 Classes

The **XnsIDrawSegment** class inherits resources and behavior from the **Object**, **RectObj**, **XmGadget**, **XnsIDrawObject** and **XnsIDrawPoly** classes.

The class pointer is **xnsIDrawSegmentClass**.

The class name is **XnsIDrawSegment**.

17.3 Description

A **XnsIDrawSegment** is similar to a **XnsIDrawPolygon** except that the object is drawn as disjointed pairs.

17.4 Resources

The **XnsIDrawSegment** widget defines no new resources.

17.4.1 Inherited Resources

XnsIDrawSegment inherits the following resources from the named superclass. For a complete description of each resource, refer to the man page for that superclass.

XnsIDrawPoly Resource Set

Name Class	Default Type	Access
XnsINdrawResizeArrows XnsICDrawResizeArrows	True XtRBoolean	CSG
XnsINdrawArrowDir XnsICDrawArrowDir	XnsIDRAW_NO_ARROW XnsIRDrawArrowDir	CSG
XnsINdrawArrowStyle XnsICDrawArrowStyle	XnsIDRAW_ARROW_FILLED XnsIRDrawArrowStyle	CSG
XnsINdrawArrowWidth XnsICDrawArrowWidth	5 XnsIRInt	CSG

XnsIDrawPoly Resource Set

Name Class	Default Type	Access
XnsINdrawArrowHeight XnsICDrawArrowHeight	15 XnsIRInt	CSG
XnsINdrawFromWidget XnsICDrawFromWidget	NULL XnsIRDrawLinkWidget	CSG
XnsINdrawXFrom XnsICDrawXFrom	NULL XtRFloat	CSG
XnsINdrawYFrom XnsICDrawYFrom	NULL XtRFloat	CSG
XnsINdrawUpdateFrom XnsICDrawUpdateFrom	XnsIDRAW_AUTO_UPDATE XnsIRDrawUpdateLinkPolicy	CSG
XnsINdrawToWidget XnsICDrawToWidget	NULL XnsIRDrawLinkWidget	CSG
XnsINdrawXTo XnsICDrawXTo	NULL XtRFloat	CSG
XnsINdrawYTo XnsICDrawYTo	NULL XtRFloat	CSG
XnsINdrawUpdateTo XnsICDrawUpdateTo	XnsIDRAW_AUTO_UPDATE XnsIRDrawUpdateLinkPolicy	CSG
XnsINdrawUpdateLinkCallback XnsICDrawUpdateLinkCallback	NULL XtRCallback	C

XnsIDrawObject Resource Set

Name Class	Default Type	Access
XnsINdrawActivateCallback XnsICDrawActivateCallback	NULL XtRCallback	C
XnsINdrawActivateCallbackList XnsICDrawActivateCallbackList	NULL XnsIRDrawActivateCallbackList	CSG
XnsINdrawActivateCondition XnsICDrawActivateCondition	XnsIDRAW_INSIDE XnsIRDrawSelectCondition	CSG
XnsINdrawAddPoint XnsICDrawAddPoint	NULL XnsIRDrawAddPoint	CSG

XnsIDrawObject Resource Set

Name Class	Default Type	Access
XnsINdrawAlignment XnsICDrawAlignment	XnsIDRAW_MIDDLE XnsIRDrawAlignment	CSG
XnsINdrawAngle XnsICDrawAngle	NULL XtRFloat	CSG
XnsINdrawAngleUnits XnsICDrawAngleUnits	XnsIRADIANS XnsIRDrawAngleUnits	CSG
XnsINdrawBackground XnsICDrawPixel	-1 XnsIRDrawPixel	CSG
XnsINdrawBackgroundPixmap XnsICDrawBackgroundPixmap	-1 XnsIRDrawImage	CSG
XnsINdrawBlinkTime XnsICDrawBlinkTime	0 XtRInt	CSG
XnsINdrawBottomMargin XnsICDrawBottomMargin	5 XtRInt	CSG
XnsINdrawClipContents XnsICDrawClipContents	False XtRBoolean	CSG
XnsINdrawCloseMode XnsICDrawCloseMode	XnsIDRAW_OPEN XnsIRDrawCloseMode	CSG
XnsINdrawDashes XnsICDrawDashes	0 XnsIRDrawDashes	CSG
XnsINdrawDefaultFontName XnsICdrawDefaultFontName	“Fixed*” String	CSG
XnsINdrawEventMask XnsICDrawEventMask	BUTTON1DOWN XnsIRDrawEventMask	CSG
XnsINdrawFillDir XnsICDrawFillDir	XnsIDRAW_FILL_BOTTOM XnsIRDrawFillDir	CSG
XnsINdrawFillMode XnsICDrawFillMode	XnsIDRAW_NON_FILLED XnsIRDrawFillMode	CSG
XnsINdrawFont XnsICDrawFont	-1 XnsIRDrawFont	CSG
XnsINdrawForeground XnsICDrawPixel	-1 XnsIRDrawPixel	CSG
XnsINdrawFrameBackground XnsICDrawPixel	-1 XnsIRDrawPixel	CSG

XnsIDrawObject Resource Set

Name Class	Default Type	Access
XnsINdrawFrameDashes XnsICDrawDashes	0 XnsIRDrawDashes	CSG
XnsINdrawFrameForeground XnsICDrawPixel	-1 XnsIRDrawPixel	CSG
XnsINdrawFrameFunction XnsICDrawFunction	GXcopy XnsIRDrawFunction	CSG
XnsINdrawFrameLineWidth XnsICDrawLineWidth	0 XnsIRInt	CSG
XnsINdrawFrameStipple XnsICDrawStipple	-1 XnsIRDrawStipple	CSG
XnsINdrawFrameXtOrigin XnsICDrawXTSOrigin	0 XnsIRInt	CSG
XnsINdrawFrameYtOrigin XnsICDrawYTSOrigin	0 XnsIRInt	CSG
XnsINdrawFunction XnsICDrawFunction	GXcopy XnsIRDrawFunction	CSG
XnsINdrawGeoAttributes XnsICDrawGeoAttributes	NULL XnsIRDrawGeoAttributes	G
XnsINdrawGlobalVariables XnsICdrawGlobalVariables	True Boolean	CSG
XnsINdrawGraphAttributes XnsICDrawGraphAttributes	NULL XnsIRDrawGraphAttributes	G
XnsINdrawHeight XtCHeight	MAX_DIM XtRDimension	CSG
XnsINdrawInheritAttributes XnsICDrawInheritAttributes	XnsIDRAW_INHERIT_ATTRIBUTES XnsIRDrawInheritAttributes	CSG
XnsINdrawLeftMargin XnsICDrawLeftMargin	5 XtRInt	CSG
XnsINdrawLineWidth XnsICDrawLineWidth	0 XnsIRInt	CSG
XnsINdrawModifyCallback XnsICDrawModifyCallback	NULL XtRCallback	C
XnsINdrawModifyVerifyCallback XnsICDrawModifyVerifyCallback	NULL XtRCallback	C

XnslDrawObject Resource Set

Name Class	Default Type	Access
XnslNdrawName XnslCDrawName	NULL XtRString	CSG
XnslNdrawNthUserData XnslCDrawNthUserData	0 XnslRInt	CSG
XnslNdrawNumPoints XnslCDrawNumPoints	0 XnslRInt	CSG
XnslNdrawNumReplacePoints XnslCDrawNumReplacePoints	0 XnslRInt	CSG
XnslNdrawObjectTolerance XtCWidth	4 XtRInt	CSG
XnslNdrawOpaque XnslCDrawOpaque	False XtRBoolean	CSG
XnslNdrawParent XnslCDrawParent	NULL XnslRObject	CSG
XnslNdrawPctFill XnslCDrawPctFill	0 XnslRInt	CSG
XnslNdrawPoints XnslCDrawPoints	NULL XnslRDrawPoints	CSG
XnslNdrawRecomputeSize XnslCDrawRecomputeSize	True XtRBoolean	CSG
XnslNdrawRegular XnslCDrawRegular	False XtRBoolean	CSG
XnslNdrawRemovePoint XnslCDrawRemovePoint	-1 XnslRInt	CSG
XnslNdrawReplacePoints XnslCDrawReplacePoints	NULL XnslRDrawReplacePoints	CSG
XnslNdrawResizeCallback XnslCDrawResizeCallback	NULL XtRCallback	C
XnslNdrawResizeLineWidth XnslCDrawResizeLineWidth	True XtRBoolean	CSG
XnslNdrawResizePolicy XnslCDrawResizePolicy	XnslDRAW_ALIGN XnslRDrawResizePolicy	CSG
XnslNdrawRightMargin XnslCDrawRightMargin	5 XtRInt	CSG

XnsIDrawObject Resource Set

Name Class	Default Type	Access
XnsINdrawScriptEnabled XnsICDrawScriptEnabled	True XtRBoolean	CSG
XnsINdrawSelectCondition XnsICDrawSelectCondition	XnsIDRAW_ON_BORDER XnsIRDrawSelectCondition	CSG
XnsINdrawSelectMode XnsICDrawSelectMode	XnsIDRAW_TRACK_POINTS XnsIRDrawSelectMode	CSG
XnsINdrawShowBorder XnsICDrawShowBorder	False XtRBoolean	CSG
XnsINdrawShowFrame XnsICDrawShowFrame	False XtRBoolean	CSG
XnsINdrawStipple XnsICDrawStipple	-1 XnsIRDrawStipple	CSG
XnsINdrawStrictFrame XnsICDrawStrictFrame	False XtRBoolean	CSG
XnsINdrawStringValue XnsICDrawStringValue	NULL XtRString	CSG
XnsINdrawTopMargin XnsICDrawTopMargin	5 XtRInt	CSG
XnsINdrawTrackMode XnsICDrawTrackMode	XnsIDRAW_SELECT_POINTS XnsIRDrawTrackMode	CSG
XnsINdrawUnits XnsICDrawUnits	XnsIPIXELS XnsIRDrawUnits	CSG
XnsINdrawUserData XnsICDrawUserData	NULL XnsIRDrawUserData	CSG
XnsINdrawUseRgbCube XnsICDrawUseRgbCube	False XtRBoolean	CG
XnsINdrawWidth XtCWidth	MAX_DIM XtRDimension	CSG
XnsINdrawX XtCPosition	MAX_POS XtRPosition	CSG
XnsINdrawXCenter XnsICDrawXCenter	NULL XtRFloat	CSG
XnsINdrawXPostScale XnsICDrawXPostScale	NULL XtRFloat	CSG

XnslDrawObject Resource Set

Name Class	Default Type	Access
XnslNdrawXPreScale XnslCDrawXPreScale	NULL XtRFloat	CSG
XnslNdrawXTranslation XnslCDrawXTranslation	NULL XtRFloat	CSG
XnslNdrawXtSOrigin XnslCDrawXTSOrigin	0 XnslRInt	CSG
XnslNdrawY XtCPosition	MAX_POS XtRPosition	CSG
XnslNdrawYCenter XnslCDrawYCenter	NULL XtRFloat	CSG
XnslNdrawYPostScale XnslCDrawYPostScale	NULL XtRFloat	CSG
XnslNdrawYPreScale XnslCDrawYPreScale	NULL XtRFloat	CSG
XnslNdrawYTranslation XnslCDrawYTranslation	NULL XtRFloat	CSG
XnslNdrawYtSOrigin XnslCDrawYTSOrigin	0 XnslRInt	CSG

RectObj Resource Set

Name Class	Default Type	Access
XmNancestorSensitive XmCSensitive	dynamic Boolean	G
XmNborderWidth XmCBorderWidth	0 Dimension	CSG
XmNheight XmCHeight	dynamic Dimension	CSG
XmNsensitive XmCSensitive	True Boolean	CSG
XmNwidth XmCWidth	dynamic Dimension	CSG

RectObj Resource Set

Name Class	Default Type	Access
XmNx XmCPosition	0 Position	CSG
XmNy XmCPosition	0 Position	CSG

Object Resource Set

Name Class	Default Type	Access
XmNdestroyCallback XmCCallback	NULL XtCallbackList	C

CHAPTER 18: The XnsIDrawSpline Class

18.1 Synopsis

```
#include <XnsI/DSpline.h>
```

18.2 Classes

The **XnsIDrawSpline** class inherits resources and behavior from the **Object**, **RectObj**, **XmGadget**, **XnsIDrawObject** and **XnsIDrawPoly** classes.

The class pointer is **xnsIDrawSplineClass**.

The class name is **XnsIDrawSpline**.

18.3 Description

Spline objects are defined by sets of four points which are the control points of the spline. The curve passes through the first and last point of each set. The curve is computed using the **Casteljau** subdivision algorithm.

18.4 Resources

The **XnsIDrawSpline** widget defines the following resources.

XnsIDrawSpline Resource Set

Name Class	Default Type	Access
XnsINdrawSplineConstrainedHandles XnsICDrawSplineConstrainedHandles	False XtRBoolean	CSG
XnsINdrawSplineStrictRegion XnsICDrawSplineStrictRegion	True XtRBoolean	CSG

XnsINdrawSplineConstrainedHandles

If **True**, the control points on either side of a point are constrained to be on the same line. If **False** these points are not constrained.

XnsINdrawSplineStrictRegion

If **True**, the object frame is the smallest rectangle enclosing the curve. If **False**, the object frame is the smallest rectangle enclosing the control points.

18.4.1 Inherited Resources

XnsIDrawSpline inherits the following resources from the named superclass. For a complete description of each resource, refer to the man page for that superclass.

XnsIDrawPoly Resource Set

Name Class	Default Type	Access
XnsINdrawResizeArrows XnsICDrawResizeArrows	True XtRBoolean	CSG
XnsINdrawArrowDir XnsICDrawArrowDir	XnsIDRAW_NO_ARROW XnsIRDrawArrowDir	CSG
XnsINdrawArrowStyle XnsICDrawArrowStyle	XnsIDRAW_ARROW_FILLED XnsIRDrawArrowStyle	CSG
XnsINdrawArrowWidth XnsICDrawArrowWidth	5 XnsIRInt	CSG
XnsINdrawArrowHeight XnsICDrawArrowHeight	15 XnsIRInt	CSG
XnsINdrawFromWidget XnsICDrawFromWidget	NULL XnsIRDrawLinkWidget	CSG
XnsINdrawXFrom XnsICDrawXFrom	NULL XtRFloat	CSG
XnsINdrawYFrom XnsICDrawYFrom	NULL XtRFloat	CSG
XnsINdrawUpdateFrom XnsICDrawUpdateFrom	XnsIDRAW_AUTO_UPDATE XnsIRDrawUpdateLinkPolicy	CSG
XnsINdrawToWidget XnsICDrawToWidget	NULL XnsIRDrawLinkWidget	CSG
XnsINdrawXTo XnsICDrawXTo	NULL XtRFloat	CSG
XnsINdrawYTo XnsICDrawYTo	NULL XtRFloat	CSG
XnsINdrawUpdateTo XnsICDrawUpdateTo	XnsIDRAW_AUTO_UPDATE XnsIRDrawUpdateLinkPolicy	CSG
XnsINdrawUpdateLinkCallback XnsICDrawUpdateLinkCallback	NULL XtRCallback	C

XnslDrawObject Resource Set

Name Class	Default Type	Access
XnslNdrawActivateCallback XnslCDrawActivateCallback	NULL XtRCallback	C
XnslNdrawActivateCallbackList XnslCDrawActivateCallbackList	NULL XnslRDrawActivateCallbackList	CSG
XnslNdrawActivateCondition XnslCDrawActivateCondition	XnslDRAW_INSIDE XnslRDrawSelectCondition	CSG
XnslNdrawAddPoint XnslCDrawAddPoint	NULL XnslRDrawAddPoint	CSG
XnslNdrawAlignment XnslCDrawAlignment	XnslDRAW_MIDDLE XnslRDrawAlignment	CSG
XnslNdrawAngle XnslCDrawAngle	NULL XtRFloat	CSG
XnslNdrawAngleUnits XnslCDrawAngleUnits	XnslRADIANS XnslRDrawAngleUnits	CSG
XnslNdrawBackground XnslCDrawPixel	-1 XnslRDrawPixel	CSG
XnslNdrawBackgroundPixmap XnslCDrawBackgroundPixmap	-1 XnslRDrawImage	CSG
XnslNdrawBlinkTime XnslCDrawBlinkTime	0 XtRInt	CSG
XnslNdrawBottomMargin XnslCDrawBottomMargin	5 XtRInt	CSG
XnslNdrawClipContents XnslCDrawClipContents	False XtRBoolean	CSG
XnslNdrawCloseMode XnslCDrawCloseMode	XnslDRAW_OPEN XnslRDrawCloseMode	CSG
XnslNdrawDashes XnslCDrawDashes	0 XnslRDrawDashes	CSG
XnslNdrawDefaultFontName XnslCdrawDefaultFontName	“Fixed*” String	CSG
XnslNdrawEventMask XnslCDrawEventMask	BUTTON1DOWN XnslRDrawEventMask	CSG

XnsIDrawObject Resource Set

Name Class	Default Type	Access
XnsINdrawFillDir XnsICDrawFillDir	XnsIDRAW_FILL_BOTTOM XnsIRDrawFillDir	CSG
XnsINdrawFillMode XnsICDrawFillMode	XnsIDRAW_NON_FILLED XnsIRDrawFillMode	CSG
XnsINdrawFont XnsICDrawFont	-1 XnsIRDrawFont	CSG
XnsINdrawForeground XnsICDrawPixel	-1 XnsIRDrawPixel	CSG
XnsINdrawFrameBackground XnsICDrawPixel	-1 XnsIRDrawPixel	CSG
XnsINdrawFrameDashes XnsICDrawDashes	0 XnsIRDrawDashes	CSG
XnsINdrawFrameForeground XnsICDrawPixel	-1 XnsIRDrawPixel	CSG
XnsINdrawFrameFunction XnsICDrawFunction	GXcopy XnsIRDrawFunction	CSG
XnsINdrawFrameLineWidth XnsICDrawLineWidth	0 XnsIRInt	CSG
XnsINdrawFrameStipple XnsICDrawStipple	-1 XnsIRDrawStipple	CSG
XnsINdrawFrameXtSOrigin XnsICDrawXTSOrigin	0 XnsIRInt	CSG
XnsINdrawFrameYtSOrigin XnsICDrawYTSOrigin	0 XnsIRInt	CSG
XnsINdrawFunction XnsICDrawFunction	GXcopy XnsIRDrawFunction	CSG
XnsINdrawGeoAttributes XnsICDrawGeoAttributes	NULL XnsIRDrawGeoAttributes	G
XnsINdrawGlobalVariables XnsICdrawGlobalVariables	True Boolean	CSG
XnsINdrawGraphAttributes XnsICDrawGraphAttributes	NULL XnsIRDrawGraphAttributes	G
XnsINdrawHeight XtCHeight	MAX_DIM XtRDimension	CSG

XnsIDrawObject Resource Set

Name Class	Default Type	Access
XnsINdrawInheritAttributes XnsICDrawInheritAttributes	XnsIDRAW_INHERIT_ATTRIBUTES XnsIRDrawInheritAttributes	CSG
XnsINdrawLeftMargin XnsICDrawLeftMargin	5 XtRInt	CSG
XnsINdrawLineWidth XnsICDrawLineWidth	0 XnsIRInt	CSG
XnsINdrawModifyCallback XnsICDrawModifyCallback	NULL XtRCallback	C
XnsINdrawModifyVerifyCallback XnsICDrawModifyVerifyCallback	NULL XtRCallback	C
XnsINdrawName XnsICDrawName	NULL XtRString	CSG
XnsINdrawNthUserData XnsICDrawNthUserData	0 XnsIRInt	CSG
XnsINdrawNumPoints XnsICDrawNumPoints	0 XnsIRInt	CSG
XnsINdrawNumReplacePoints XnsICDrawNumReplacePoints	0 XnsIRInt	CSG
XnsINdrawObjectTolerance XtCWidth	4 XtRInt	CSG
XnsINdrawOpaque XnsICDrawOpaque	False XtRBoolean	CSG
XnsINdrawParent XnsICDrawParent	NULL XnsIRObject	CSG
XnsINdrawPctFill XnsICDrawPctFill	0 XnsIRInt	CSG
XnsINdrawPoints XnsICDrawPoints	NULL XnsIRDrawPoints	CSG
XnsINdrawRecomputeSize XnsICDrawRecomputeSize	True XtRBoolean	CSG
XnsINdrawRegular XnsICDrawRegular	False XtRBoolean	CSG
XnsINdrawRemovePoint XnsICDrawRemovePoint	-1 XnsIRInt	CSG

XnsIDrawObject Resource Set

Name Class	Default Type	Access
XnsINdrawReplacePoints XnsICDrawReplacePoints	NULL XnsIRDrawReplacePoints	CSG
XnsINdrawResizeCallback XnsICDrawResizeCallback	NULL XtRCallback	C
XnsINdrawResizeLineWidth XnsICDrawResizeLineWidth	True XtRBoolean	CSG
XnsINdrawResizePolicy XnsICDrawResizePolicy	XnsIDRAW_ALIGN XnsIRDrawResizePolicy	CSG
XnsINdrawRightMargin XnsICDrawRightMargin	5 XtRInt	CSG
XnsINdrawScriptEnabled XnsICDrawScriptEnabled	True XtRBoolean	CSG
XnsINdrawSelectCondition XnsICDrawSelectCondition	XnsIDRAW_ON_BORDER XnsIRDrawSelectCondition	CSG
XnsINdrawSelectMode XnsICDrawSelectMode	XnsIDRAW_TRACK_POINTS XnsIRDrawSelectMode	CSG
XnsINdrawShowBorder XnsICDrawShowBorder	False XtRBoolean	CSG
XnsINdrawShowFrame XnsICDrawShowFrame	False XtRBoolean	CSG
XnsINdrawStipple XnsICDrawStipple	-1 XnsIRDrawStipple	CSG
XnsINdrawStrictFrame XnsICDrawStrictFrame	False XtRBoolean	CSG
XnsINdrawStringValue XnsICDrawStringValue	NULL XtRString	CSG
XnsINdrawTopMargin XnsICDrawTopMargin	5 XtRInt	CSG
XnsINdrawTrackMode XnsICDrawTrackMode	XnsIDRAW_SELECT_POINTS XnsIRDrawTrackMode	CSG
XnsINdrawUnits XnsICDrawUnits	XnsIPIXELS XnsIRDrawUnits	CSG
XnsINdrawUserData XnsICDrawUserData	NULL XnsIRDrawUserData	CSG

XnslDrawObject Resource Set

Name Class	Default Type	Access
XnslNdrawUseRgbCube XnslCDrawUseRgbCube	False XtRBoolean	CG
XnslNdrawWidth XtCWidth	MAX_DIM XtRDimension	CSG
XnslNdrawX XtCPosition	MAX_POS XtRPosition	CSG
XnslNdrawXCenter XnslCDrawXCenter	NULL XtRFloat	CSG
XnslNdrawXPostScale XnslCDrawXPostScale	NULL XtRFloat	CSG
XnslNdrawXPreScale XnslCDrawXPreScale	NULL XtRFloat	CSG
XnslNdrawXTranslation XnslCDrawXTranslation	NULL XtRFloat	CSG
XnslNdrawXtSOrigin XnslCDrawXTSOrigin	0 XnslRInt	CSG
XnslNdrawY XtCPosition	MAX_POS XtRPosition	CSG
XnslNdrawYCenter XnslCDrawYCenter	NULL XtRFloat	CSG
XnslNdrawYPostScale XnslCDrawYPostScale	NULL XtRFloat	CSG
XnslNdrawYPreScale XnslCDrawYPreScale	NULL XtRFloat	CSG
XnslNdrawYTranslation XnslCDrawYTranslation	NULL XtRFloat	CSG
XnslNdrawYtSOrigin XnslCDrawYTSOrigin	0 XnslRInt	CSG

RectObj Resource Set

Name Class	Default Type	Access
XmNancestorSensitive XmCSensitive	dynamic Boolean	G
XmNborderWidth XmCBorderWidth	0 Dimension	CSG
XmNheight XmCHeight	dynamic Dimension	CSG
XmNsensitive XmCSensitive	True Boolean	CSG
XmNwidth XmCWidth	dynamic Dimension	CSG
XmNx XmCPosition	0 Position	CSG
XmNy XmCPosition	0 Position	CSG

Object Resource Set

Name Class	Default Type	Access
XmNdestroyCallback XmCCallback	NULL XtCallbackList	C

CHAPTER 19: The XnsIDrawText Class

19.1 Synopsis

`#include <XnsI/DText.h>`

19.2 Classes

The **XnsIDrawText** inherits resources and behavior from the **Object**, **RectObj**, **XmGadget**, **XnsIDrawObject** and **XnsIDrawRect** classes.

The class pointer is **xnsIDrawTextClass**.

The class name is **XnsIDrawText**.

19.3 Description

The **XnsIDrawText** widget allows text to be displayed and/or edited. The text can consist of several paragraphs. Each paragraph has its own justification mode, ruler (margins and tabulation positions) and line spacing. Paragraphs in the **XnsIDrawText** widget are separated by the `newline` character.

Different segments of the text can be displayed with specific foreground and background colors, and with specific fonts. Text segments can be underlined and/or overlined. The text can contain subscripts and superscripts.

This widget can also load and/or save ASCII text and files. It can read and save **DTIF** format files. The **DTIF**¹ format is a human readable format that describes the text and the various attributes used to display the text.

1. **Draw Text Interchange Format** (See “DTIF grammar” on page 230.)

19.4 Resources

The **XnsIDrawText** defines the following resources::

XnsIDrawText Resource Set

Name Class	Default Type	Access
XnsINdrawTextActivateCallback XnsICDrawTextActivateCallback	NULL XtRCallback	C
XnsINdrawTextAlignment XnsICDrawTextAlignment	XnsIDRAW_TEXT_ALIGN_LEFT XnsIRDrawTextAlignment	CSG
XnsINdrawTextAsciiFile XnsICDrawTextAsciiFile	NULL XnsINdrawTextAsciiFile	CS
XnsINdrawTextAsciiText XnsICDrawTextAsciiText	NULL XnsIRDrawTextAsciiText	CSG
XnsINdrawTextBegSelectionPosition XnsICDrawTextBegSelectionPosition	0 XnsIRDrawTextPosition	G
XnsINdrawTextDtiffFile XnsICDrawTextDtiffFile	NULL XnsIRDrawTextDtiffFile	CS
XnsINdrawTextDtiffText XnsICDrawTextDtiffText	NULL XnsIRDrawTextDtiffText	CSG
XnsINdrawTextEndSelectionPosition XnsICDrawTextEndSelectionPosition	0 XnsIRDrawTextPosition	G
XnsINdrawTextHyperCursor XtCCursor	“hand1” XtRCursor	CSG
XnsINdrawTextHyperMode XnsICDrawTextHyperMode	False XtRBoolean	CSG
XnsINdrawTextIsModified XnsICDrawTextIsModified	False XtRBoolean	CSG
XnsINdrawTextIsSelected XnsICDrawTextIsSelected	False XtRBoolean	G
XnsINdrawTextMoveCallback XnsICDrawTextMoveCallback	NULL XtRCallback	C
XnsINdrawTextParagraphSpacing XnsICDrawTextParagraphSpacing	-1 XtRInt	CSG
XnsINdrawTextReadOnly XnsICDrawTextReadOnly	False XtRBoolean	CSG

XnslDrawText Resource Set

Name Class	Default Type	Access
XnslNdrawTextResizePolicy XnslCDrawTextResizePolicy	XnslDRAW_TEXT_CHANGE_RULER XnslRDrawTextResizePolicy	CSG
XnslNdrawTextShowCursor XnslCDrawTextShowCursor	True XtRBoolean	CSG
XnslNdrawTextTransformText XnslCDrawTextTransformText	True XtRBoolean	CSG
XnslNdrawTextVerifyCallback XnslCDrawTextVerifyCallback	NULL XtRCallback	C
XnslNdrawTextVerticalMargin XnslCDrawTextVerticalMargin	5 XtRInt	CSG

The **XnslDrawText** also overrides the following resources:

Name Class	Default Type	Access
XnslNdrawEventMask XnslCDrawEventMask	BUTTON1DOWN BUTTON2DOWN BUTTON1UP BUTTON2UP KEYPRESS KEYRELEASE BUTTON1MOTION XnslRDrawEventMask	CSG
XnslNdrawSelectCondition XnslCDrawSelectCondition	XnslDRAW_ON_BORDER XnslRDrawSelectCondition	CSG
XnslNdrawShowBorder XnslCDrawShowBorder	True XtRBoolean	CSG

XnslNdrawTextActivateCallback

This callback is invoked when the **XnslNdrawTextHyperMode** resource is set to True and when the user clicks the left mouse button on a text segment that lies between two text markers, with attributes **XnslDrawTextBEG_HYPERTEXT** (to the left of the text) and **XnslDrawTextEND_HYPERTEXT** (to the right). The user is informed that the mouse pointer is over such text segments by the changing cursor; the cursor becomes the one specified by the **XnslNdrawText-HyperCursor** resource. The reason for this callback is **XnslCR_DRAW_TEXT_ACTIVATE**.

XnslNdrawTextAlignment

Specifies the current value for text alignment. Any new paragraph created is aligned in conformance with the setting of this resource. Acceptable values are:

- **XnsIDRAW_TEXT_ALIGN_LEFT** flush new paragraphs left.
- **XnsIDRAW_TEXT_ALIGN_RIGHT** flush new paragraphs right.
- **XnsIDRAW_TEXT_ALIGN_CENTER** center new paragraphs .
- **XnsIDRAW_TEXT_JUSTIFY** justify new paragraphs except for the last line.
- **XnsIDRAW_TEXT_JUSTIFY_FULL** justify new paragraphs.

XnsINdrawTextAsciiFile

When this resource is set to a file name, the widget replaces its content with the file text. This resource is ignored when the **XnsINdrawTextDtifFile** or **XnsINdrawTextDtifText** resources are set.

XnsINdrawTextAsciiText

When this resource is set to a string, the widget replaces its content with this string. This resource is ignored when the **XnsINdrawTextDtifFile**, the **XnsINdrawTextDtifText** or the **XnsINdrawTextAsciiFile** resources are set.

XnsINdrawTextBegSelectionPosition

Specifies the starting position of the selection.

XnsINdrawTextDtifFile

When this resource is set to a file name, the widget replaces its content with the text in the file. This text is computed by the **DTIF** parser.

XnsINdrawTextDtifText

When this resource is set to a string, the widget replaces its content with the text in the string. This text is computed by the **DTIF** parser. This resource is ignored when the **XnsINdrawTextDtifFile** resource is set.

XnsINdrawTextEndSelectionPosition

Specifies the ending position of the selection.

XnslNdrawTextHorizontalMargin

Specifies the margin in pixels for new paragraphs.

XnslNdrawTextHyperCursor

Specifies the mouse cursor that should be displayed when the mouse pointer is over a hypertext segment. The cursor will change to the one specified in this resource if the **XnslNdrawTextHyperMode** resource is set to **True**, and the mouse pointer lies over a text segment which has a marker with attribute **XnslDrawTextBEG_HYPertext** to its left and a marker with attribute **XnslDrawTextEND_HYPertext** to its right.

XnslNdrawTextHyperMode

If **True**, indicates that hypertext mode is enabled.

XnslNdrawTextInsertionPosition

Indicates the position in the text where the insertion point is located. It is the number of characters from the beginning of the text, the position of the first character being 0.

XnslNdrawTextIsModified

If **True**, indicates that the widget has been modified by user interaction.

XnslNdrawTextIsSelected

If **True**, indicates that a text segment is selected.

XnslNdrawTextMoveCallback

Specifies the list of callbacks invoked when an event causes the position of the insertion point to change. The reason for the callback is **XnslCR_DRAW_TEXT_MOVE**.

XnslNdrawTextParagraphSpacing

Specifies, in pixels, the spacing between paragraphs. If set to -1 the paragraph spacing is computed from the font information of the previous paragraph.

XnslNdrawTextReadOnly

If this resources is set to **True**, the user can not edit the widget.

XnsIDrawTextResizePolicy

Specifies how the widget reacts when its size changes, and where the text is to be displayed within the widget. Acceptable values are:

- **XnsIDRAW_TEXT_CHANGE_ALL** The bounding box of the text is centered in the widget. When the widget is resized, the default ruler is also resized to fit in the new width of the widget.
- **XnsIDRAW_TEXT_CHANGE_NONE** The first character of the text is displayed in the upper left corner of the widget. When the widget is resized, the text position remains unchanged.
- **XnsIDRAW_TEXT_CHANGE_RULER** The first character of the text is displayed in the upper left corner of the widget. When the widget is resized, the default ruler is also resized to fit in the new width of the widget.
- **XnsIDRAW_TEXT_CHANGE_NONE_ON_TRANSFORM** The first character of the text is displayed in the upper left corner of the widget. When the widget is resized, the default ruler is also resized to fit in the new width of the widget. If the widget is transformed, the text position remains unchanged.

XnsIDrawTextShowCursor

If **True**, the insertion point cursor is visible.

XnsIDrawTextTransformText

If **False**, no geometric transformations are applied to the text for Postscript output.

XnsIDrawTextVerifyCallback

Specifies the list of callbacks invoked when an event requests text modifications. The structure whose address is passed to this callback is **XnsIDrawTextVerifyCallbackStruct**.

The reason for this callback is **XnsICR_DRAW_TEXT_INSERT_VERIFY** when a user attempts to insert characters. The **string** field of the structure is set to the string that is about to be inserted; it is null terminated. If the **doit** field is set to **False**, the string is not inserted. The **pos** field gives the current insertion position.

The reason is **XnsICR_DRAW_TEXT_INSERT** when characters have been inserted; the **pos** field gives the new insertion position.

The reason is **XnsICR_DRAW_TEXT_DELETE_VERIFY** when a user attempts to remove characters. The **count** field is set to the number of charac-

ters that are about to be removed. If the `doit` field is set to `False`, no characters are removed. The `pos` field gives the current insertion position.

The reason is `XnsICR_DRAW_TEXT_DELETE` when characters have been removed. The `pos` field gives the new insertion position.

XnsINdrawTextVerticalMargin

Specifies the distance, in pixels, from the top of the widget to the top of the first line of text.

19.4.1 Inherited Resources

XnsIDrawText inherits the following resources from the named superclass. For a complete description of each resource, refer to the man page for that superclass.

XnsIDrawRect Resource Set

Name Class	Default Type	Access
XnsINdrawRadius XnsICDrawRadius	0.0 XtRFloat	CSG

XnsIDrawObject Resource Set

Name Class	Default Type	Access
XnsINdrawActivateCallback XnsICDrawActivateCallback	NULL XtRCallback	C
XnsINdrawActivateCallbackList XnsICDrawActivateCallbackList	NULL XnsIRDrawActivateCallbackList	CSG
XnsINdrawActivateCondition XnsICDrawActivateCondition	XnsIDRAW_INSIDE XnsIRDrawSelectCondition	CSG
XnsINdrawAddPoint XnsICDrawAddPoint	NULL XnsIRDrawAddPoint	CSG
XnsINdrawAlignment XnsICDrawAlignment	XnsIDRAW_MIDDLE XnsIRDrawAlignment	CSG
XnsINdrawAngle XnsICDrawAngle	NULL XtRFloat	CSG

XnsIDrawObject Resource Set

Name Class	Default Type	Access
XnsINdrawAngleUnits XnsICDrawAngleUnits	XnsIRADIANS XnsIRDrawAngleUnits	CSG
XnsINdrawBackground XnsICDrawPixel	-1 XnsIRDrawPixel	CSG
XnsINdrawBackgroundPixmap XnsICDrawBackgroundPixmap	-1 XnsIRDrawImage	CSG
XnsINdrawBlinkTime XnsICDrawBlinkTime	0 XtRInt	CSG
XnsINdrawBottomMargin XnsICDrawBottomMargin	5 XtRInt	CSG
XnsINdrawClipContents XnsICDrawClipContents	False XtRBoolean	CSG
XnsINdrawCloseMode XnsICDrawCloseMode	XnsIDRAW_OPEN XnsIRDrawCloseMode	CSG
XnsINdrawDashes XnsICDrawDashes	0 XnsIRDrawDashes	CSG
XnsINdrawDefaultFontName XnsICdrawDefaultFontName	“Fixed*” String	CSG
XnsINdrawEventMask XnsICDrawEventMask	BUTTON1DOWN XnsIRDrawEventMask	CSG
XnsINdrawFillDir XnsICDrawFillDir	XnsIDRAW_FILL_BOTTOM XnsIRDrawFillDir	CSG
XnsINdrawFillMode XnsICDrawFillMode	XnsIDRAW_NON_FILLED XnsIRDrawFillMode	CSG
XnsINdrawFont XnsICDrawFont	-1 XnsIRDrawFont	CSG
XnsINdrawForeground XnsICDrawPixel	-1 XnsIRDrawPixel	CSG
XnsINdrawFrameBackground XnsICDrawPixel	-1 XnsIRDrawPixel	CSG
XnsINdrawFrameDashes XnsICDrawDashes	0 XnsIRDrawDashes	CSG
XnsINdrawFrameForeground XnsICDrawPixel	-1 XnsIRDrawPixel	CSG

XnslDrawObject Resource Set

Name Class	Default Type	Access
XnslNdrawFrameFunction XnslCDrawFunction	GXcopy XnslRDrawFunction	CSG
XnslNdrawFrameLineWidth XnslCDrawLineWidth	0 XnslRInt	CSG
XnslNdrawFrameStipple XnslCDrawStipple	-1 XnslRDrawStipple	CSG
XnslNdrawFrameXtSOrigin XnslCDrawXTSOrigin	0 XnslRInt	CSG
XnslNdrawFrameYtSOrigin XnslCDrawYTSOrigin	0 XnslRInt	CSG
XnslNdrawFunction XnslCDrawFunction	GXcopy XnslRDrawFunction	CSG
XnslNdrawGeoAttributes XnslCDrawGeoAttributes	NULL XnslRDrawGeoAttributes	G
XnslNdrawGlobalVariables XnslCDrawGlobalVariables	True Boolean	CSG
XnslNdrawGraphAttributes XnslCDrawGraphAttributes	NULL XnslRDrawGraphAttributes	G
XnslNdrawHeight XtCHeight	MAX_DIM XtRDimension	CSG
XnslNdrawInheritAttributes XnslCDrawInheritAttributes	XnslDRAW_INHERIT_ATTRIBUTES XnslRDrawInheritAttributes	CSG
XnslNdrawLeftMargin XnslCDrawLeftMargin	5 XtRInt	CSG
XnslNdrawLineWidth XnslCDrawLineWidth	0 XnslRInt	CSG
XnslNdrawModifyCallback XnslCDrawModifyCallback	NULL XtRCallback	C
XnslNdrawModifyVerifyCallback XnslCDrawModifyVerifyCallback	NULL XtRCallback	C
XnslNdrawName XnslCDrawName	NULL XtRString	CSG
XnslNdrawNthUserData XnslCDrawNthUserData	0 XnslRInt	CSG

XnsIDrawObject Resource Set

Name Class	Default Type	Access
XnsINdrawNumPoints XnsICDrawNumPoints	0 XnsIRInt	CSG
XnsINdrawNumReplacePoints XnsICDrawNumReplacePoints	0 XnsIRInt	CSG
XnsINdrawObjectTolerance XtCWidth	4 XtRInt	CSG
XnsINdrawOpaque XnsICDrawOpaque	False XtRBoolean	CSG
XnsINdrawParent XnsICDrawParent	NULL XnsIRObject	CSG
XnsINdrawPctFill XnsICDrawPctFill	0 XnsIRInt	CSG
XnsINdrawPoints XnsICDrawPoints	NULL XnsIRDrawPoints	CSG
XnsINdrawRecomputeSize XnsICDrawRecomputeSize	True XtRBoolean	CSG
XnsINdrawRegular XnsICDrawRegular	False XtRBoolean	CSG
XnsINdrawRemovePoint XnsICDrawRemovePoint	-1 XnsIRInt	CSG
XnsINdrawReplacePoints XnsICDrawReplacePoints	NULL XnsIRDrawReplacePoints	CSG
XnsINdrawResizeCallback XnsICDrawResizeCallback	NULL XtRCallback	C
XnsINdrawResizeLineWidth XnsICDrawResizeLineWidth	True XtRBoolean	CSG
XnsINdrawResizePolicy XnsICDrawResizePolicy	XnsIDRAW_ALIGN XnsIRDrawResizePolicy	CSG
XnsINdrawRightMargin XnsICDrawRightMargin	5 XtRInt	CSG
XnsINdrawScriptEnabled XnsICDrawScriptEnabled	True XtRBoolean	CSG
XnsINdrawSelectCondition XnsICDrawSelectCondition	XnsIDRAW_ON_BORDER XnsIRDrawSelectCondition	CSG

XnslDrawObject Resource Set

Name Class	Default Type	Access
XnslNdrawSelectMode XnslCDrawSelectMode	XnslDRAW_TRACK_POINTS XnslRDrawSelectMode	CSG
XnslNdrawShowBorder XnslCDrawShowBorder	False XtRBoolean	CSG
XnslNdrawShowFrame XnslCDrawShowFrame	False XtRBoolean	CSG
XnslNdrawStipple XnslCDrawStipple	-1 XnslRDrawStipple	CSG
XnslNdrawStrictFrame XnslCDrawStrictFrame	False XtRBoolean	CSG
XnslNdrawStringValue XnslCDrawStringValue	NULL XtRString	CSG
XnslNdrawTopMargin XnslCDrawTopMargin	5 XtRInt	CSG
XnslNdrawTrackMode XnslCDrawTrackMode	XnslDRAW_SELECT_POINTS XnslRDrawTrackMode	CSG
XnslNdrawUnits XnslCDrawUnits	XnslPIXELS XnslRDrawUnits	CSG
XnslNdrawUserData XnslCDrawUserData	NULL XnslRDrawUserData	CSG
XnslNdrawUseRgbCube XnslCDrawUseRgbCube	False XtRBoolean	CG
XnslNdrawWidth XtCWidth	MAX_DIM XtRDimension	CSG
XnslNdrawX XtCPosition	MAX_POS XtRPosition	CSG
XnslNdrawXCenter XnslCDrawXCenter	NULL XtRFloat	CSG
XnslNdrawXPostScale XnslCDrawXPostScale	NULL XtRFloat	CSG
XnslNdrawXPreScale XnslCDrawXPreScale	NULL XtRFloat	CSG
XnslNdrawXTranslation XnslCDrawXTranslation	NULL XtRFloat	CSG

XnslDrawObject Resource Set

Name Class	Default Type	Access
XnslNdrawXtSOrigin XnslCDrawXTSOrigin	0 XnslRInt	CSG
XnslNdrawY XtCPosition	MAX_POS XtRPosition	CSG
XnslNdrawYCenter XnslCDrawYCenter	NULL XtRFloat	CSG
XnslNdrawYPostScale XnslCDrawYPostScale	NULL XtRFloat	CSG
XnslNdrawYPreScale XnslCDrawYPreScale	NULL XtRFloat	CSG
XnslNdrawYTranslation XnslCDrawYTranslation	NULL XtRFloat	CSG
XnslNdrawYtSOrigin XnslCDrawYTSOrigin	0 XnslRInt	CSG

RectObj Resource Set

Name Class	Default Type	Access
XmNancestorSensitive XmCSensitive	dynamic Boolean	G
XmNborderWidth XmCBorderWidth	0 Dimension	CSG
XmNheight XmCHeight	dynamic Dimension	CSG
XmNsensitive XmCSensitive	True Boolean	CSG
XmNwidth XmCWidth	dynamic Dimension	CSG
XmNx XmCPosition	0 Position	CSG
XmNy XmCPosition	0 Position	CSG

Object Resource Set

Name Class	Default Type	Access
XmNdestroyCallback XmCCallback	NULL XtCallbackList	C

19.5 Callback Information

19.5.1 XnsIDrawTextVerifyCallback

A pointer to the following structure is passed to the **XnsIDrawTextVerifyCallback**. Not all fields are relevant for each reason.

```
typedef struct {
    int
    reason;
    XEvent
    *event;
    Boolean
    doit;
    XnsIDrawTextPosition
    pos;
    int
    count;
    String
    string;
} XnsIDrawTextVerifyCallbackStruct, *XnsIDrawTextVerifyCallback;
```

- *reason* Indicates why the callback was invoked.
- *event* Pointer to the **XEvent** that triggered the callback.
- *doit* Indicates whether the action that invoked the callback is performed. Setting *doit* to False disables the action.
- *pos* Position of the insertion point.
- *count* Number of characters that are about to be removed.
- *string* Pointer to the string that is about to be inserted.

19.5.2 XnsIDrawTextMoveCallback

A pointer to the following structure is passed to the **XnsIDrawTextMoveCallback**.

```
typedef struct {
    int
    XEvent
    XnsIDrawTextPosition
} XnsIDrawTextMoveCallbackStruct, *XnsIDrawTextMoveCallback;
```

- *reason* Indicates why the callback was invoked.
- *event* Pointer to the **XEvent** that triggered the callback.

- *pos* Position of the insertion point.

19.6 Bindings

The **XnsIDrawText** widget uses the following bindings:

- m **<Key>OsfBackSpace** Remove the character to the left of the insertion point.
- m **<Key>OsfCancel** Cancel the selection
- m **<Key>OsfClear** Clear the contents of the widget.
- m **<Key>OsfCut** Cut the selection.
- m **<Key>OsfDelete** Remove the character to the right of the insertion point.
- m **<Key>OsfDown** Move the insertion point to the next line.
- m **<Key>OsfLeft** Move the insertion point one character to the left.
- m **<Key>OsfPaste** Paste the last selection removed.
- m **<Key>OsfRight** Move the insertion point one character to the right.
- m **<Key>OsfUp** Move the insertion point to the previous line.
- m **<Button1Down>** Set the insertion point according to the pointer coordinates.
- m **<Button1Motion>** Extend the selection.
- m **Shift <Button1Up>** Extend the selection according to the pointer coordinates.
- m **<Button1Up>** Set the selection if there is one.
- m **<Button2Down>** Paste the selection at the insertion point.

CHAPTER 20: The XnsldrawTextWidget Class

20.1 Synopsis

#include <Xnsldraw.h>

20.2 Classes

The **XnsldrawTextWidget** inherits resources and behavior from the **Core** and **XmPrimitive** classes.

The class pointer is **XnsldrawTextWidgetClass**.

The class name is **XnsldrawTextWidget**.

20.3 Description

The **XnsldrawTextWidget** widget allows text to be displayed and/or edited. The text can consist of several paragraphs. Each paragraph has its own justification mode, ruler (margins and tabulation positions) and line spacing. Paragraphs in the **XnsldrawTextWidget** widget are separated by the `newline` character.

Different segments of the text can be displayed with specific foreground and background colors, and with specific fonts. Text segments can be underlined and/or overlined. The text can contain subscripts and superscripts.

This widget can also load and/or save ASCII text and files. It can read and save **DTIF** format files. The **DTIF**¹ format is a human readable format that describes the text and the various attributes used to display the text.

This widget provides several callback lists to verify and manage text modification and movement of the insertion position.

When the parent of the **XnsldrawTextWidget** is an **XmScrolledWindow** the **scrollingPolicy** is forced to *application_defined*, the widget manages its own scrolling.

The user can select text segments, delete text, and move text using keys.

1. **Draw Text Interchange Format** (See “DTIF grammar” on page 230.)

Text segments can be instructed to invoke a callback when the user clicks on them with the left mouse button. This implements hypertext behavior.

20.4 Resources

The **XnsIDrawTextWidget** widget class defines the new set of resources listed in the table below.

- C indicates that the resource can be set at creation time;
- S indicates that the resource can be set using XtSetValues;
- G indicates that the resource can be retrieved using XtGetValues

XnsIDrawTextWidget Resource Set

Name Class	Default Type	Access
XnsINdrawBackground XnsICDrawPixel	-1 XnsIRDrawPixel	CSG
XnsINdrawFont XnsICDrawFont	-1 XnsIRDrawFont	CSG
XnsINdrawForeground XnsICDrawPixel	-1 XnsIRDrawPixel	CSG
XnsINdrawTextActivateCallback XnsICDrawTextActivateCallback	NULL XtRCallback	C
XnsINdrawTextAlignment XnsICDrawTextAlignment	XnsIDRAW_TEXT_ALIGN_LEFT XnsIRDrawTextAlignment	CSG
XnsINdrawTextAsciiFile XnsICDrawTextAsciiFile	NULL XnsINdrawTextAsciiFile	CS
XnsINdrawTextAsciiText XnsICDrawTextAsciiText	NULL XnsIRDrawTextAsciiText	CSG
XnsINdrawTextBegSelectionPosition XnsICDrawTextBegSelectionPosition	0 XnsIRDrawTextPosition	G
XnsINdrawTextDtiffFile XnsICDrawTextDtiffFile	NULL XnsIRDrawTextDtiffFile	CS
XnsINdrawTextDtiffText XnsICDrawTextDtiffText	NULL XnsIRDrawTextDtiffText	CSG
XnsINdrawTextEndSelectionPosition XnsICDrawTextEndSelectionPosition	0 XnsIRDrawTextPosition	G

XnsIDrawTextWidget Resource Set

Name Class	Default Type	Access
XnsINdrawTextHorizontalMargin XnsICDrawTextHorizontalMargin	5 XtRInt	CSG
XnsINdrawTextHyperCursor XtCCursor	“hand1” Cursor	CSG
XnsINdrawTextHyperMode XnsICDrawTextHyperMode	False XtRBoolean	CSG
XnsINdrawTextInsertionPosition XnsICDrawTextInsertionPosition	0 XnsIRDrawTextPosition	CSG
XnsINdrawTextIsModified XnsICDrawTextIsModified	False XtRBoolean	CSG
XnsINdrawTextIsSelected XnsICDrawTextIsSelected	False XtRBoolean	G
XnsINdrawTextMoveCallback XnsICDrawTextMoveCallback	NULL XtRCallback	C
XnsINdrawTextParagraphSpacing XnsICDrawTextParagraphSpacing	-1 XtRInt	CSG
XnsINdrawTextReadOnly XnsICDrawTextReadOnly	False XtRBoolean	CSG
XnsINdrawTextResizePolicy XnsICDrawTextResizePolicy	XnsIDRAW_TEXT_CHANGE_RULER XnsIRDrawTextResizePolicy	CSG
XnsINdrawTextShowCursor XnsICDrawTextShowCursor	True XtRBoolean	CSG
XnsINdrawTextStandBy XnsICDrawTextStandBy	False XtRBoolean	SG
XnsINdrawTextTransformText XnsICDrawTextTransformText	True XtRBoolean	CSG
XnsINdrawTextVerifyCallback XnsICDrawTextVerifyCallback	NULL XtRCallback	C
XnsINdrawTextVerticalMargin XnsICDrawTextVerticalMargin	5 XtRInt	CSG

XnsIDrawBackground

Specifies the widget's default text background color. Its value is the index returned by the function **XnsIDrawGetColor**. If this resource is not set, the default text background color is the widget's background color.

XnsIDrawFont

Specifies the default text font used by the widget. It is the value returned by the function **XnsIDrawGetFont**. If this resource is not set, the default font is implementation dependent, according to the X library.

XnsIDrawForeground

Specifies the widget's default text foreground color. Its value is the index returned by the function **XnsIDrawGetColor**. If this resource is not set the default text foreground color is the widget's foreground.

XnsIDrawTextActivateCallback

This callback is invoked when the **XnsIDrawTextHyperMode** resource is set to True and when the user clicks the left mouse button on a text segment that lies between two text markers, with attributes **XnsIDrawTextBEG_HYPertext** (to the left of the text) and **XnsIDrawTextEND_HYPertext** (to the right). The user is informed that the mouse pointer is over such text segments by the changing cursor; the cursor becomes the one specified by the **XnsIDrawTextHyperCursor** resource. The reason for this callback is **XnsICR_DRAW_TEXT_ACTIVATE**.

XnsIDrawTextAlignment

Specifies the current value for text alignment. Any new paragraph created is aligned in conformance with the setting of this resource. Acceptable values are:

- **XnsIDRAW_TEXT_ALIGN_LEFT** flush new paragraphs left.
- **XnsIDRAW_TEXT_ALIGN_RIGHT** flush new paragraphs right.
- **XnsIDRAW_TEXT_ALIGN_CENTER** center new paragraphs .
- **XnsIDRAW_TEXT_JUSTIFY** justify new paragraphs except for the last line.
- **XnsIDRAW_TEXT_JUSTIFY_FULL** justify new paragraphs.

XnsIDrawTextAsciiFile

When this resource is set to a file name, the widget replaces its content with the file text. This resource is ignored when the **XnsIDrawTextDtifFile** or **XnsIDrawTextDtifText** resources are set.

XnsINdrawTextAsciiText

When this resource is set to a string, the widget replaces its content with this string. This resource is ignored when the **XnsINdrawTextDtifFile**, the **XnsINdrawTextDtifText** or the **XnsINdrawTextAsciiFile** resources are set.

XnsINdrawTextBegSelectionPosition

Specifies the starting position of the selection; it is the number of characters from the beginning of the text, the first character being at position 0.

XnsINdrawTextDtifFile

When this resource is set to a file name, the widget replaces its content with the text in the file. This text is computed by the **DTIF** parser.

XnsINdrawTextDtifText

When this resource is set to a string, the widget replaces its content with the text in the string. This text is computed by the **DTIF** parser. This resource is ignored when the **XnsINdrawTextDtifFile** resource is set.

XnsINdrawTextEndSelectionPosition

Specifies the ending position of the selection; it is the number of characters from the beginning of the text, the first character being at position 0.

XnsINdrawTextHorizontalMargin

Specifies the margin in pixels for new paragraphs.

XnsINdrawTextHyperCursor

Specifies the mouse cursor that should be displayed when the mouse pointer is over a hypertext segment. The cursor will change to the one specified in this resource if the **XnsINdrawTextHyperMode** resource is set to True, and the mouse pointer lies over a text segment which has a marker with attribute **XnsIDrawTextBEG_HYPertext** to its left and a marker with attribute **XnsIDrawTextEND_HYPertext** to its right.

XnsINdrawTextHyperMode

If **True**, indicates that hypertext mode is enabled.

XnsINdrawTextInsertionPosition

Indicates the position in the text where the insertion point is located. It is the number of characters from the beginning of the text, the position of the first character being 0.

XnsIDrawTextIsModified

If **True**, indicates that the widget has been modified by user interaction.

XnsIDrawTextIsSelected

If **True**, indicates that a text segment is selected.

XnsIDrawTextMoveCallback

Specifies the list of callbacks invoked when an event causes the position of the insertion point to change. The reason for the callback is **XnsIDRAW_TEXT_MOVE**.

XnsIDrawTextParagraphSpacing

Specifies, in pixels, the spacing between paragraphs. If set to -1 the paragraph spacing is computed from the font information of the previous paragraph.

XnsIDrawTextReadOnly

If set to **True**, the user is not allowed to edit the widget's contents.

XnsIDrawTextResizePolicy

Specifies how the widget reacts when its size changes, and where the text is to be displayed within the widget. Acceptable values are:

- **XnsIDRAW_TEXT_CHANGE_ALL** The bounding box of the text is centered in the widget. When the widget is resized, the default ruler is also resized to fit in the new width of the widget.
- **XnsIDRAW_TEXT_CHANGE_NONE** The first character of the text is displayed in the upper left corner of the widget. When the widget is resized, the text position remains unchanged.
- **XnsIDRAW_TEXT_CHANGE_RULER** The first character of the text is displayed in the upper left corner of the widget. When the widget is resized, the default ruler is resized to fit in the new width of the widget.
- **XnsIDRAW_TEXT_CHANGE_NONE_ON_TRANSFORM** The first character of the text is displayed in the upper left corner of the widget. When the widget is resized, the default ruler is resized to fit in the new width of the widget. If the widget is transformed, the text position remains unchanged.

XnsIDrawTextShowCursor

If set to **True**, the insertion point cursor is visible.

XnsIDrawTextStandBy

When this resource is set to **True**, the widget does not do any internal calculations, does not accept user input, does not refresh. Use this resource to optimize text modifications such as adding text, changing fonts, changing colors, etc Set it to **True** before issuing text modification commands, do the commands and set the resource to **False** to apply all the commands at once.

XnsINdrawTextTransformText

If **False**, no geometric transformations are applied to the text for Postscript output.

XnsINdrawTextVerifyCallback

Specifies the list of callbacks invoked when an event requests text modifications. The structure whose address is passed to this callback is **XnsIDrawTextVerifyCallbackStruct**.

The reason for this callback is **XnsICR_DRAW_TEXT_INSERT_VERIFY** when a user attempts to insert characters. The **string** field of the structure is set to the string that is about to be inserted; it is null terminated. If the **doit** field is set to **False**, the string will not be inserted. The **pos** field gives the current insertion position.

The reason is **XnsICR_DRAW_TEXT_INSERT** when characters have been inserted; the **pos** field gives the new insertion position.

The reason is **XnsICR_DRAW_TEXT_DELETE_VERIFY** when a user attempts to remove characters. The **count** field is set to the number of characters that are about to be removed. If the **doit** field is set to **False**, no characters will be removed. The **pos** field gives the current insertion position.

The reason is **XnsICR_DRAW_TEXT_DELETE** when characters have been removed. The **pos** field gives the new insertion position.

XnsINdrawTextVerticalMargin

Specifies the distance, in pixels, from the top of the widget to the top of the first line of text.

20.4.1 Inherited Resources

XnsIDrawTextWidget inherits the following resources from the named superclass. For a complete description of each resource, refer to the man page for that superclass

XmPrimitive Resource Set

Name Class	Default Type	Access
XmNbottomShadowColor XmCBottomShadowColor	dynamic Pixel	CSG
XmNbottomShadowPixmap XmCBottomShadowPixmap	XmUNSPECIFIED_PIXMAP Pixmap	CSG
XmNforeground XmCForeground	dynamic Pixel	CSG
XmNhelpCallback XmCCallback	NULL XtCallbackList	C
XmNhighlightColor XmCHighlightColor	dynamic Pixel	CSG
XmNhighlightOnEnter XmCHighlightOnEnter	False Boolean	CSG
XmNhighlightPixmap XmCHighlightPixmap	dynamic Pixmap	CSG
XmNhighlightThickness XmCHighlightThickness	2 Dimension	CSG
XmNnavigationType XmCNavigationType	XmTAB_GROUP XmNavigationType	CSG
XmNshadowThickness XmCShadowThickness	2 Dimension	CSG
XmNtopShadowColor XmCTopShadowColor	dynamic Pixel	CSG
XmNtopShadowPixmap XmCTopShadowPixmap	dynamic Pixmap	CSG
XmNtraversalOn XmCTraversalOn	True Boolean	CSG
XmNunitType XmCUnitType	unsigned char dynamic	CSG
XmNuserData XmCUserData	NULL XtPointer	CSG

Core Resource Set

Name Class	Default Type	Access
XmNaccelerators XmCAccelerators	dynamic XtAccelerators	N/A
XmNancestorSensitive XmCSensitive	dynamic Boolean	G
XmNbackground XmCBackground	dynamic Pixel	CSG
XmNbackgroundPixmap XmCPixmap	XmUNSPECIFIED_PIXMAP Pixmap	CSG
XmNborderColor XmCBorderColor	XtDefaultForeground Pixel	CSG
XmNborderPixmap XmCPixmap	XmUNSPECIFIED_PIXMAP Pixmap	CSG
XmNborderWidth XmCBorderWidth	0 Dimension	CSG
XmNcolormap XmCColormap	dynamic Colormap	CG
XmNdepth XmCDepth	dynamic int	CG
XmNdestroyCallback XmCCallback	NULL XtCallbackList	C
XmNheight XmCHeight	dynamic Dimension	CSG
XmNinitialResourcesPersistent XmCInitialResourcesPersistent	True Boolean	C
XmNmappedWhenManaged XmCMappedWhenManaged	True Boolean	CSG
XmNscreen XmCScreen	dynamic Screen	CG
XmNsensitive XmCSensitive	True Boolean	CSG
XmNtranslations XmCTranslations	dynamic XtTranslations	CSG

Core Resource Set

Name Class	Default Type	Access
XmNwidth XmCWidth	dynamic Dimension	CSG
XmNx XmCPosition	0 Position	CSG
XmNy XmCPosition	0 Position	CSG

20.5 Callback Information

20.5.1 XnsINdrawTextActivateCallback

A pointer to the following structure is passed to the **XnsINdrawTextActivateCallback**.

```
typedef struct {
    int                reason;
    XEvent             *event;
    XnsIDrawTextPosition from;
    XnsIDrawTextPosition to;
    XtPointer          hyper_data;
} XnsIDrawTextActivateCallbackStruct;
```

- *reason* Indicates why the callback was invoked.
- *event* Pointer to the **XEvent** that triggered the callback.
- *from* Starting position of the text segment.
- *to* Ending position of the text segment
- *hyper_data* **user_data** field of the **XnsIDrawTextMark** structure at *from* position. This mark contains the **XnsIDrawTextBEG_HYPertext** attribute.

20.5.2 XnsINdrawTextVerifyCallback

A pointer to the following structure is passed to the **XnsINdrawTextVerifyCallback**. Not all fields are relevant for each reason.

```
typedef struct {
```

```

        int                reason;
        XEvent             *event;
        Boolean            doit;
        XnsIDrawTextPosition pos;
        int                count;
        String             string;
    } XnsIDrawTextVerifyCallbackStruct;

```

- *reason* Indicates why the callback was invoked.
- *event* Pointer to the **XEvent** that triggered the callback.
- *doit* Indicates whether the action that invoked the callback is to be performed. Setting *doit* to False disables the action.
- *pos* Position of the insertion point.
- *count* Number of characters that are about to be removed.
- *string* Pointer to the string that is about to be inserted.

20.5.3 XnsIDrawTextMoveCallback

A pointer to the following structure is passed to the **XnsIDrawTextMoveCallback**.

```
typedef struct {
    int                    reason;
    XEvent                 *event;
    XnsIDrawTextPosition   pos;
} XnsIDrawTextMoveCallbackStruct;
```

- *reason* Indicates why the callback was invoked.
- *event* Pointer to the **XEvent** that triggered the callback.
- *pos* Position of the insertion point.

20.6 Bindings

The **XnsIDrawTextWidget** widget uses the following bindings:

- m **<Key>OsFBackSpace** Remove the character to the left of the insertion point.
- m **<Key>OsFCancel** Cancel the selection.
- m **<Key>OsFClear** Clear the contents of the widget.
- m **<Key>OsFCut** Cut the selection.
- m **<Key>OsFDelete** Remove the character to the right of the insertion point.
- m **<Key>OsFLeft** Move the insertion point one character to the left.
- m **<Key>OsFPaste** Paste the last selection removed.
- m **<Key>OsFRight** Move the insertion point one character to the right.
- m **<Key>OsFPageUp** Move the insertion point to the top of the previous page.
- m **<Key>OsFPageDown** Move the insertion point to the top of the next page.
- m **<Key>OsFUp** Move the insertion point to the previous line.
- m **<Key>OsFDown** Move the insertion point to the next line.
- m **<Button1Down>** Set the insertion point according to the pointer coordinates.
- m **<Button1Motion>** Extend the selection.
- m **Shift <Button1Up>** Extend the selection according to the pointer coordinates.
- m **<Button1Up>** Set the selection if there is one.
- m **<Button2Down>** Paste the selection at the insertion point.

20.7 Translations

The **XnsIDrawTextWidget** translations are listed below:

<FocusIn>:	focus-in()
<FocusOut>:	focus-out()
<Enter>:	enter()
<Leave>:	leave()

20.8 Actions

focus-in: Draws the insertion cursor as solid. If the shell's keyboard focus policy is **XmEXPLICIT**, highlights the widget and gives it the focus.

focus-out: Displays the insertion cursor as a stippled I-beam. If the shell's keyboard focus policy is **XmEXPLICIT**, un-highlights the widget and removes the focus.

enter: If the shell's keyboard focus policy is not **XmEXPLICIT**, draws the insertion cursor as solid.

leave: If the shell's keyboard focus policy is not **XmEXPLICIT**, displays the insertion cursor as a stippled I-beam.

20.9 Structures

20.9.1 The XnsIDrawTextMark Structure

To draw text with different attributes, the **XnsIDrawTextWidget** widget uses marks. A mark is attached at a specific position in the text and contains information on the foreground and background color, on the font, etc. that are to be used to draw the text starting from that position.

For example, to change the font of the text between position 10 and position 25, there must be a mark at position 10 with the information to use a new font, and a mark at position 25 with the information to revert to the previous font.

```
typedef struct _XnsIDrawTextMark{
    XnsIDrawTextPosition    position;
    String                  name;
    XtPointer               user_data;
    unsigned long           mask;
    int                     num_font;
    int                     num_fg;
    int                     num_bg;
    int                     superscript_offset;
    int                     subscript_offset;
    int                     underline_offset;
    int                     overline_offset;
    struct _XnsIDrawTextMark * end_mark;
} XnsIDrawTextMark;
```

- *position* The position of the mark; an offset in number of characters from the beginning of the text.
- *name* The name of the mark.
- *user_data* Reserved for the application. This field is not used internally by the widget.
- *mask* The contents mask. It is the logical OR of the following values
 - **XnsIDrawTextDEFAULT** Do nothing special;
 - **XnsIDrawTextFONT** use the value of the *num_font* field for drawing;
 - **XnsIDrawTextFOREGROUND** use the value of the *num_fg* field for drawing;
 - **XnsIDrawTextBACKGROUND** use the value of the *num_bg* field for

drawing;

- **XnslDrawTextBEG_SUPERSCRIPT** start to draw the text in superscript mode at the *superscript_offset* from the top of the line;
- **XnslDrawTextEND_SUPERSCRIPT** stop drawing the text in superscript;
- **XnslDrawTextBEG_SUBSCRIPT** start to draw the text in subscript mode at the *subscript_offset* from the baseline;
- **XnslDrawTextEND_SUBSCRIPT** stop drawing the text in subscript;
- **XnslDrawTextBEG_UNDERLINE** start to underline the text at *underline_offset* from the baseline;
- **XnslDrawTextEND_UNDERLINE** stop underlining;
- **XnslDrawTextBEG_OVERLINE** start to overline the text at *overline_offset* from the top of the line;
- **XnslDrawTextEND_OVERLINE** stop overlining;
- **XnslDrawTextBEG_HYPertext** Start a hypertext text segment;
- **XnslDrawTextEND_HYPertext** End a hypertext segment;
- *num_font* The index of the font to use (cf **XnslDrawGetFont**) to draw text. To specify the default font, use the value **XnslDrawTextDEFAULT_FONT**.
- *num_fg* The index of the foreground color to use (cf **XnslDrawStoreColor**, **XnslDrawGetColor**, **XnslDrawIndexPixel**) to draw text. To specify the default foreground color, use the value **XnslDrawTextDEFAULT_FOREGROUND**.
- *num_bg* The index of the background color to use (cf **XnslDrawStoreColor**, **XnslDrawGetColor**, **XnslDrawIndexPixel**) to draw text. To specify the default background color, use the value **XnslDrawTextDEFAULT_BACKGROUND**.
- *superscript_offset* The offset, in pixels from the baseline, for the superscript.
- *subscript_offset* The offset, in pixels from the baseline, for the subscript.
- *underline_offset* The offset, in pixels from the baseline, for underlining.
- *overline_offset* The offset, in pixels from the baseline, for overlining.
- *end_mark* Reserved for future use.

20.9.2 The XnsIDrawTextParAttributes Structure

This structure is used to specify the paragraph attributes.

```
typedef struct {
    unsigned long    mask;
    int              alignment;
    int              ruler;
    int              line_spacing;
    Boolean          enumeration;
} XnsIDrawTextParAttributes;
```

- *mask* The contents mask. It is the logical OR of the following values:
 - **DTAlignment** set the paragraph alignment to the value of the *alignment* field.
 - **DTRuler** set the paragraph margins and tabulation according to the *ruler* field.
 - **DTLineSpacing** set the paragraph line spacing according to the value of the *line_spacing* field.
 - **DTEnumeration** reserved for future use.
- *alignment* The alignment of new paragraphs. This field can take the following values:
 - **XnsIDRAW_TEXT_ALIGN_LEFT**
 - **XnsIDRAW_TEXT_ALIGN_RIGHT**
 - **XnsIDRAW_TEXT_ALIGN_CENTER**
 - **XnsIDRAW_TEXT_JUSTIFY**
 - **XnsIDRAW_TEXT_JUSTIFY_FULL**
- *ruler* Contains the index of the ruler to be used.
- *line_spacing* Contains the new line spacing.
- *enumeration* reserved for future use.

20.10 DTIF grammar

```
file          : widgetdefs rulerdefs pardefs
               | rulerdefs pardefs

widgetdefs : widgetdefs widgetdef
            | widgetdef
```

```

widgetdef : <HorizontalMargin> <=> [int] <\n>
          | <VerticalMargin> <=> [int] <\n>
          | <ParagraphSpacing> <=> [int] <\n>
          | <DefaultAlignment> <=> [string] <\n>
          | <ReadOnly> <=> [string] <\n>
          | <ResizePolicy> <=> [string] <\n>
          | <ShowCursor> <=> [string] <\n>

rulerdefs : <BeginRulerDefinition> <\n> nrul rlist
<EndRulerDefinition> <\n>

nrul      : <NumberOfRuler> <=> [int]

rlist     : rulerlist ruler
          | ruler

ruler     : rulerdescr numlist <\n>
          | rulerdescr <\n>

rulerdescr : <From> <=> [int] <To> <=> [int] <NumTab> <=>
[int]

numlist   : numlist [int]
          | [int]

pardefs   : <BeginParagraphsDefinition> <\n> parlist
<EndParagraphsDefinition> <\n>
          | <BeginParagraphsDefinition> <\n>
<EndParagraphsDefinition> <\n>

parlist   : parlist pardef
          | pardef

pardef    : <BeginParagraph> <\n> paratts parflow
<EndParagraph> <\n>

paratts   : paratts paratt
          | paratt

paratt    : <Alignment> <=> [string] <\n>
          | <Enumeration> <=> [string] <\n>
          | <Linespacing> <=> [int] <\n>

```

```
        | <Ruler> <=> [int] <\n>

    parflow      : <BeginParagraphText> <\n> doc_text
<EndParagraphText> <\n>
        | <BeginParagraphText> <\n> <EndParagraphText>
<\n>

    doc_text     : doc_text text
        | text

    text         : textflow
        | <BeginMark> <\n>

    textflow     : <TextFlow> <\n> <EndText> <\n>
        | <BeginText> <\n> <EndText> <\n>

    markdescpr  : markdescpr markatt
        | markatt

    markatt      : <Font> <=> [string] <\n>
        | <Foreground> <=> [string] <\n>
        | <Background> <=> [string] <\n>
        | <SuperScriptOffset> <=> [int] <\n>
        | <EndSuperScript> <\n>
        | <SubScriptOffset> <=> [int] <\n>
        | <EndSubScript> <\n>
        | <UnderlineOffset> <=> [int] <\n>
        | <EndUnderline> <\n>
        | <OverlineOffset> <=> [int] <\n>
        | <EndOverline> <\n>
        | <BeginHyperText> <\n>
        | <EndHyperText> <\n>
```

CHAPTER 21: Structures

21.1 Object Representation

Graphic objects are defined by a set of **XnslDrawPoint** points. These points are passed through a matrix containing rotation, scale and translation values, to give **XPoints** suitables for drawing. The set of **XnslDrawPoint** is contained in a **XnslDrawPointList** structure. The **XPoints** are stored in a **XnslDrawXPointList** structure.

21.1.1 XnslDrawPoint

```
typedef struct {  
    double          x;  
    double          y;  
} XnslDrawPoint;
```

- *x* X coordinate.
- *y* Y coordinate.

21.1.2 XnslDrawPointList

The **XnslDrawPointList** is used to store a list of **XnslDrawPoint**. *p* is a pointer to an array of points. *np* is the number of valid points in the array. *mp* is the number of allocated points. A point can be added to the array with the **XnslDrawAddDrawPoint** function.

```
typedef struct {  
    XnslDrawPoint  *p;  
    int             np;  
    int             mp;  
} XnslDrawPointList;
```

- *p* XnslDrawPoint list.
- *np* Number of valid points.
- *mp* Size of the list.

21.1.3 XnsIDrawXPointList

```
typedef struct {  
    XPoint          *p;  
    int             np;  
    int             mp;  
} XnsIDrawXPointList;
```

- *p* XPoint list.
- *np* Number of valid points.
- *mp* Size of the list.

21.1.4 XnsIDrawNumberedPoint

The **XnsIDrawNumberedPoint** is used to add a point or to replace a point. It is used by some resources **XnsINdrawAddPoint**, **XnsINdrawReplacePoints**, **XnsINdrawAddSlot**, **XnsINdrawReplaceSlots**.

```
typedef struct {  
    XnsIDrawPoint    p;  
    int              n;  
} XnsIDrawNumberedPoint;
```

- *p* XnsIDrawPoint to replace or add.
- *n* index of the point in the list.

21.1.5 XnsIDrawActivateCallbackList

The **XnsIDrawActivateCallbackList** structure is used to provide a list of **Face** scripts to the **XnsINdrawActivateCallbackList** resource.

```
typedef struct _XnsIDrawActivateCallbackList {
    struct _XnsIDrawActivateCallbackList*next;
    int event;
    XrmQuark q_event;
    char *translation_text;
    int location;
    XrmQuark q_location;
    String script;
    XnsIDrawScriptType script_type
    struct Prog *prog;
    XtPointer closure;
    XtPointer client_data;
    Boolean doit;
    Widget widget;
} XnsIDrawActivateCallbackList;
```

- *next* Used to link the structures. The last item in the list must have the **next** field set to **NULL**;
- *event* Event to match in an int form. (e.g **BUTTON1DOWN**).
- *q_event* The quark corresponding to the event in a string form (e.g XrmStringToQuark("Button1Down")).
- *translation_text* The string corresponding to an action. (e.g "<Key-Press>a").
- *location* The location required to launch the script (e.g XnsIDRAW_INSIDE).
- *q_location* The quark corresponding to the location in a string form.(e.g XrmStringToQuark ("draw_inside"))
- *script* The **Face** script body.
- *script_type* Indicates if the **prog** field is a **Face** prog or a C function. Valid values are:
 - **XnsIDRAW_FACE_SCRIPT**
 - **XnsIDRAW_C_FUNCTION**

The prototype of the C function is the following:

```
void func (Widget w, XtPointer client_data, XEvent *e)
```

- *prog* Either a **Face** prog returned by the **XnsIDrawFace-
ParseProg** function. For example:

`XnslDrawFaceParseProg ("printf (\\"script launched\\");")`

or a pointer to a C function.

- *closure* Internal use.
- *client_data* Client data passed to the C function.
- *doit* Internal use.
- *widget* Internal use.

21.1.6 XnslDrawDoit

The `XnslDrawDoit` enumeration lists the acceptable values for the `doit` field of modify verify callback structures.

```
typedef enum {  
    XnslYes,  
    XnslNo,  
    XnslAlmost  
} XnslDrawDoit;
```

- **XnslYes** Enables the modification.
- **XnslNo** Disables the modification.
- **XnslAlmost** Enables the modification as modified by the application.

21.2 Object Structures

Graphic objects can be created and manipulated by the application by calling functions or by modifying object resources.

Graphic attributes and geometric transformations are defined and stored in structures. They are valid for any object. The first element of these structures is a mask indicating which field is valid in the structure. A resource is defined for each field of these structures.

21.3 Geometric Structures

```
#define      DAXTranslation    (1L << 0)
#define      DAYTranslation    (1L << 1)
#define      DARotation        (1L << 2)
#define      DAXPreScale       (1L << 3)
#define      DAYPreScale       (1L << 4)
#define      DAXPostScale      (1L << 5)
#define      DAYPostScale      (1L << 6)
#define      DAXCenter         (1L << 7)
#define      DAYCenter         (1L << 8)
```

```
typedef struct {
    unsigned long      mask;
    XnslDrawPoint      translation;
    double             angle;
    XnslDrawPoint      pre_scale;
    XnslDrawPoint      post_scale;
    XnslDrawPoint      center;
} XnslDrawGeoAttributes;
```

This structure describes the transformation applied to an object. The transformation includes translation, rotation, scale and definition of the center of an object. The center of an object is the reference point for rotation and scale.

- *mask* The mask is a set of bits indicating which of the fields of the structure are valid. You can affect any combination of the **DAXTranslation**, **DAYTranslation**, **DARotation**, **DAXPreScale**, **DAYPreScale**, **DAXPostScale**, **DAYPostScale**, **DAXCenter**, and **DAYCenter** bits.

- *translation* Value of the translation applied to the object.
- *angle* Value of the rotation applied to the object in radians.
- *pre_scale* Value of the pre scale applied to the object.
- *post_scale* Value of the post scale applied to the object.
- *center* Center of the object. The above transformations are applied using this center as a reference point. If no center is defined, an automatic center is computed which is the center of the smallest rectangle in which the object fits.

21.4 Graphic Structures

The *XnsIDrawGraphAttributes* affects the look of the object. Some graphics attributes are directly taken from a **XGCValues** structure¹.

```
#define      DAForeground      (1L << 0)
#define      DABackground     (1L << 1)
#define      DAFunction       (1L << 2)
#define      DALineWidth      (1L << 3)
#define      DAFont           (1L << 4)
#define      DASTipple        (1L << 5)
#define      DADashes         (1L << 6)
#define      DAXtsOrigin      (1L << 7)
#define      DAYtsOrigin      (1L << 8)
#define      DACloseMode      (1L << 9)
#define      DAFillMode       (1L << 10)
#define      DAFillDir        (1L << 11)
#define      DAPctFill        (1L << 12)

typedef struct {
    unsigned long      mask;
    int                foreground;
    int                background;
    char               function;
    int                lineWidth;
    int                font;
    int                stipple;
    char               *dashes;
    int                xTsOrigin;
    int                yTsOrigin;
    char               closeMode;
    char               fillMode;
    char               fillDir;
    int                pctFill;
    int                sizedLineWidth;
} XnsIDrawGraphAttributes;
```

1. See the Xlib reference manual.

This structure defines the visual aspect of the object.

- *mask* Indicates which fields are valid in the structure.
- *foreground* Foreground color with which the object is drawn. It is an index in the table of colors.
- *background* Background color with which the object is drawn. It is an index in the table of colors.
- *function* Drawing function. Any X graphics function is valid.
- *lineWidth* Width of lines (if the object has lines).
- *font* Font used to draw strings of text (if the object has some). It is an index in a table of fonts.
- *stipple* Stipple to fill the object. If the object defines a background color, the stipple is opaque, otherwise, the stipple is transparent.
- *dashes* Dashes to draw lines. It is a **NULL** terminated string.
- *xTsOrigin*
- *yTsOrigin* If the object defines a stipple, these fields define the origin of the stipple. This is an index in the table of Bitmaps.
- *closeMode* Define the way the object is closed. Possible values are:
 - **XnsIDRAW_OPEN** The object is not closed.
 - **XnsIDRAW_OBJECT_CLOSED** The object is fully closed.
 - **XnsIDRAW_LINE_CLOSED** The end points (Ellipse) are joined.
 - **XnsIDRAW_CENTER_CLOSED** The end points are joined to the center.
- *fillMode* Define the way the object is filled.
 - **XnsIDRAW_NON_FILLED** The object is not filled.
 - **XnsIDRAW_FILLED** The object is filled
 - **XnsIDRAW_PCT_FILLED** The object is partially filled.
- *fillDir* A direction for a partial fill.
 - **XnsIDRAW_FILL_BOTTOM**
 - **XnsIDRAW_FILL_TOP**
 - **XnsIDRAW_FILL_LEFT**
 - **XnsIDRAW_FILL_RIGHT**
- *pctFill* Percent of fill. Range from 0 to 100.
- *sizedLineWidth* Internal use.

For color and Pixmap attributes (Foreground, Background, Stipple, Icon Pixmap, etc...), the library maintains tables of already used values. These tables are maintained when specifying a resource for graphic objects, and can be accessed by special functions in the application. The functions return an index in a table, which is used as the value for the attribute.

21.5 Object Specific Structures

In addition, some object types have particular attributes: Arrows are treated as special attributes for the polygonal objects.

21.5.1 Arrows Structures

```
#define      DAArrowDir      (1L << 0)
#define      DAArrowStyle    (1L << 1)
#define      DAArrowWidth    (1L << 2)
#define      DAArrowHeight   (1L << 3)

typedef struct {
    unsigned      mask;
    char          arrowDir;
    char          arrowStyle;
    int           arrowWidth;
    int           arrowHeight;
} XnsIDrawArrowAttributes;
```

- *mask* Indicates which fields are valid in the structure.
- *arrowDir* Specify the direction of the arrow. Possible values are:
 - **XnsIDRAW_NO_ARROW** No arrow is drawn.
 - **XnsIDRAW_FROM_ARROW** An arrow is drawn at the first point of the object.
 - **XnsIDRAW_TO_ARROW** An arrow is drawn at the last point of the object.
 - **XnsIDRAW_FROM_TO_ARROW** The above two arrows are drawn.
- *arrowStyle* Style of the arrow(s). Possible values are:
 - **XnsIDRAW_ARROW_OPEN** Two segments are drawn to form the arrow.
 - **XnsIDRAW_ARROW_CLOSED** A polygon is drawn to form the arrow.
 - **XnsIDRAW_ARROW_FILLED** A filled polygon is drawn to form the arrow.
- *arrowHeight*
- *arrowWidth* Specify the size of the arrows. The **width** is the length of the arrow. The **height** is the thickness of the arrow.

21.6 Draw Structures

The **Page attributes** structure is used in association with the **XnslNdrawPageSize** resource to give a page description. A convenient way to set the contents of this structure is to use the converter between **String** and **XnslRDrawPageSize**.

```
typedef struct {  
    char *           name;  
    double           width;  
    double           height;  
    double           v_margin;  
    double           h_margin;  
} XnslDrawPageSize;
```

- *name* The name of the page. The converter knows the following page names:
 - a3
 - a4
 - a5
 - a6
 - b4
 - b5
 - tabloid
 - executive
 - legal
 - letter
 - custom
- *width* The width of the page in inches.
- *height* The height of the page in inches.
- *v_margin* The vertical margin in inches.
- *h_margin* The horizontal margin in inches.

If the converter is used, the **v_margin** and the **h_margin** are set to 0.24 inches. If the name of the page is *custom*, the content of the structure is ignored and the **XnslNdrawPageWidth**, **XnslNdrawPageHeight** resources are considered.

21.7 XnslDrawText Structures

21.7.1 The XnsIDrawTextMark Structure

To draw text with different attributes, the **XnsIDrawText** widget uses marks. A mark is attached at a specific position in the text and contains information on the foreground and background color, on the font, etc. that are to be used to draw the text starting from that position.

For example, to change the font of the text between position 10 and position 25, there must be a mark at position 10 with the information to use a new font, and a mark at position 25 with the information to revert to the previous font.

```
typedef struct _XnsIDrawTextMark{
    XnsIDrawTextPosition    position;
    String                  name;
    XtPointer                user_data;
    unsigned long           mask;
    int                     num_font;
    int                     num_fg;
    int                     num_bg;
    int                     superscript_offset;
    int                     subscript_offset;
    int                     underline_offset;
    int                     overline_offset;
    struct _XnsIDrawTextMark * end_mark;
} XnsIDrawTextMark;
```

- *position* The position of the mark; an offset in number of characters from the beginning of the text.
- *name* The name of the mark.
- *user_data* Reserved for the application.
- *mask* The contents mask. It is the OR of the following values
 - **XnsIDrawTextDEFAULT** Do nothing special;
 - **XnsIDrawTextFONT** use the value of the *num_font* field for drawing;
 - **XnsIDrawTextFOREGROUND** use the value of the *num_fg* field for drawing;
 - **XnsIDrawTextBACKGROUND** use the value of the *num_bg* field for drawing;
 - **XnsIDrawTextBEG_SUPERSCRIPT** start to draw the text in superscript mode at the *superscript_offset* from the top of the line;
 - **XnsIDrawTextEND_SUPERSCRIPT** stop drawing the text in super-

script;

- **XnsIDrawTextBEG_SUBSCRIPT** start to draw the text in subscript mode at the *subscript_offset* from the baseline;
- **XnsIDrawTextEND_SUBSCRIPT** stop drawing the text in subscript;
- **XnsIDrawTextBEG_UNDERLINE** start to underline the text at *underline_offset* from the baseline;
- **XnsIDrawTextEND_UNDERLINE** stop underlining;
- **XnsIDrawTextBEG_OVERLINE** start to overline the text at *overline_offset* from the top of the line;
- **XnsIDrawTextEND_OVERLINE** stop overlining;
- **XnsIDrawTextBEG_HYPertext** Reserved for future use;
- **XnsIDrawTextEND_HYPertext** Reserved for future use;
- *num_font* The index of the font to use (cf **XnsIDrawGetFont**). To return to the default font, use the value **XnsIDrawTextDEFAULT_FONT**.
- *num_fg* The index of the foreground color to use (cf **XnsIDrawStoreColor**, **XnsIDrawGetColor**, **XnsIDrawIndexPixel**). To return to the default foreground color, use the value **XnsIDrawTextDEFAULT_FOREGROUND**.
- *num_bg* The index of the background color to use (cf **XnsIDrawStoreColor**, **XnsIDrawGetColor**, **XnsIDrawIndexPixel**). To return to the default foreground use the value **XnsIDrawTextDEFAULT_BACKGROUND**.
- *superscript_offset* The offset, in pixels from the baseline, for the superscript.
- *subscript_offset* The offset, in pixels from the baseline, for the subscript.
- *underline_offset* The offset, in pixels from the baseline, for underlining.
- *overline_offset* The offset, in pixels from the baseline, for overlining.
- *end_mark* Reserved for future use.

21.7.2 The XnsIDrawTextParAttributes Structure

This structure is used to specify the paragraph attributes.

typedef struct {

unsigned long	<i>mask</i> ;
int	<i>alignment</i> ;
int	<i>ruler</i> ;
int	<i>line_spacing</i> ;

```
        Boolean          enumeration;
} XnsIDrawTextParAttributes;
```

- *mask* The contents mask. It is the OR of the following values:
 - **DTAlignment** set the paragraph alignment to the value of the *alignment* field.
 - **DTRuler** set the paragraph margins and tabulation according to the *ruler* field.
 - **DTLineSpacing** set the paragraph line spacing according to the value of the *line_spacing* field.
 - **DTEnumeration** reserved for future use.
- *alignment* The alignment of new paragraphs. This field can take the following values:
 - **XnsIDRAW_TEXT_ALIGN_LEFT**
 - **XnsIDRAW_TEXT_ALIGN_RIGHT**
 - **XnsIDRAW_TEXT_ALIGN_CENTER**
 - **XnsIDRAW_TEXT_JUSTIFY**
 - **XnsIDRAW_TEXT_JUSTIFY_FULL**
- *ruler* Contains the index of the ruler to be used.
- *line_spacing* Contains the new line spacing.
- *enumeration* reserved for future use.

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